

StaffSync – Smart Workforce Management Application



Project ID: Fall-2025-51

Session: BSCS Fall 2022 to 2026

Project Advisor: Sir Arslan Bodla

Submitted By

Ahmad Raza	70134670
Ali Hassnan Amjad	70135148

Department of Computer Science & IT
The University of Lahore
Lahore, Pakistan

Declaration

We have read the project guidelines and we understand the meaning of academic dishonesty, in particular plagiarism and collusion. We hereby declare that the work we submitted for our final year project, entitled **StaffSync** is original work and has not been printed, published or submitted before as final year project, research work, publication or any other documentation.

Group Member 1 Name:

SAP No:

Signature:

Group Member 2 Name:

SAP No:

Signature:

Statement of Submission

This is to certify that **Ahmad Raza** Roll No. **70134670**, **Ali Hassnan** Roll No. **70135148** have successfully submitted the final project named as: **StaffSync**, at Computer Science & IT Department, The University of Lahore, Lahore Pakistan, to fulfill the partial requirement of the degree of **BS in Computer Science**.

Supervisor Name:

Signature:

Date:

Dedication

This project is dedicated to our parents, whose constant prayers, support, and sacrifices have been the foundation of our success. Their encouragement and belief in us motivated us to overcome challenges and remain focused throughout our academic journey.

We also dedicate this project to our teachers and supervisors who guided us with their knowledge, patience, and valuable feedback, helping us shape this project into a meaningful learning experience.

Acknowledgement

We would like to express our sincere gratitude to our project supervisor **Mr. Arsalan Bodla** for his continuous guidance, support, and valuable suggestions throughout the development of this project. His encouragement and feedback played a vital role in shaping our ideas and improving the quality of our work.

We are also thankful to the faculty members of the Department of Computer Science & IT, The University of Lahore, for providing us with the knowledge and resources required to complete this project. Finally, we would like to thank our friends and families for their motivation, patience, and moral support during this journey.

Date:

Dec 23, 2025

Abstract

The purpose of this project is to develop **StaffSync**, a smart workforce management application that helps organizations digitally manage their staff operations. The system is designed to improve efficiency, transparency, and coordination in workplaces that rely on shift-based and field staff.

Many organizations face problems such as manual attendance, inefficient shift scheduling, lack of real-time monitoring, and errors in payroll calculation. These issues lead to poor productivity, increased administrative workload, and communication gaps between management and staff.

To address these problems, StaffSync is developed as a mobile and web-based application that provides features such as smart attendance tracking, role-based access, shift management, task monitoring, and multilingual communication. Modern development tools and cloud-based technologies are used to design and implement the system.

The results of this project include a functional workforce management system that automates attendance and basic staff management processes. The system reduces manual effort, improves accuracy, and provides better control over workforce activities.

In conclusion, StaffSync offers an effective and user-friendly solution for managing staff operations. The project demonstrates how digital and intelligent systems can improve workforce productivity and support organizations across multiple industries.

Area of the Project

The StaffSync project falls under multiple areas of computer science, including **Web Application Development**, **Mobile Application Development**, **Artificial Intelligence**, and **Cloud Computing**. The system also relates to **Workforce Management Systems** and **Human Resource Automation**, as it focuses on managing staff attendance, scheduling, and performance through digital tools.

Technologies used

The StaffSync project is developed using modern and reliable technologies. The backend of the system is built using **.NET Core**, while **Firebase** is used for authentication, real-time database, and cloud services. The mobile application interface is developed using **Flutter**, providing cross-platform support for Android devices.

Additional tools and technologies include **SignalR** for real-time communication, **ML.NET** for implementing basic AI features, **Git and GitHub** for version control, and **Figma** for UI/UX design and prototyping.

List of Figures

Figure 1 Use Case Diagram	27
Figure 2 Use Case Diagram Login.....	29
Figure 3 Use Case Diagram Logout.....	30
Figure 4 Use Case Diagram Manage Staff.....	31
Figure 5 Use Case Diagram Mark Attendance	32
Figure 6 Use Case Diagram Manage Staff.....	33
Figure 7 Use Case Diagram Manage Shifts	34
Figure 8 Use Case Diagram Apply For Leave	35
Figure 9 Use Case Diagram Approve / Reject Leave	36
Figure 10 Use Case Diagram Assign Tasks.....	37
Figure 11 Use Case Diagram Verify Tasks	38
Figure 12 Use Case Manage Payroll.....	39
Figure 13 Use Case Generate Reports	40
Figure 14 Use Case Send Notifications	41
Figure 15 Use Case View Dashboard Diagram	42
Figure 16 Architecture Diagram	44
Figure 17 ERD	45
Figure 18 Level 0 DFD	50
Figure 19 Level 1 DFD	51
Figure 20 Class Diagram.....	52
Figure 21 Activity Diagram – Login.....	54
Figure 22 Activity Diagram – Logout.....	55
Figure 23 Activity Diagram – Manage Staff	56
Figure 24 Activity Diagram – Mark Attendance	56
Figure 25 Activity Diagram – View Attendance	57
Figure 26 Activity Diagram – Manage Shifts.....	58
Figure 27 Activity Diagram – Apply for Leave.....	59
Figure 28 Activity Diagram 8: Approve / Reject Leave	60
Figure 29 Activity Diagram – Assign Task	61
Figure 30 Activity Diagram – Verify Task.....	62
Figure 31 Activity Diagram – Manage Payroll.....	63
Figure 32 Activity Diagram – Generate Reports	64
Figure 33 Activity Diagram – Send Notifications	65
Figure 34 Activity Diagram 14: View Dashboard.....	66
Figure 35 Sequence Diagram for Login Process	67
Figure 36 Sequence Diagram for Logout Process	68

Figure 37 Sequence Diagram for Manage Staff.....	69
Figure 38 Sequence Diagram for Mark Attendance	70
Figure 39 Sequence Diagram for View Attendance	71
Figure 40 Sequence Diagram for Manage Shifts	72
Figure 41 Sequence Diagram for Apply For Leave	73
Figure 42 Sequence Diagram for Approve / Reject Leave	74
Figure 43 Sequence Diagram for Assign Task	75
Figure 44 Sequence Diagram for Verify Task	76
Figure 45 Sequence Diagram for Manage Payrolls	77
Figure 46 Sequence Diagram for Generate Reports	78
Figure 47 Sequence Diagram for Send Notifications	79
Figure 48 Sequence Diagram for View Dashboard	80
Figure 49 Collaboration Diagram For Login Process	81
Figure 50 Collaboration Diagram For Logout Process	82
Figure 51 Collaboration Diagram For Manage staff.....	83
Figure 52 Collaboration Diagram For Mark Attendance.....	84
Figure 53 Collaboration Diagram for View Attendance.....	85
Figure 54 Collaboration Diagram for Manage Shift.....	86
Figure 55 Collaboration Diagram for Apply Leave.....	87
Figure 56 Collaboration Diagram for Approve / Leave Reject	88
Figure 57 Collaboration Diagram for Assign Task.....	89
Figure 58 Collaboration Diagram for Verify Task	90
Figure 59 Collaboration Diagram for Manage Payroll	90
Figure 60 Collaboration Diagram for Generating Report	91
Figure 61 Collaboration Diagram for Sending Notifications.....	92
Figure 62 Collaboration Diagram for View Dashboard.....	93
Figure 63 Abbreviated Collaboration Diagram	93
Figure 64 State Transition Diagram.....	94
Figure 65 Component Diagram.....	95
Figure 66 Deployment Diagram	96

List of Tables

Table 1 Functional Requirement Create Account.....	9
Table 2 Functional Requirement Login Account.....	10
Table 3 Functional Requirement Logout Account.....	10
Table 4 Functional Requirement Reset Password.....	11
Table 5 Functional Requirement Add Staff	11
Table 6 Functional Requirement View Staff	11
Table 7 Functional Requirement Update Staff	12
Table 8 Functional Requirement Delete Staff.....	12
Table 9 Functional Requirement Mark Attendance	13
Table 10 Functional Requirement View Attendance	13
Table 11 Functional Requirement Update Attendance	14
Table 12 Functional Requirement Generate Attendance Report	14
Table 13 Functional Requirement Create Shift.....	15
Table 14 Functional Requirement Assign Shift.....	15
Table 15 Functional Requirement Update Shift	15
Table 16 Functional Requirement Delete Shift.....	16
Table 17 Functional Requirement Apply for Leave	16
Table 18 Functional Requirement Approve / Reject Leave.....	17
Table 19 Functional Requirement View Leave Status.....	17
Table 20 Functional Requirement Assign Task	18
Table 21 Functional Requirement Submit Task Proof.....	18
Table 22 : Functional Requirement Verify Task	19
Table 23 Functional Requirement Calculate Salary	19
Table 24 Functional Requirement Generate Pay slip.....	19
Table 25 Functional Requirement View Payroll Report.....	20
Table 26 Functional Requirement Send Notification	20
Table 27 Functional Requirement View Notification.....	21
Table 28 Functional Requirement Offline Data Sync.....	21
Table 29 Functional Requirement Absentee Prediction	22
Table 30 Functional Requirement AI-Based Message Translation	22
Table 31 Functional Requirement Multilingual Voice Announcement.....	23
Table 32 Functional Requirement View Dashboard.....	23
Table 33 Functional Requirement Generate Reports.....	23
Table 34 Functional Requirement Export Data	24
Table 35 Usecase Aggregated Diagram Table.....	28
Table 36 Use-Case Login Table.....	29
Table 37 Use Case Logout	30

Table 38 Use-Case Manage Staff Table	31
Table 39 Use-Case Mark Attendance Table	32
Table 40 Use-Case View Attendance Table	33
Table 41 Use-Case Manage Shifts Table	34
Table 42 Use-Case Apply For Leave Table.....	35
Table 43 Use-Case Approve / Reject Leave Table	36
Table 44 Use-Case Assign Tasks Table.....	37
Table 45 Use-Case Verify Tasks Table	38
Table 46 Use-Case Manage Payroll Table.....	39
Table 47 Use-Case Generate Reports Table	40
Table 48 Use-Case Send Notifications Table	41
Table 49 Use- View Dashboard Table	42
Table 50 User	46
Table 51 Staff.....	46
Table 52 Shift.....	47
Table 53 Attendance	47
Table 54 Leave Request.....	47
Table 55 Task.....	48
Table 56 Task Proof.....	48
Table 57 Payroll	48
Table 58 Notification	49
Table 59 User Roles.....	98

Table of Content

<i>Declaration</i>	i
<i>Statement of Submission</i>	ii
<i>Dedication</i>	iii
<i>Acknowledgement</i>	iv
<i>Abstract</i>	v
List of Figures	viii
List of Tables	x
Chapter 1: Introduction to the Problem	1
1.1 Introduction	1
1.2 Purpose	1
1.3 Objective	1
1.4 Existing Solution	1
1.5 Proposed Solution	2
Chapter 2: Software Requirement Specification	3
2.1 Introduction	3
2.1.1 Purpose	3
2.1.2 Scope	3
2.1.3 Definitions, acronyms, and abbreviations	4
2.2 Overall description	5
2.2.1 Product perspective	6
2.2.2 Product functions	6
2.2.3 User characteristics	7
2.2.4 Constraints	7
Technical Constraints	7
2.2.5 Assumptions and dependencies	8
2.2.6 Apportioning of requirements	8
2.3 Specific requirements	8

2.3.1 Functional Requirement.....	9
2.3.6 Non-functional Requirements.....	24
Chapter 3: Use Case Analysis.....	26
3.1 Introduction	26
3.2 Use Case Diagrams.....	26
3.2.1 Aggregated Use Case Diagram.....	27
3.2.2 Use Case Diagrams.....	28
3.2.2.1 UC-01 (Login)	28
3.2.2.2 UC-02 (Logout)	30
3.2.2.3 UC-03 (Manage Staff)	31
3.2.2.4 UC-04 (Mark Attendance)	31
3.2.2.5 UC-05 (View Attendance)	32
3.2.2.6 UC-06 (Manage Shifts).....	33
3.2.2.7 UC-07 (Apply for Leave)	34
3.2.2.8 UC-08 (Approve / Reject Leave).....	35
3.2.2.9 UC-09 (Assign Tasks)	36
3.2.2.10 UC-10 (Verify Tasks)	37
3.2.2.11 UC-11 (Manage Payroll)	38
3.2.2.12 UC-12 (Generate Reports).....	39
3.2.2.13 UC-13 (Send Notifications).....	40
3.2.2.14 UC-14 (View Dashboard).....	41
Chapter 4: Design	43
4.1 Architecture Diagram	43
4.1.1 Architecture Layers Explanation	44
4.2 ERD with data dictionary	45
4.3 Data Flow diagram	49
4.3.1The level 0	49
4.3.2 The level 1	50
4.4 Class Diagram.....	52
4.5 Activity Diagram	53
4.6 Sequence Diagram.....	67
4.7 Collaboration Diagram	81

4.8 State Transition Diagram	94
4.9 Component Diagram.....	95
4.10 Deployment Diagram.....	96
Chapter 5: User Manual	97
5.1 Introduction	97
5.2 Intended Audience	97
5.3 System Requirements	97
5.3.1 Hardware Requirements	97
5.3.2 Software Requirements.....	97
5.4 User Roles and Access Levels	97
5.5 System Login	98
5.5.1 Purpose	98
5.5.2 Pre-requisites	98
5.5.3 Steps to Login.....	98
5.5.4 Input Fields Description	98
5.5.5 System Response	99
5.5.6 Error Handling.....	99
5.5.7 Screenshot Placeholder	99
5.6 Logout.....	100
5.6.1 Purpose	100
5.6.2 Pre-requisites	100
5.6.3 Steps to Logout.....	100
5.6.4 System Response	101
5.6.5 Security Considerations	101
5.6.6 Screenshot Placeholder	101
5.7 Dashboard Overview	102
5.7.1 Purpose	102
5.7.2 Dashboard Access.....	102
5.7.3 Dashboard Components.....	102
5.7.4 Role-Based Dashboard View.....	102
Administrator Dashboard.....	102
Supervisor Dashboard.....	102

Staff Dashboard	103
5.7.5 System Response	103
5.7.6 Screenshot Placeholder	103
5.8 Manage Staff.....	104
5.8.1 Purpose	104
5.8.2 Access Rights.....	104
5.8.3 Steps to Add New Staff	104
5.8.4 Steps to Update Staff Information	105
5.8.5 Steps to Delete Staff	105
5.8.6 System Response	105
5.8.7 Screenshot Placeholder	105
5.9 Mark Attendance	105
5.9.1 Purpose	106
5.9.2 Access Rights.....	106
5.9.3 Steps to Mark Attendance.....	106
5.9.4 Attendance Validation	106
5.9.5 System Response	106
5.9.8 Screenshot Placeholder	106
5.10 View Attendance	107
5.10.1 Purpose	107
5.10.2 Access Rights.....	107
5.10.3 Steps to View Attendance.....	108
5.10.4 Attendance Details Displayed.....	108
5.10.5 System Response	108
5.10.6 Screenshot Placeholder	108
5.12 Apply for Leave.....	109
5.12.1 Purpose	109
5.12.2 Access Rights.....	109
5.12.3 Steps to Apply for Leave	109
5.12.4 Leave Request Validation.....	109
5.12.5 System Response	109
5.12.6 Screenshot Placeholder	110

5.13 Approve / Reject Leave	110
5.13.1 Purpose	110
5.13.2 Access Rights.....	110
5.13.3 Steps to Approve or Reject Leave	111
5.13.4 System Response	111
5.13.5 Notifications	111
5.13.6 Screenshot Placeholder	111
5.14 Assign Task	112
5.14.1 Purpose	112
5.14.2 Access Rights.....	112
5.14.3 Steps to Assign a Task.....	112
5.14.4 System Response	112
5.14.5 Notifications	112
5.14.6 Screenshot Placeholder	113
5.15 Verify Task	113
5.15.1 Purpose	113
5.15.2 Access Rights.....	113
5.15.3 Steps to Verify a Task.....	113
5.15.4 System Response	114
5.15.5 Notifications	114
5.15.6 Screenshot Placeholder	114
5.16 Manage Payroll.....	114
5.16.1 Purpose	114
5.16.2 Access Rights.....	114
5.16.3 Steps to Process Payroll.....	115
5.16.4 Payroll Details Included	115
5.16.5 System Response	115
5.16.6 Screenshot Placeholder	115
5.17 Generate Reports	115
5.17.1 Purpose	116
5.17.2 Access Rights.....	116
5.17.3 Steps to Generate Reports.....	116

5.17.4 Report Output Options.....	116
5.17.5 System Response	116
5.17.6 Screenshot Placeholder.....	117
5.18 Send Notifications	117
5.18.1 Purpose	117
5.18.2 Access Rights.....	117
5.18.3 Steps to Send Notifications.....	117
5.18.4 Notification Delivery	118
5.18.5 System Response	118
5.18.6 Screenshot Placeholder.....	118
References	119
Appendix	120
Appendix A: Tools and Technologies Used	120
Appendix B: User Roles	120
Appendix C: System Screenshots.....	120

Chapter 1: Introduction to the Problem

1.1 Introduction

In many organizations, especially those that rely on shift-based and field staff, managing attendance and daily workforce activities is a difficult task. Most businesses still use manual methods such as paper registers or simple spreadsheets, which often lead to errors, delays, and lack of proper monitoring.

With the advancement of technology, there is a need for a digital system that can efficiently manage staff attendance, work schedules, and basic workforce operations. StaffSync is proposed as a smart workforce management application that provides a centralized and reliable platform for managing staff-related activities in an organized and efficient manner.

1.2 Purpose

The purpose of this project is to develop **StaffSync**, a smart workforce management application that helps organizations manage staff attendance and basic workforce operations digitally. The system is designed to reduce manual work, minimize errors, and improve overall efficiency in managing shift-based and field staff.

StaffSync aims to provide a centralized platform that allows management to monitor staff activities, maintain records, and improve coordination between staff and administrators. The system focuses on simplicity, reliability, and ease of use to support organizations from different industries.

1.3 Objective

The main objectives of the StaffSync project are as follows:

- To develop a digital system for managing staff attendance and work schedules
- To reduce manual effort and errors in workforce management
- To provide a centralized platform for maintaining staff records
- To improve coordination between management and staff
- To design a user-friendly system suitable for shift-based and field staff

1.4 Existing Solution

At present, many organizations manage their workforce using manual methods such as attendance registers, spreadsheets, or basic biometric systems. These solutions are often not integrated and require a lot of manual effort to maintain records and prepare reports.

Existing workforce management systems available in the market are mostly expensive and designed for foreign markets. They often lack features such as real-time monitoring, offline support, and local language support. As a result, small and medium-sized organizations face difficulties in adopting these systems and continue to rely on inefficient traditional methods

1.5 Proposed Solution

StaffSync is proposed as a smart and centralized workforce management application that overcomes the limitations of existing systems. The system provides a digital platform for managing staff attendance, work schedules, and basic workforce operations in an efficient manner.

The proposed solution focuses on simplicity, accuracy, and ease of use. StaffSync allows organizations to maintain staff records, monitor attendance, and improve coordination between management and employees. By offering a user-friendly and scalable solution, StaffSync helps organizations manage their workforce more effectively compared to traditional methods

Chapter 2: Software Requirement Specification

2.1 Introduction

This chapter provides an overview of the Software Requirement Specification (SRS) for the StaffSync project. It describes the overall requirements of the system and outlines what the software is expected to accomplish.

The SRS serves as a foundational document that helps define system behavior, constraints, and expectations. It ensures that all stakeholders have a clear and common understanding of the system before development and further implementation phases.

2.1.1 Purpose

The purpose of this Software Requirement Specification (SRS) document is to clearly define the functional and non-functional requirements of the **StaffSync** system. This document is intended to describe what the system will do and the constraints under which it will operate.

The SRS is prepared for project supervisors, developers, and future stakeholders to ensure a common understanding of system requirements. It serves as a reference throughout the development lifecycle and helps in system design, implementation, and testing.

2.1.2 Scope

The StaffSync system is a smart workforce management application designed to support organizations that rely on shift-based and field staff. The scope of this project includes managing staff attendance, work schedules, and basic workforce records through a centralized digital platform.

The system allows administrators to register staff members, monitor attendance, and manage work-related information efficiently. StaffSync is intended to be used by different types of organizations such as construction companies, restaurants, cafés, and healthcare facilities. The system focuses on providing a user-friendly, scalable, and reliable solution suitable for real-world operational environments.

This project does not include advanced financial accounting or third-party enterprise integrations in its initial phase. However, the system is designed in a modular way to allow future enhancements and feature expansion.

2.1.3 Definitions, acronyms, and abbreviations

Definitions

StaffSync:

StaffSync is a smart workforce management application developed to manage staff-related operations in a digital environment.

It provides the following functionalities:

- Centralized management of staff attendance and records
- Digital shift scheduling and monitoring
- Reduction of manual paperwork and human errors
- Improved coordination between management and employees
- Support for organizations with shift-based and field staff

Workforce Management:

Workforce management refers to the systematic process of managing employees within an organization to ensure smooth operations.

It includes:

- Planning and managing staff schedules
- Tracking employee attendance and working hours
- Monitoring workforce productivity
- Maintaining staff records and reports
- Supporting decision-making for management

Attendance:

Attendance is the process of recording the presence, absence, or working hours of staff members.

In StaffSync, attendance management helps to:

- Track daily staff presence accurately
- Maintain digital attendance records
- Support payroll and performance evaluation
- Reduce manipulation and errors in manual attendance systems

Shift:

A shift is a specific time duration assigned to a staff member for performing work.

Shift management involves:

- Assigning staff to specific working hours
- Managing multiple shifts (morning, evening, night)
- Avoiding staff overlap or shortages
- Ensuring proper workload distribution

Administrator (Admin):

An administrator is a user with full system access and control.

The responsibilities of an administrator include:

- Managing staff accounts and roles
- Monitoring attendance and schedules
- Maintaining system data and settings
- Generating reports for management purposes

Employee / Staff:

An employee or staff member is a registered system user who performs assigned work.

Staff members can:

- Mark their attendance
- View assigned shifts
- Follow organizational rules and schedules
- Interact with the system according to assigned permissions

Acronyms

- **SRS:** Software Requirement Specification
- **DFD:** Data Flow Diagram
- **ERD:** Entity Relationship Diagram
- **UI:** User Interface
- **API:** Application Programming Interface

Abbreviations

- **UOL:** The University of Lahore
- **DB:** Database
- **AI:** Artificial Intelligence

2.2 Overall description

This section provides a high-level description of the StaffSync system and explains how the system operates as a whole. It outlines the general characteristics of the system, its users, and the environment in which it will function.

StaffSync is designed as a centralized workforce management application that helps organizations manage staff attendance, scheduling, and basic operational activities. The system is intended to be simple, reliable, and adaptable so that it can be used by different types of organizations with minimal training

2.2.1 Product perspective

System Overview

StaffSync is an independent and self-contained workforce management system developed to manage staff attendance, scheduling, and basic workforce operations. The system is designed as a new solution and does not rely on any existing workforce management software.

System Architecture

The system follows a client–server architecture. The mobile application developed using **Flutter** acts as the client, while backend services developed using **.NET Core** handle business logic and data processing. **Firestore** is used for authentication and real-time data storage to ensure secure and reliable system operations.

Scalability and Modularity

StaffSync is designed with scalability and modularity in mind. The system structure allows new features and modules to be added in the future without affecting existing functionalities. This design approach supports future enhancements such as advanced reporting, analytics, and system integrations

2.2.2 Product functions

The StaffSync system provides a set of high-level product functions that describe the core capabilities of the application. These product functions represent grouped system functionalities and define what the system is designed to achieve without going into implementation-level details.

The detailed operations and actions related to each product function are described later in the **Functional Requirements section**.

Primary Product Functions

1. **User and Role Management**
Supports secure user authentication and role-based access control within the system.
2. **Workforce Attendance and Scheduling Management**
Provides mechanisms to record staff attendance and manage work schedules efficiently.
3. **Workforce Monitoring and Performance Support**
Enables monitoring of staff activities, task completion, and overall workforce performance.
4. **Workforce Planning and Decision Support**
Assists management in staff planning, leave handling, and workload balancing.
5. **Communication and Notification Management**
Facilitates effective communication between management and staff through system notifications.
6. **System Data Management and Reporting**
Maintains staff-related data and generates reports to support administrative decision-making.

Each of these product functions serves as a foundation for defining detailed functional requirements, which are documented in the following sections using structured requirement templates

2.2.3 User characteristics

Administrators

Administrators are responsible for managing the overall system. They are expected to have basic computer and smartphone usage skills. Administrators should be familiar with managing staff records, monitoring attendance, and viewing system reports. No advanced technical knowledge is required to operate the system.

Supervisors / Managers

Supervisors or managers oversee daily staff activities and monitor workforce performance. They are expected to have basic understanding of mobile applications and internet usage. Supervisors interact with attendance records, shift schedules, and task-related information through the system.

Staff / Employees

Staff members are the end users of the system who mark attendance and follow assigned work schedules. They may have limited technical knowledge; therefore, the system is designed to be simple and user-friendly. Basic smartphone usage ability is sufficient for staff members to use the application effectively.

2.2.4 Constraints

The development and operation of the StaffSync system are subject to certain constraints that may limit design or implementation choices.

Technical Constraints

- The system depends on stable internet connectivity for real-time features.
- The mobile application requires Android-compatible devices.
- Cloud-based services such as Firebase impose usage limits based on service plans.

Resource Constraints

- The project is developed within a limited academic timeline.
- Development is carried out by student developers with limited industry experience.

Operational Constraints

- Users may have varying levels of technical knowledge.
- Some organizations may have limited access to advanced hardware or devices.

2.2.5 Assumptions and dependencies

This subsection lists the assumptions made during the development of the StaffSync system and the dependencies that may affect system requirements.

Assumptions

- Users will have access to a smartphone to use the mobile application.
- Basic internet connectivity will be available for most system operations.
- Organizations will follow defined work schedules and roles within the system.
- Users will enter accurate information during registration and attendance marking.

Dependencies

- The system depends on cloud services for data storage and authentication.
- Application performance depends on device capabilities and network quality.
- Future enhancements may depend on availability of additional resources and tools.

2.2.6 Apportioning of requirements

Some system requirements are planned to be implemented in later versions of the StaffSync application. These requirements are identified to allow phased development and future enhancement of the system without affecting the core functionality.

The following features may be deferred to future releases:

- Advanced AI-based analytics and detailed performance insights
- Integration with third-party accounting or HR systems
- Advanced reporting dashboards and data visualization
- Support for additional platforms and extended device compatibility

The initial version of StaffSync focuses on core workforce management functionalities, while the above requirements are planned for future system versions based on resource availability and user feedback.

2.3 Specific requirements

This section describes the detailed requirements of the StaffSync system at a level sufficient for system design and verification. It includes both functional and non-functional requirements that define how the system should behave and what quality attributes it must satisfy.

The functional requirements specify the individual operations that the system must perform, while the non-functional requirements describe the performance, usability, reliability, and other constraints of the system. These requirements are defined to ensure that the StaffSync system meets user expectations and project objectives.

2.3.1 Functional Requirement

Functional requirements describe the specific actions and operations that the StaffSync system must perform. These requirements define how users interact with the system and how the system responds to those interactions.

Each functional requirement is documented using a structured template that includes an identifier, name, description, inputs, outputs, and basic workflow. This structured approach ensures clarity, consistency, and ease of understanding for system development and testing.

Field	Description
Id	FR_01
Name	Create Account
Description	This functional requirement allows a new user to create an account in the StaffSync system. The system registers the user by collecting basic information and assigning a specific role.
Input	Name, Email Address, Password, User Role
Output	User account successfully created and confirmation message displayed
Requirements	Internet connectivity required, valid and unique email address
Basic Work Flow	1. User opens the registration screen. 2. User enters required details. 3. User submits the form. 4. System validates the information. 5. System stores data in the database. 6. System displays confirmation message.

Table 1 Functional Requirement Create Account

Field	Description
Id	FR_02
Name	Login Account
Description	This functional requirement allows a registered user to log in to the StaffSync system using valid credentials. The system verifies the user's identity and grants access based on the assigned role.

Input	Email Address, Password
Output	User successfully logged into the system and redirected to the dashboard
Requirements	Internet connectivity required, valid registered account
Basic Work Flow	1. User opens the login screen. 2. User enters email and password. 3. User clicks the login button. 4. System validates credentials. 5. System grants access and redirects user to dashboard.

Table 2 Functional Requirement Login Account

Field	Description
Id	FR_03
Name	Logout Account
Description	This functional requirement allows a logged-in user to securely log out of the StaffSync system. The system ends the active session and prevents unauthorized access.
Input	Logout request (Logout button click)
Output	User successfully logged out and redirected to the login screen
Requirements	Active user session required
Basic Work Flow	1. User clicks the logout option. 2. System terminates the active session. 3. System clears session data. 4. User is redirected to the login screen.

Table 3 Functional Requirement Logout Account

Field	Description
Id	FR_04
Name	Reset Password
Description	This functional requirement allows a registered user to reset their account password in case they forget it. The system verifies the user and enables secure password recovery.
Input	Registered Email Address
Output	Password reset link or verification message sent to user
Requirements	Internet connectivity required, valid registered email
Basic Work Flow	1. User clicks on “Forgot Password”.

	2. User enters registered email address. 3. System verifies the email. 4. System sends password reset link or code. 5. User sets a new password. 6. System updates the password successfully.
--	---

Table 4 Functional Requirement Reset Password

Field	Description
Id	FR_05
Name	Add Staff
Description	This functional requirement allows an administrator to add a new staff member into the StaffSync system. The system stores staff details and assigns appropriate roles.
Input	Staff Name, Email, Contact Number, Role, Assigned Shift
Output	Staff member successfully added and confirmation message displayed
Requirements	Administrator access required, internet connectivity
Basic Work Flow	1. Administrator opens the Add Staff screen. 2. Administrator enters staff details. 3. Administrator submits the form. 4. System validates the entered information. 5. System saves staff data in the database. 6. System displays confirmation message.

Table 5 Functional Requirement Add Staff

Field	Description
Id	FR_06
Name	View Staff
Description	This functional requirement allows an administrator or authorized user to view the list of all registered staff members along with their basic details.
Input	View request (open staff list screen)
Output	List of staff members displayed with relevant details
Requirements	Authorized access required
Basic Work Flow	1. User navigates to the Staff List screen. 2. System fetches staff records from the database. 3. System displays staff details in a structured list or table..

Table 6 Functional Requirement View Staff

Field	Description
Id	FR_07
Name	Update Staff
Description	This functional requirement allows an administrator to update existing staff member information in the StaffSync system.
Input	Staff ID, Updated Staff Details
Output	Staff record successfully updated
Requirements	Administrator access required
Basic Work Flow	1. Administrator selects a staff member from the staff list. 2. Administrator edits the required details. 3. Administrator submits the update request. 4. System validates the updated information. 5. System saves the changes in the database.

Table 7 Functional Requirement Update Staff

Field	Description
Id	FR_08
Name	Delete Staff
Description	This functional requirement allows an administrator to remove a staff member from the StaffSync system. The system ensures that only authorized users can perform this action.
Input	Staff ID / Staff Selection
Output	Staff member record successfully deleted
Requirements	Administrator access required
Basic Work Flow	1. Administrator selects a staff member from the staff list. 2. Administrator clicks the delete option. 3. System asks for confirmation. 4. Administrator confirms the action. 5. System removes the staff record from the database.

Table 8 Functional Requirement Delete Staff

Field	Description
Id	FR_09
Name	Mark Attendance
Description	This functional requirement allows a staff member to mark their attendance in the StaffSync system. Attendance is recorded against the assigned shift and date.

Input	Staff ID, Attendance Status (Check-in/Check-out), Date & Time
Output	Attendance record successfully saved
Requirements	Active user account, internet connectivity
Basic Work Flow	1. Staff member opens the attendance screen. 2. Staff selects check-in or check-out option. 3. System captures date and time. 4. System validates attendance entry. 5. System saves attendance record in the database. 6. System displays confirmation message.

Table 9 Functional Requirement Mark Attendance

Field	Description
Id	FR_10
Name	View Attendance
Description	This functional requirement allows staff members and administrators to view attendance records for a selected date or time period. The system displays attendance details in a structured format.
Input	Staff ID (optional), Date / Date Range
Output	Attendance records displayed
Requirements	Authorized access required
Basic Work Flow	1. User opens the attendance view screen. 2. User selects date or date range. 3. System retrieves attendance data from the database. 4. System displays attendance records in a list or table format. 6. System displays confirmation message.

Table 10 Functional Requirement View Attendance

Field	Description
Id	FR_11
Name	Update Attendance
Description	This functional requirement allows an administrator or supervisor to update an existing attendance record in case of errors or adjustments.
Input	Staff ID, Date, Updated Attendance Status
Output	Attendance record successfully updated
Requirements	Administrator or Supervisor access required

Basic Work Flow	1. Authorized user opens the attendance management screen. 2. User selects the staff member and date. 3. User edits the attendance status. 4. System validates the changes. 5. System updates the attendance record in the database. 6. System displays confirmation message.
------------------------	--

Table 11 Functional Requirement Update Attendance

Field	Description
Id	FR_12
Name	Generate Attendance Report
Description	This functional requirement allows administrators and supervisors to generate attendance reports for a selected time period. The system compiles attendance data into a structured report for review and record keeping.
Input	Date Range, Staff ID (optional), Report Type
Output	Attendance report generated and displayed/downloaded
Requirements	Authorized access required
Basic Work Flow	1. Authorized user opens the report generation screen. 2. User selects date range and report parameters. 3. System retrieves attendance data from the database. 4. System generates the attendance report. 5. System displays or allows download of the report.

Table 12 Functional Requirement Generate Attendance Report

Field	Description
Id	FR_13
Name	Create Shift
Description	This functional requirement allows an administrator to create new work shifts in the StaffSync system. Shifts define working hours for staff members.
Input	Shift Name, Start Time, End Time, Shift Type
Output	Shift successfully created
Requirements	Administrator access required
Basic Work Flow	1. Administrator opens the shift management screen. 2. Administrator selects the create shift option. 3. Administrator enters shift details.

	4. System validates shift information. 5. System saves the shift in the database. 6. System displays confirmation message.
--	--

Table 13 Functional Requirement Create Shift

Field	Description
Id	FR_14
Name	Assign Shift
Description	This functional requirement allows an administrator or supervisor to assign an existing shift to a staff member. The system links staff with defined working hours.
Input	Staff ID, Shift ID, Assignment Date
Output	Shift successfully assigned to staff member
Requirements	Authorized (Admin/Supervisor) access required
Basic Work Flow	1. Authorized user opens shift assignment screen. 2. User selects staff member. 3. User selects available shift. 4. User confirms assignment. 5. System saves assignment in the database. 6. System displays confirmation message.

Table 14 Functional Requirement Assign Shift

Field	Description
Id	FR_15
Name	Update Shift
Description	This functional requirement allows an administrator to modify existing shift details in the StaffSync system to accommodate operational changes.
Input	Shift ID, Updated Shift Details (Start Time, End Time, Shift Type)
Output	Shift information successfully updated
Requirements	Administrator access required
Basic Work Flow	1. Administrator opens the shift management screen. 2. Administrator selects an existing shift. 3. Administrator edits shift details. 4. System validates the updated information. 5. System saves changes in the database. 6. System displays confirmation message.

Table 15 Functional Requirement Update Shift

Field	Description
Id	FR_16
Name	Delete Shift
Description	This functional requirement allows an administrator to delete an existing shift from the StaffSync system. The system ensures that shifts currently assigned to staff are handled appropriately before deletion.
Input	Shift ID / Shift Selection
Output	Shift successfully deleted
Requirements	Administrator access required
Basic Work Flow	1. Administrator opens shift management screen. 2. Administrator selects a shift. 3. Administrator clicks delete option. 4. System checks for assigned staff. 5. System deletes the shift. 6. System displays confirmation message.

Table 16 Functional Requirement Delete Shift

Field	Description
Id	FR_17
Name	Apply for Leave
Description	This functional requirement allows a staff member to apply for leave by submitting a leave request with relevant details.
Input	Staff ID, Leave Type, Start Date, End Date, Reason
Output	Leave request successfully submitted
Requirements	Active staff account, internet connectivity
Basic Work Flow	1. Staff opens the leave application screen. 2. Staff selects leave type and dates. 3. Staff enters reason for leave. 4. Staff submits the request. 5. System validates the information. 6. System records the leave request and notifies supervisor/admin.

Table 17 Functional Requirement Apply for Leave

Field	Description
Id	FR_18
Name	Approve / Reject Leave

Description	This functional requirement allows an administrator or supervisor to review submitted leave requests and approve or reject them based on staff availability and organizational policies.
Input	Leave Request ID, Approval Decision (Approve/Reject), Remarks (optional)
Output	Leave request status updated and notification sent to staff
Requirements	Administrator or Supervisor access required
Basic Work Flow	1. Authorized user opens leave requests list. 2. User selects a leave request. 3. User reviews leave details. 4. User approves or rejects the request. 5. System updates leave status in the database. 6. System notifies the staff member of the decision.

Table 18 Functional Requirement Approve / Reject Leave

Field	Description
Id	FR_19
Name	View Leave Status
Description	This functional requirement allows a staff member to view the current status of their submitted leave requests. The system displays whether the leave is pending, approved, or rejected.
Input	Staff ID / View request
Output	Active staff account required
Requirements	Administrator or Supervisor access required
Basic Work Flow	1. Staff member opens the leave status screen. 2. System retrieves leave requests related to the staff member. 3. System displays leave details and current status.

Table 19 Functional Requirement View Leave Status

Field	Description
Id	FR_20
Name	Assign Task
Description	This functional requirement allows an administrator or supervisor to assign tasks to staff members with defined details and deadlines.
Input	Staff ID, Task Title, Task Description, Due Date

Output	Task successfully assigned and notification sent to staff
Requirements	Administrator or Supervisor access required
Basic Work Flow	1. Authorized user opens the task management screen. 2. User selects staff member. 3. User enters task details and due date. 4. User assigns the task. 5. System saves task information in the database. 6. System notifies the assigned staff member.

Table 20 Functional Requirement Assign Task

Field	Description
Id	FR_21
Name	Submit Task Proof
Description	This functional requirement allows a staff member to submit proof of task completion (such as images or notes) for review by the administrator or supervisor.
Input	Task ID, Proof Attachment (Image/File), Remarks (optional)
Output	Task proof successfully submitted
Requirements	Active staff account, assigned task
Basic Work Flow	1. Staff member opens assigned task details. 2. Staff uploads proof of completion and adds remarks (if any). 3. Staff submits the task proof. 4. System validates the submission. 5. System stores the proof in the database. 6. System notifies supervisor/admin for review.

Table 21 Functional Requirement Submit Task Proof

Field	Description
Id	FR_22
Name	Verify Task
Description	This functional requirement allows an administrator or supervisor to review and verify the task proof submitted by a staff member. The task can be approved or rejected based on verification.
Input	Task ID, Verification Decision (Approve/Reject), Remarks (optional)
Output	Task verification status updated and notification sent to staff

Requirements	Administrator or Supervisor access required
Basic Work Flow	1. Authorized user opens the task verification screen. 2. User reviews submitted task proof. 3. User approves or rejects the task. 4. System updates task status in the database. 5. System notifies the staff member of the verification result.

Table 22 : Functional Requirement Verify Task

Field	Description
Id	FR_23
Name	Calculate Salary
Description	This functional requirement allows the system to calculate staff salaries based on attendance records, assigned shifts, and defined pay rules.
Input	Staff ID, Attendance Data, Pay Rate, Pay Period
Output	Calculated salary details
Requirements	Authorized access required, valid attendance data
Basic Work Flow	1. Authorized user opens payroll calculation screen. 2. System retrieves attendance and shift data. 3. System applies pay rules and rates. 4. System calculates total salary for the pay period. 5. System displays calculated salary details.

Table 23 Functional Requirement Calculate Salary

Field	Description
Id	FR_24
Name	Generate Pay slip
Description	This functional requirement allows the system to generate a payslip for a staff member based on the calculated salary for a specific pay period.
Input	Staff ID, Pay Period
Output	Payslip generated and displayed/downloaded
Requirements	Calculated salary data available, authorized access
Basic Work Flow	1. Authorized user selects staff member and pay period. 2. System retrieves calculated salary details. 3. System generates the payslip. 4. System displays the payslip or allows download.

Table 24 Functional Requirement Generate Pay slip

Field	Description
Id	FR_25
Name	View Payroll Report
Description	This functional requirement allows administrators or authorized users to view payroll reports for a selected period. The system summarizes salary details for staff members to support review and record keeping.
Input	Pay Period, Staff ID (optional)
Output	Payroll report displayed
Requirements	Authorized access required
Basic Work Flow	1. Authorized user opens the payroll reports screen. 2. User selects the pay period and optional filters. 3. System retrieves payroll data from the database. 4. System displays payroll report in a structured format.

Table 25 Functional Requirement View Payroll Report

Field	Description
Id	FR_26
Name	Send Notification
Description	This functional requirement allows an administrator or supervisor to send notifications or announcements to staff members through the StaffSync system.
Input	Notification Title, Message Content, Target Audience (All Staff / Selected Staff)
Output	Notification successfully sent
Requirements	Authorized (Admin/Supervisor) access required
Basic Work Flow	1. Authorized user opens notification module. 2. User composes notification message. 3. User selects target audience. 4. User sends the notification. 5. System delivers notification to selected staff members.

Table 26 Functional Requirement Send Notification

Field	Description
Id	FR_27
Name	View Notification
Description	This functional requirement allows staff members to view notifications and announcements sent by administrators or supervisors.
Input	View notification request
Output	List of notifications displayed
Requirements	Active user account required
Basic Work Flow	1. User opens notifications screen. 2. System retrieves notifications for the user. 3. System displays notifications in chronological order.

Table 27 Functional Requirement View Notification

Field	Description
Id	FR_28
Name	Offline Data Synchronization
Description	This functional requirement allows the StaffSync system to operate in offline mode by storing user actions locally and synchronizing data automatically once internet connectivity is restored.
Input	User actions performed in offline mode
Output	Offline data successfully synchronized with the central database
Requirements	Device storage available, internet connectivity for synchronization
Basic Work Flow	1. User performs actions while offline. 2. System stores data locally on the device. 3. Internet connectivity becomes available. 4. System automatically syncs local data with the server. 5. System confirms successful synchronization.

Table 28 Functional Requirement Offline Data Sync

Field	Description
Id	FR_29
Name	Absentee Prediction

Description	This functional requirement allows the system to analyze past attendance patterns to predict the likelihood of staff absenteeism. The feature supports proactive workforce planning.
Input	Historical Attendance Data, Staff ID
Output	Predicted absenteeism insights displayed
Requirements	Sufficient attendance history, authorized access
Basic Work Flow	1. System collects historical attendance data. 2. System analyzes attendance patterns. 3. System predicts potential absenteeism. 4. System displays prediction results to authorized users.

Table 29 Functional Requirement Absentee Prediction

Field	Description
Id	FR_30
Name	AI-Based Message Translation
Description	This functional requirement allows the system to translate messages from English or Roman Urdu into simple Urdu to improve communication with staff members.
Input	Text Message (English / Roman Urdu)
Output	Translated message displayed
Requirements	Internet connectivity, language processing service
Basic Work Flow	1. User enters a message in English or Roman Urdu. 2. System processes the message using translation logic. 3. System converts the message into simple Urdu. 4. System displays the translated message.

Table 30 Functional Requirement AI-Based Message Translation

Field	Description
Id	FR_31
Name	Multilingual Voice Announcement
Description	This functional requirement allows administrators to broadcast voice announcements in multiple languages to ensure clear communication with diverse staff members.
Input	Voice Message, Selected Language, Target Audience
Output	Voice announcement delivered to staff
Requirements	Authorized access, audio input capability
Basic Work Flow	1. Authorized user records a voice message.

	2. User selects the announcement language. 3. User selects target audience. 4. System broadcasts the voice announcement.
--	--

Table 31 Functional Requirement Multilingual Voice Announcement

Field	Description
Id	FR_32
Name	View Dashboard
Description	This functional requirement allows authorized users to view a centralized dashboard displaying key workforce statistics such as attendance summary, staff count, pending tasks, and notifications.
Input	Dashboard view request
Output	Dashboard displayed with summarized system information
Requirements	Authorized access required
Basic Work Flow	1. User logs into the system. 2. System loads dashboard data. 3. System displays key metrics and summaries. 4. User views updated information in real time.

Table 32 Functional Requirement View Dashboard

Field	Description
Id	FR_33
Name	Generate Reports
Description	This functional requirement allows authorized users to generate system reports related to attendance, staff, payroll, and tasks for analysis and record keeping.
Input	Report Type, Date Range, Filters (optional)
Output	Generated report displayed/downloaded
Requirements	Authorized access required
Basic Work Flow	1. Authorized user opens reports module. 2. User selects report type and parameters. 3. System retrieves relevant data. 4. System generates the report. 5. System displays or allows download of the report.

Table 33 Functional Requirement Generate Reports

Field	Description
Id	FR_34
Name	Export Data
Description	This functional requirement allows authorized users to export system data such as attendance, staff, payroll, and reports into downloadable formats for offline use and record keeping.
Input	Data Type, Date Range, Export Format (PDF/Excel/CSV)
Output	Data file exported and downloaded
Requirements	Authorized access required
Basic Work Flow	1. Authorized user selects export option. 2. User chooses data type and date range. 3. User selects export format. 4. System generates the export file. 5. System provides the file for download.

Table 34 Functional Requirement Export Data

2.3.6 Non-functional Requirements

Non-functional requirements define the quality attributes of the StaffSync system. These requirements ensure that the system performs reliably, securely, and efficiently while providing a

Usability

- The system shall provide a simple and intuitive user interface.
- The mobile application shall be easy to use for users with basic smartphone skills.
- The system shall support clear navigation and readable layouts.
- Error messages shall be user-friendly and informative.

Reliability

- The system shall operate reliably under normal working conditions.
- Attendance and staff data shall be stored accurately without loss.
- The system shall recover gracefully from minor failures.
- Offline actions shall be synchronized correctly once connectivity is restored.

Performance

- The system shall operate reliably under normal working conditions.
- Attendance and staff data shall be stored accurately without loss.
- The system shall recover gracefully from minor failures.
- Offline actions shall be synchronized correctly once connectivity is restored.

Security

- The system shall require authentication for all users.
- Role-based access control shall restrict unauthorized actions.
- User data shall be protected from unauthorized access.
- Passwords shall be stored securely using encryption techniques.

Portability

- The mobile application shall run on Android-based devices.
- The system shall be deployable on cloud platforms.
- Future versions shall support additional platforms with minimal changes.

Maintainability

- The system shall be modular and well-structured.
- Code changes shall be easy to implement and test.
- The system shall support future feature enhancements without major redesign.

License Agreement

- The system shall be used according to institutional and organizational policies.
- Any third-party tools or services used shall comply with their licensing terms.

Chapter 3: Use Case Analysis

3.1 Introduction

The Use Case Analysis describes how different users interact with the StaffSync system to perform various tasks. This chapter focuses on identifying system actors and defining the interactions between users and the system through use cases.

Use cases help in understanding system behavior from the user's perspective and provide a clear representation of system functionality. This chapter serves as a bridge between system requirements and system design by translating functional requirements into user interactions.

3.2 Use Case Diagrams

The Use Case Diagram illustrates the interaction between different users and the StaffSync system. It provides a visual overview of system functionality and shows how various actors communicate with the system to perform different tasks.

The diagram defines the system boundary of StaffSync and identifies the main actors involved in system operations. Each actor interacts with specific use cases according to their role and responsibilities. This diagram helps in understanding the scope of the system and serves as a foundation for detailed use case descriptions.

The primary actors of the StaffSync system are **Administrator**, **Supervisor**, and **Staff**.

3.2.1 Aggregated Use Case Diagram

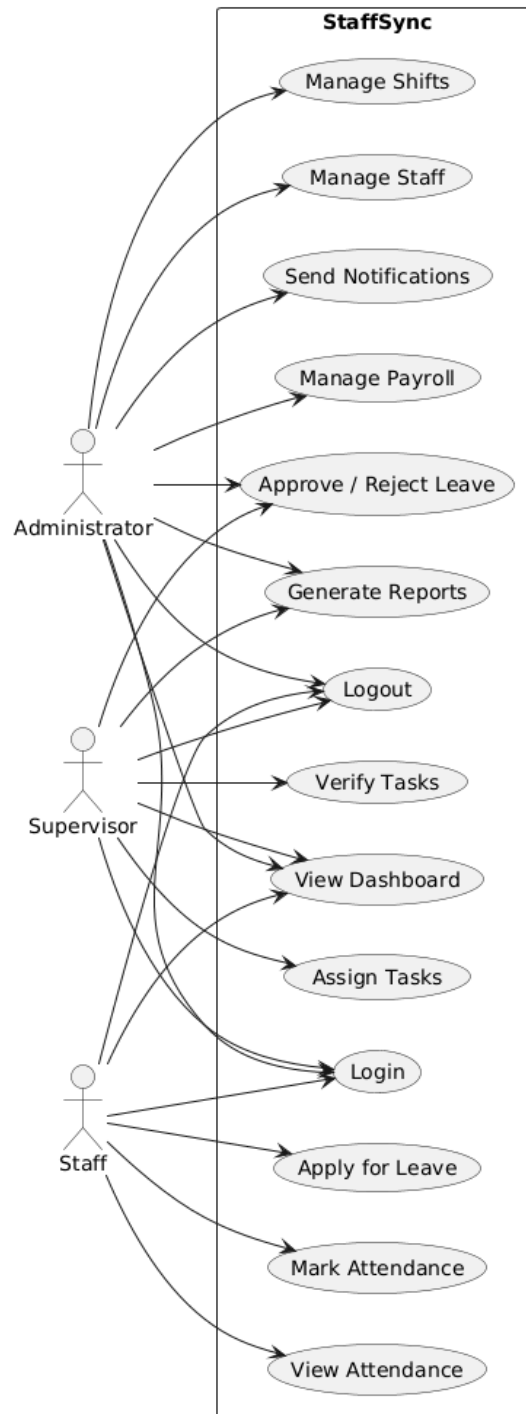


Figure 1 Use Case Diagram

Use Case ID	Use Case Name
UC_01	Login
UC_02	Logout
UC_03	Manage Staff
UC_04	Mark Attendance
UC_05	View Attendance
UC_06	Manage Shifts
UC_07	Apply for Leave
UC_08	Approve / Reject Leave
UC_09	Assign Tasks
UC_10	Verify Tasks
UC_11	Manage Payroll
UC_12	Generate Reports
UC_13	Send Notifications

Table 35 Usecase Aggregated Diagram Table

3.2.2 Use Case Diagrams

3.2.2.1 UC-01 (Login)

Field	Description
Use Case ID	UC_01
Use Case Name	Login
Primary Actor	Administrator, Supervisor, Staff
Description	This use case allows a registered user to log in to the StaffSync system using valid credentials in order to access system features based on assigned roles.
Pre-Conditions	User must be registered in the system. Internet connectivity must be available.
Post-Conditions	User is successfully logged in and redirected to the dashboard according to their role.
Main Flow	1. User opens the StaffSync application. 2. User enters email and password. 3. User clicks the login button. 4. System validates the credentials. 5. System grants access and displays the dashboard.

Alternate Flow	1. User enters invalid credentials. 2. System displays an error message and asks user to re-enter details.
Exceptions	System displays an error if the server is unavailable or internet connection is lost.

Table 36 Use-Case Login Table

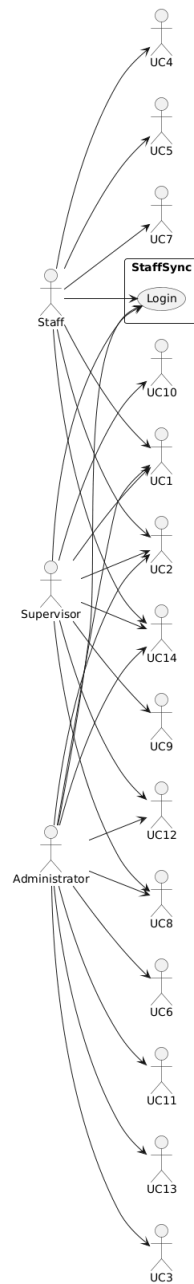


Figure 2 Use Case Diagram Login

3.2.2.2 UC-02 (Logout)

Field	Description
Use Case ID	UC_02
Use Case Name	Logout
Primary Actor	Administrator, Supervisor, Staff
Description	This use case allows a logged-in user to safely log out of the StaffSync system in order to end the current session and prevent unauthorized access.
Pre-Conditions	User must be logged into the system.
Post-Conditions	User session is terminated and the user is redirected to the login screen.
Main Flow	1. User clicks the logout option.2. System terminates the active session.3. System clears session data.4. System redirects the user to the login screen.
Alternate Flow	If the system fails to terminate the session, an error message is displayed.
Exceptions	Network or server failure during logout process.

Table 37 Use Case Logout

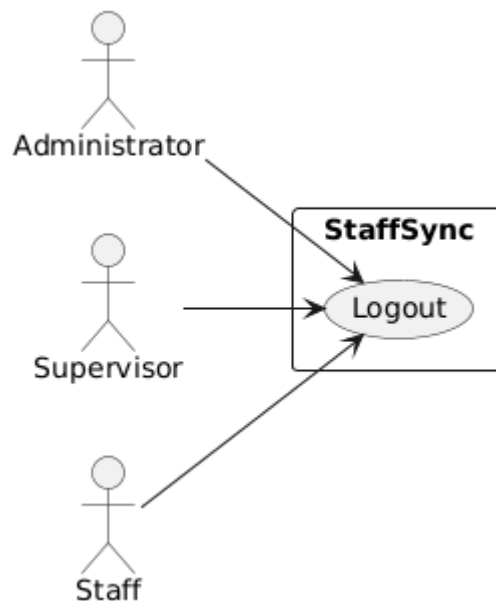


Figure 3 Use Case Diagram Logout

3.2.2.3 UC-03 (Manage Staff)

Field	Description
Use Case ID	UC_03
Use Case Name	Manage Staff
Primary Actor	Administrator
Description	This use case allows the administrator to manage staff records in the StaffSync system, including adding, viewing, updating, and deleting staff information.
Pre-Conditions	Administrator must be logged into the system.
Post-Conditions	Staff records are successfully created, updated, viewed, or deleted as requested.
Main Flow	1. Administrator opens the staff management module.2. System displays the list of staff members.3. Administrator selects an action (Add, View, Update, Delete).4. Administrator enters or modifies staff details as required.5. System validates the information.6. System saves changes and updates the staff records.
Alternate Flow	If invalid information is entered, the system displays an error message and requests correction.
Exceptions	System error or database failure while managing staff records.

Table 38 Use-Case Manage Staff Table

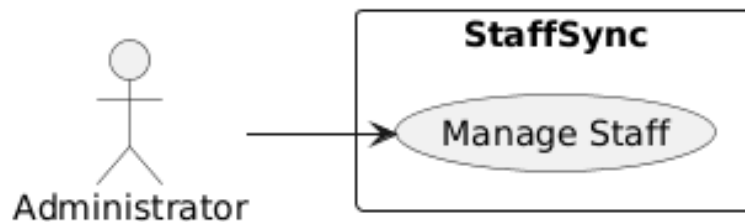


Figure 4 Use Case Diagram Manage Staff

3.2.2.4 UC-04 (Mark Attendance)

Field	Description
Use Case ID	UC_04
Use Case Name	Mark Attendance
Primary Actor	Staff
Description	This use case allows a staff member to mark their attendance (check-in/check-out) in the StaffSync system for the assigned shift and date.
Pre-Conditions	Staff must be logged into the system. Assigned shift must exist for the day.
Post-Conditions	Attendance record is successfully saved for the selected date and time.
Main Flow	1. Staff opens the attendance module.2. Staff selects Check-in or Check-out.3. System captures date and time.4. System validates the attendance entry.5. System saves the attendance record.6. System shows confirmation message.
Alternate Flow	If attendance is already marked, the system notifies the staff and prevents duplicate entry.
Exceptions	Internet or server failure during attendance submission.

Table 39 Use-Case Mark Attendance Table



Figure 5 Use Case Diagram Mark Attendance

3.2.2.5 UC-05 (View Attendance)

Field	Description
Use Case ID	UC_05
Use Case Name	View Attendance
Primary Actor	Staff, Administrator, Supervisor
Description	This use case allows authorized users to view attendance records for a selected date or date range in the StaffSync system.
Pre-Conditions	User must be logged into the system.Attendance records must exist.
Post-Conditions	Attendance records are displayed successfully.
Main Flow	1. User opens the attendance view module.2. User selects date or date range.3. System retrieves attendance data.4. System displays attendance records in a list or table.
Alternate Flow	If no records are found, the system displays a “No records available” message.
Exceptions	Database or network error while fetching attendance data.

Table 40 Use-Case View Attendance Table

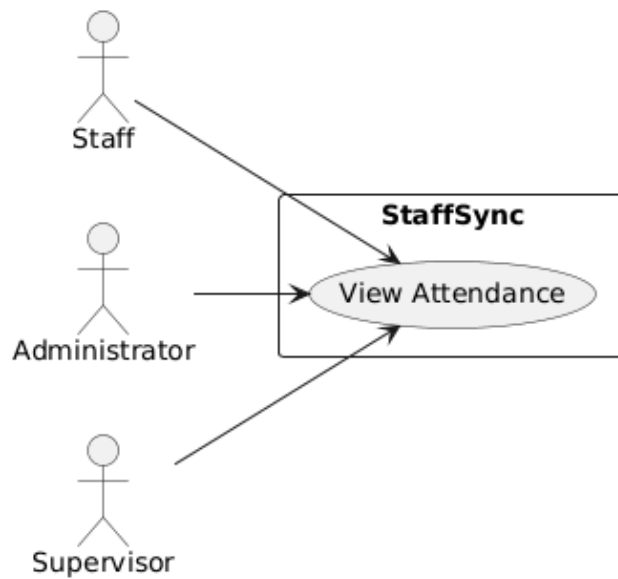


Figure 6 Use Case Diagram Manage Staff

3.2.2.6 UC-06 (Manage Shifts)

Field	Description
Use Case ID	UC_06
Use Case Name	Manage Shifts
Primary Actor	Administrator
Description	This use case allows the administrator to create, update, assign, and delete work shifts in the StaffSync system.
Pre-Conditions	Administrator must be logged into the system.
Post-Conditions	Shift information is successfully created, updated, assigned, or removed.
Main Flow	1. Administrator opens the shift management module.2. System displays existing shifts.3. Administrator selects an action (Create/Update/Assign/Delete).4. Administrator enters or modifies shift details.5. System validates the information.6. System saves changes and updates shift records.
Alternate Flow	If invalid shift details are entered, the system displays an error message and requests correction.
Exceptions	System or database error while managing shift data.

Table 41 Use-Case Manage Shifts Table

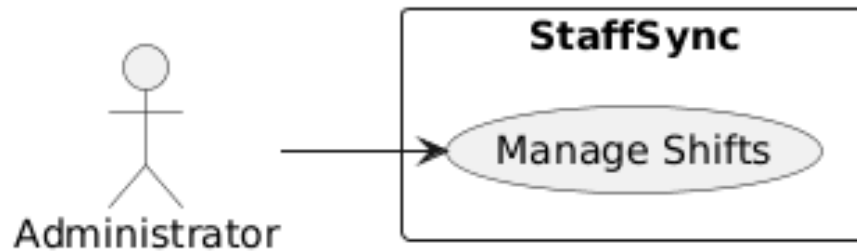


Figure 7Use Case Diagram Manage Shifts

3.2.2.7 UC-07 (Apply for Leave)

Field	Description
Use Case ID	UC_07
Use Case Name	Apply for Leave
Primary Actor	Staff
Description	This use case allows a staff member to submit a leave request by providing leave details such as type, duration, and reason.
Pre-Conditions	Staff must be logged into the system.
Post-Conditions	Leave request is successfully submitted and marked as pending for approval.
Main Flow	1. Staff opens the leave module.2. Staff selects leave type and dates.3. Staff enters reason for leave.4. Staff submits the request.5. System validates the information.6. System records the leave request and notifies the supervisor/administrator.
Alternate Flow	If required fields are missing or dates are invalid, the system prompts the staff to correct the information.
Exceptions	Network or server error during leave submission.

Table 42 Use-Case Apply For Leave Table



Figure 8 Use Case Diagram Apply For Leave

3.2.2.8 UC-08 (Approve / Reject Leave)

Field	Description
Use Case ID	UC_08
Use Case Name	Approve / Reject Leave
Primary Actor	Administrator, Supervisor
Description	This use case allows an administrator or supervisor to review leave requests submitted by staff members and approve or reject them based on organizational policies and staff availability.
Pre-Conditions	Administrator or Supervisor must be logged into the system. Pending leave requests must exist.
Post-Conditions	Leave request status is updated as approved or rejected and the staff member is notified.
Main Flow	1. Administrator/Supervisor opens the leave management module.2. System displays pending leave requests.3. Actor selects a leave request.4. Actor reviews leave details.5. Actor approves or rejects the request.6. System updates leave status and notifies the staff member.
Alternate Flow	If the leave request does not meet policy rules, the actor rejects the request with remarks.
Exceptions	System or database error while updating leave status.

Table 43 Use-Case Approve / Reject Leave Table

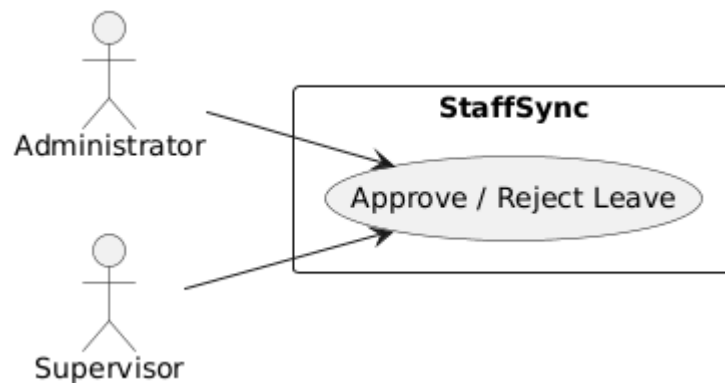


Figure 9 Use Case Diagram Approve / Reject Leave

3.2.2.9 UC-09 (Assign Tasks)

Field	Description
Use Case ID	UC_09
Use Case Name	Assign Tasks
Primary Actor	Administrator, Supervisor
Description	This use case allows the administrator or supervisor to assign tasks to staff members by defining task details and deadlines.
Pre-Conditions	Administrator or Supervisor must be logged into the system. Staff members must exist in the system.
Post-Conditions	Task is successfully assigned to the selected staff member and stored in the system.
Main Flow	1. Actor opens the task management module. 2. Actor selects a staff member. 3. Actor enters task details and deadline. 4. Actor assigns the task. 5. System validates the information. 6. System saves the task and notifies the staff member.
Alternate Flow	If required task details are missing, the system prompts the actor to provide correct information.
Exceptions	System or database error during task assignment.

Table 44 Use-Case Assign Tasks Table

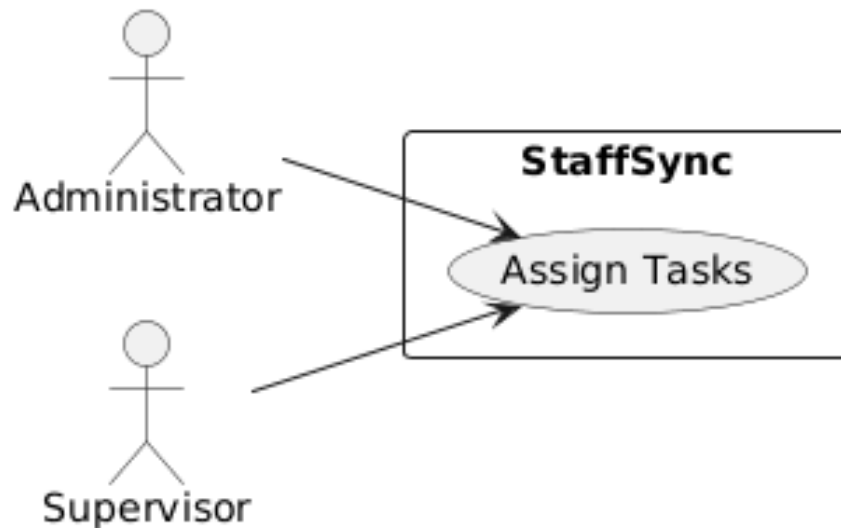


Figure 10 Use Case Diagram Assign Tasks

3.2.2.10 UC-10 (Verify Tasks)

Field	Description
Use Case ID	UC_10
Use Case Name	Verify Tasks
Primary Actor	Administrator, Supervisor
Description	This use case allows the administrator or supervisor to review the task proof submitted by staff members and verify the completion status of assigned tasks.
Pre-Conditions	Administrator or Supervisor must be logged into the system.Task proof must be submitted by staff.
Post-Conditions	Task status is updated as verified or rejected and the staff member is notified.
Main Flow	1. Actor opens the task verification module.2. System displays submitted task proofs.3. Actor selects a task to review.4. Actor verifies or rejects the task.5. System updates the task status.6. System notifies the staff member.
Alternate Flow	If the task proof is insufficient, the actor rejects the task with remarks.
Exceptions	System or database error during task verification.

Table 45 Use-Case Verify Tasks Table

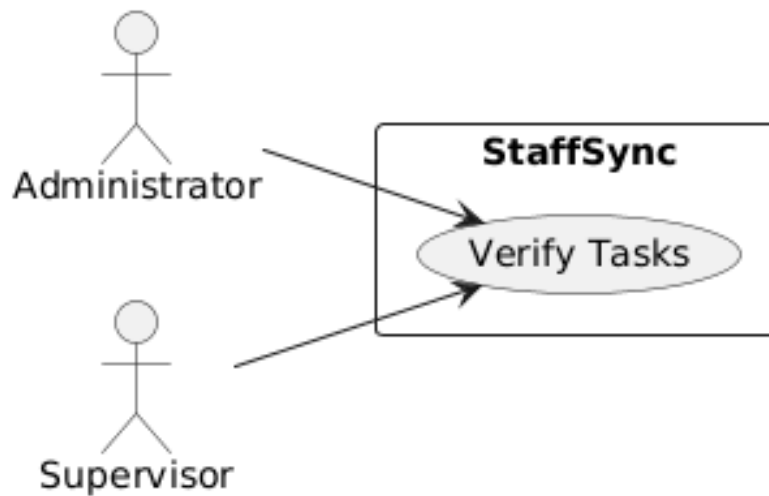


Figure 11 Use Case Diagram Verify Tasks

3.2.2.11 UC-11 (Manage Payroll)

Field	Description
Use Case ID	UC_11
Use Case Name	Manage Payroll
Primary Actor	Administrator
Description	This use case allows the administrator to manage payroll operations including salary calculation, payslip generation, and payroll record maintenance in the StaffSync system.
Pre-Conditions	Administrator must be logged into the system. Attendance and staff records must exist.
Post-Conditions	Payroll data is calculated, stored, and available for viewing or reporting.
Main Flow	1. Administrator opens the payroll module.2. System retrieves attendance and staff data.3. Administrator initiates salary calculation.4. System calculates salaries based on defined rules.5. System generates payroll records/payslips.6. System saves payroll information successfully.
Alternate Flow	If required data is missing, the system prompts the administrator to resolve data issues.
Exceptions	Calculation error or database failure during payroll processing.

Table 46 Use-Case Manage Payroll Table

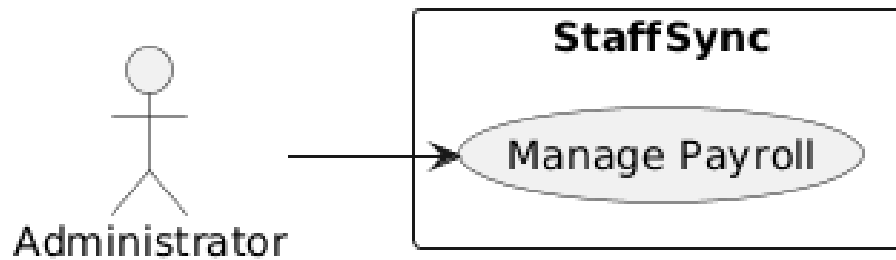


Figure 12 Use Case Manage Payroll

3.2.2.12 UC-12 (Generate Reports)

Field	Description
Use Case ID	UC_12
Use Case Name	Generate Reports
Primary Actor	Administrator, Supervisor
Description	This use case allows authorized users to generate system reports related to attendance, staff, payroll, tasks, and performance for analysis and record keeping.
Pre-Conditions	User must be logged into the system. Relevant data must exist in the system.
Post-Conditions	Selected report is generated and displayed or made available for download.
Main Flow	1. Actor opens the reports module. 2. Actor selects report type and date range. 3. System retrieves relevant data. 4. System generates the report. 5. System displays the report or provides download options.
Alternate Flow	If no data matches the selected criteria, the system displays a “No data available” message.
Exceptions	System or database error during report generation.

Table 47 Use-Case Generate Reports Table

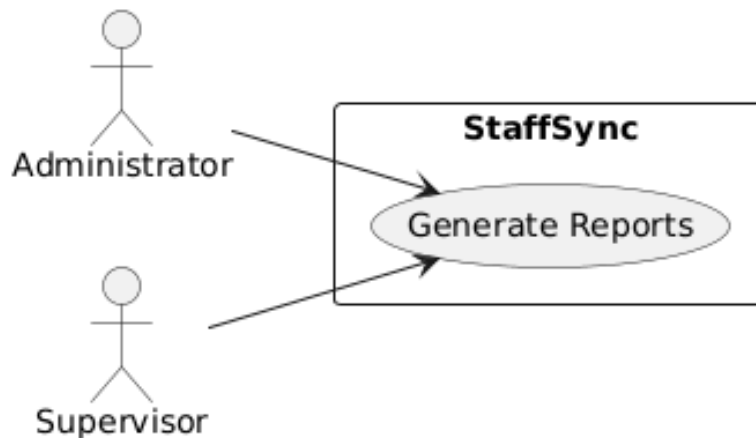


Figure 13 Use Case Generate Reports

3.2.2.13 UC-13 (Send Notifications)

Field	Description
Use Case ID	UC_13
Use Case Name	Send Notifications
Primary Actor	Administrator, Supervisor
Description	This use case allows the administrator or supervisor to send notifications or announcements to staff members through the StaffSync system for communication and updates.
Pre-Conditions	Administrator or Supervisor must be logged into the system.
Post-Conditions	Notification is successfully sent and delivered to the selected staff members.
Main Flow	1. Actor opens the notification module.2. Actor composes the notification message.3. Actor selects the target audience (all staff or specific staff).4. Actor sends the notification.5. System delivers the notification to recipients.
Alternate Flow	If the message content is empty, the system prompts the actor to enter valid content.
Exceptions	Network or server failure while sending notifications.

Table 48 Use-Case Send Notifications Table

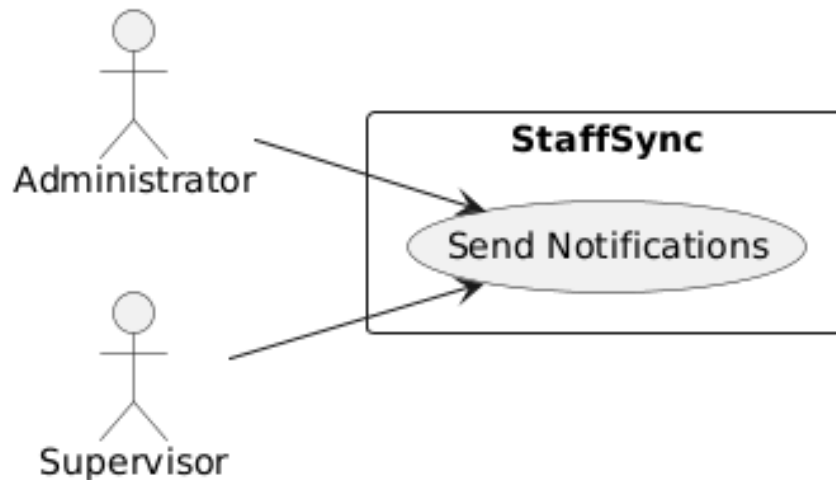


Figure 14 Use Case Send Notifications

3.2.2.14 UC-14 (View Dashboard)

Field	Description
Use Case ID	UC_14
Use Case Name	View Dashboard
Primary Actor	Administrator, Supervisor, Staff
Description	This use case allows authorized users to view the StaffSync dashboard, which presents a summarized view of key system information such as attendance status, tasks, notifications, and reports based on user roles.
Pre-Conditions	User must be logged into the system.
Post-Conditions	Dashboard is displayed with up-to-date and role-specific information.
Main Flow	1. User logs into the StaffSync system.2. System loads dashboard components based on user role.3. System retrieves relevant data (attendance, tasks, notifications).4. System displays the dashboard to the user.
Alternate Flow	If some data is unavailable, the system displays partial information with appropriate messages.
Exceptions	System or database error while loading dashboard data.

Table 49 Use- View Dashboard Table

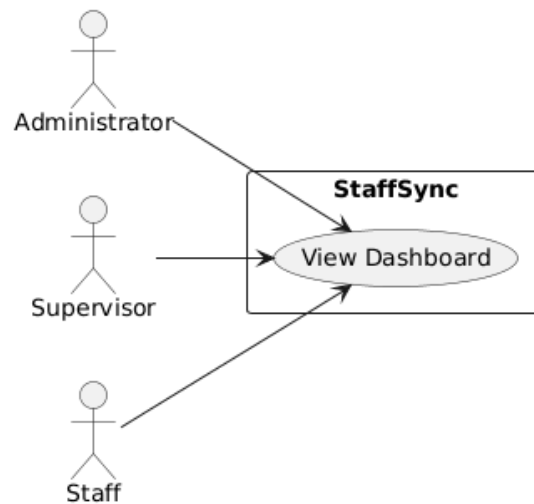


Figure 15 Use Case View Dashboard Diagram

Chapter 4: Design

System Design defines the overall architecture and structural components of the StaffSync system. This chapter focuses on how the system is organized, how different components interact with each other, and how system requirements are translated into a technical solution.

The purpose of this chapter is to provide a clear understanding of system architecture, data flow, and internal structure. It explains the design decisions taken to ensure scalability, maintainability, and efficient performance of the system.

This chapter includes architectural design, database design, and UML-based diagrams to visually represent system components and their relationships.

In this section, we provide the design analysis of our modules including the following designs

1. Architecture Diagram
2. ERD with data dictionary
3. Data Flow diagram
4. Class Diagram
5. Activity Diagram
6. Sequence Diagram
7. Collaboration Diagram
8. State Transition Diagram
9. Component Diagram
10. Deployment Diagram

4.1 Architecture Diagram

The system architecture of StaffSync defines the high-level structure of the system and explains how different components interact with each other. StaffSync follows a layered architecture approach to ensure scalability, maintainability, and efficient system performance.

The system is designed as a mobile-based application integrated with a centralized backend server and database. The mobile application is responsible for user interaction, while the backend handles business logic, data processing, authentication, and communication with the database.

The architecture consists of three main layers: **Presentation Layer**, **Application Layer**, and **Data Layer**. Each layer has a specific responsibility and communicates with other layers through well-defined interfaces.

4.1.1 Architecture Layers Explanation

Presentation Layer:

This layer represents the user interface of the StaffSync system. It includes the mobile application used by administrators, supervisors, and staff members to interact with the system. This layer handles user input, displays information, and sends requests to the backend server.

Application Layer:

The application layer contains the core business logic of the system. It processes requests received from the presentation layer, applies system rules, manages authentication, handles attendance, tasks, payroll, and generates reports.

Data Layer:

The data layer is responsible for data storage and management. It includes the database where all system data such as user accounts, attendance records, staff information, and payroll data are stored securely.

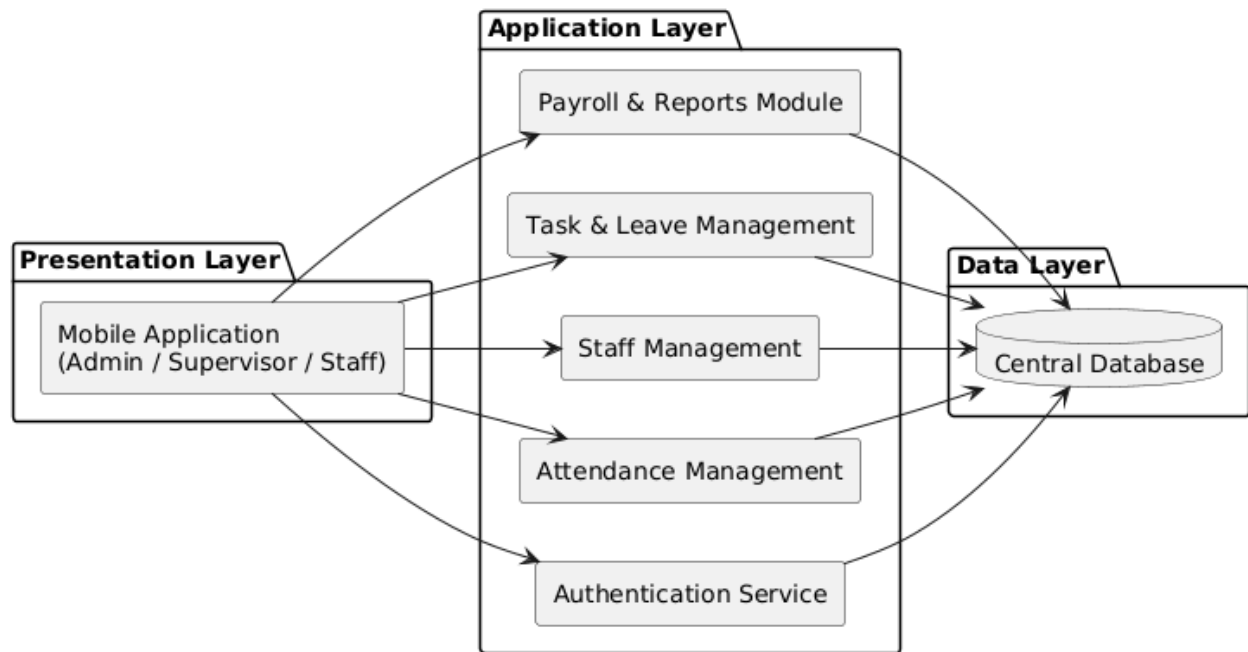


Figure 16 Architecture Diagram

4.2 ERD with data dictionary

Entity The database design of the StaffSync system defines how data is stored, organized, and managed within the system. A well-structured database ensures data integrity, consistency, and efficient data retrieval.

StaffSync uses a centralized relational database to store information related to users, staff members, attendance records, leave requests, tasks, payroll, and notifications. The database design is based on real-world entities and their relationships to support system functionality.

An Entity Relationship Diagram (ERD) is used to visually represent the database structure, followed by a data dictionary that describes each entity and its attributes in detail.

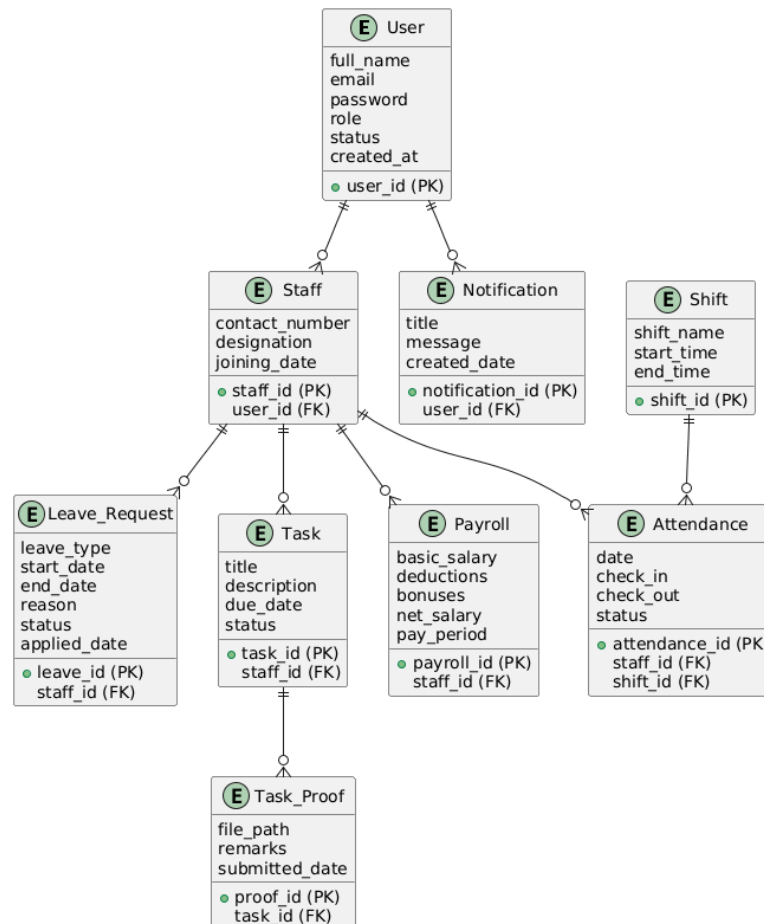


Figure 17 ERD

Data Dictionary

The data dictionary provides a detailed description of each entity used in the StaffSync database. It defines the attributes, data types, and purpose of each field to ensure clarity, consistency, and proper database implementation.

Attribute Name	Data Type	Description
user_id (PK)	Integer	Unique identifier for each user
full_name	Varchar(100)	Full name of the user
email	Varchar(100)	User's email address (unique)
password	Varchar(255)	Encrypted password
role	Varchar(20)	Role of user (Admin, Supervisor, Staff)
status	Varchar(20)	Account status (Active/Inactive)
created_at	DateTime	Account creation date

Table 50 User

Attribute Name	Data Type	Description
staff_id (PK)	Integer	Unique identifier for staff member
user_id (FK)	Integer	References User table
contact_number	Varchar(20)	Staff contact number
designation	Varchar(50)	Job title/designation
joining_date	Date	Date of joining

Table 51 Staff

Attribute Name	Data Type	Description
----------------	-----------	-------------

shift_id (PK)	Integer	Unique identifier for shift
shift_name	Varchar(50)	Name of the shift
start_time	Time	Shift start time
end_time	Time	Shift end time

Table 52 Shift

Attribute Name	Data Type	Description
attendance_id (PK)	Integer	Unique attendance record ID
staff_id (FK)	Integer	References Staff table
shift_id (FK)	Integer	References Shift table
date	Date	Attendance date
check_in	Time	Check-in time
check_out	Time	Check-out time
status	Varchar(20)	Present / Absent / Late

Table 53 Attendance

Attribute Name	Data Type	Description
leave_id (PK)	Integer	Unique leave request ID
staff_id (FK)	Integer	References Staff table
leave_type	Varchar(30)	Type of leave
start_date	Date	Leave start date
end_date	Date	Leave end date
reason	Text	Reason for leave
status	Varchar(20)	Pending / Approved / Rejected
applied_date	Date	Date of leave application

Table 54 Leave Request

Attribute Name	Data Type	Description
task_id (PK)	Integer	Unique task identifier
staff_id (FK)	Integer	Assigned staff member
Title	Varchar(100)	Task title
Description	Text	Task details
due_date	Date	Task deadline
status	Varchar(20)	Assigned / Completed / Verified

Table 55 Task

Attribute Name	Data Type	Description
proof_id (PK)	Integer	Unique proof ID
task_id (FK)	Integer	References Task table
file_path	Varchar(255)	Proof file location
remarks	Text	Staff remarks
submitted_date	Date	Proof submission date

Table 56 Task Proof

Attribute Name	Data Type	Description
payroll_id (PK)	Integer	Unique payroll ID
staff_id (FK)	Integer	References Staff table
basic_salary	Decimal	Base salary
deductions	Decimal	Salary deductions
bonuses	Decimal	Additional bonuses
net_salary	Decimal	Final salary
pay_period	Varchar(30)	Salary period

Table 57 Payroll

Attribute Name	Data Type	Description
notification_id (PK)	Integer	Unique notification ID
user_id (FK)	Integer	References User table
title	Varchar(100)	Notification title
message	Text	Notification content
created_date	Date	Notification date

Table 58 Notification

4.3 Data Flow diagram

A Data Flow Diagram (DFD) represents the flow of data within the StaffSync system. It illustrates how data moves between external entities, system processes, and the database. DFDs help in understanding system functionality from a data processing perspective.

The StaffSync system uses Data Flow Diagrams to clearly show how user inputs are processed, how data is stored, and how information is retrieved. This section includes a **Level 0 (Context Diagram)** and a **Level 1 DFD** to provide both high-level and detailed views of the system.

4.3.1 The level 0

What this diagram shows:

- **One single process:** StaffSync System
- **External Entities:**
 - Administrator
 - Supervisor
 - Staff
- **Data flows** (Login data, Attendance data, Reports, Notifications, etc.)

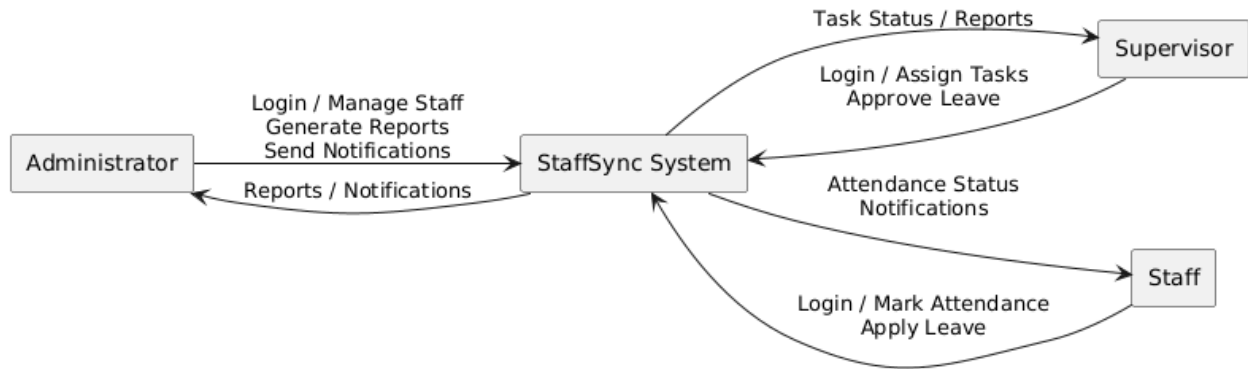


Figure 18 Level 0 DFD

4.3.2 The level 1

What this diagram shows:

- **Multiple internal processes:**
 - Authentication
 - Attendance Management
 - Leave Management
 - Task Management
 - Payroll & Reports

One central database

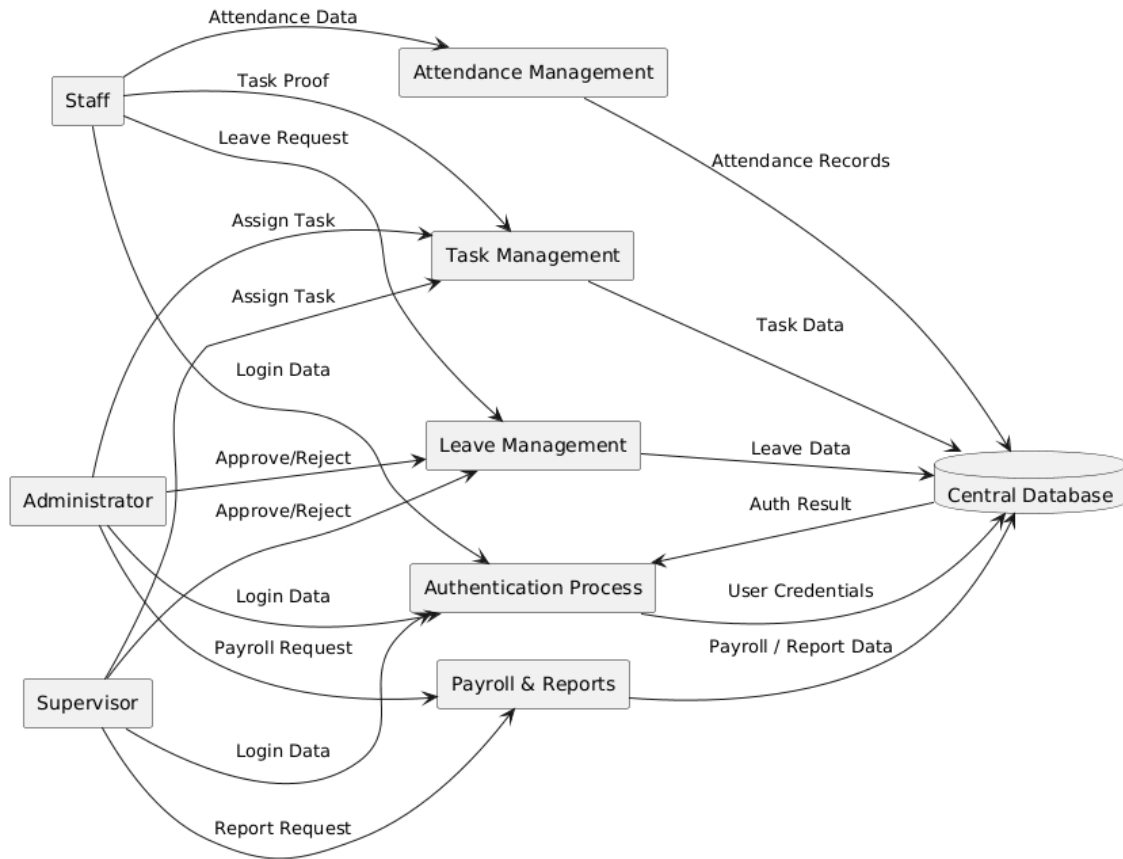


Figure 19 Level 1 DFD

4.4 Class Diagram

The class diagram represents the object-oriented structure of the StaffSync system. It shows the system classes, their attributes, methods, and relationships. This diagram helps in understanding how data and behavior are encapsulated within different system components.

The StaffSync class diagram is designed based on system requirements and use cases. It defines core entities such as users, staff, attendance, leave, tasks, payroll, and notifications, along with their associations. The diagram supports modularity and reusability of system components. Objects.

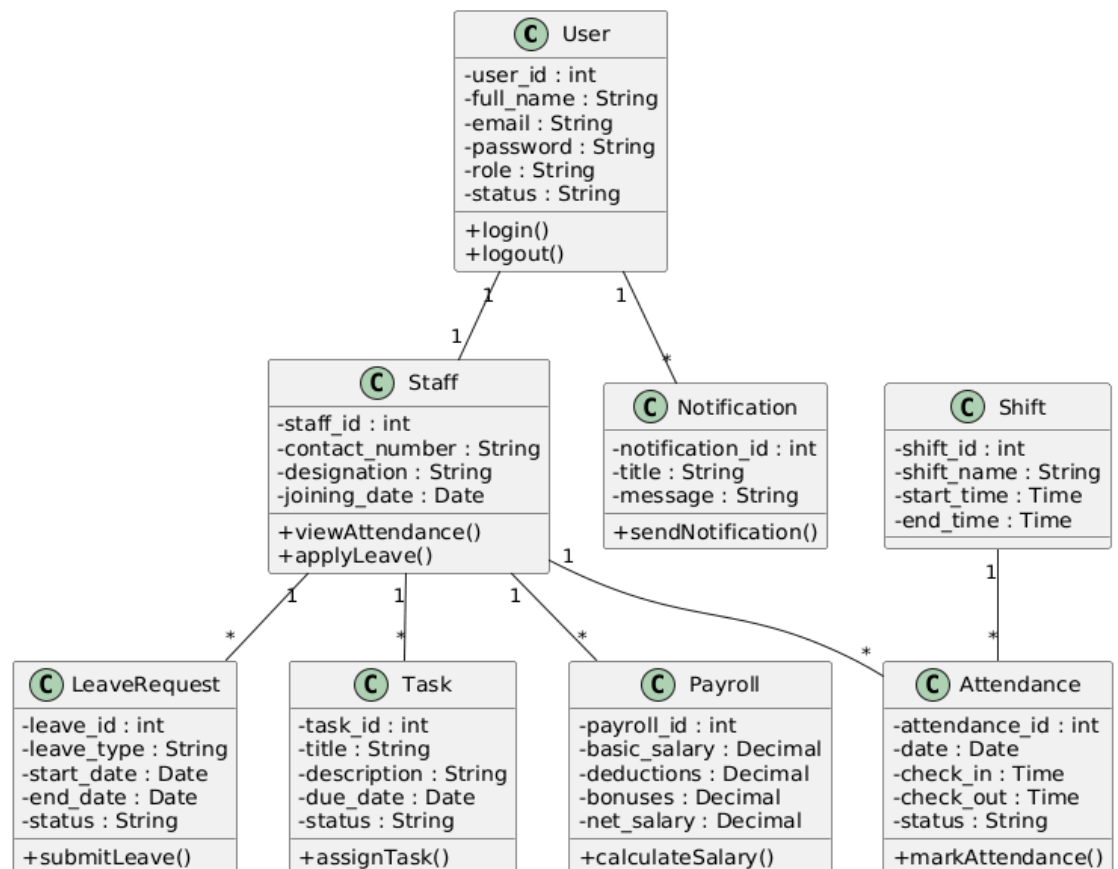


Figure 20 Class Diagram

4.5 Activity Diagram

This Activity diagrams represent the dynamic behavior of the StaffSync system by illustrating the flow of activities involved in performing a specific task. These diagrams help in understanding the sequence of actions, decision points, and parallel activities within the system.

In this section, activity diagrams are used to model key system processes to provide a clear understanding of user workflows. The diagrams focus on common and critical operations performed by users during system interaction.

Activity Diagram – Login

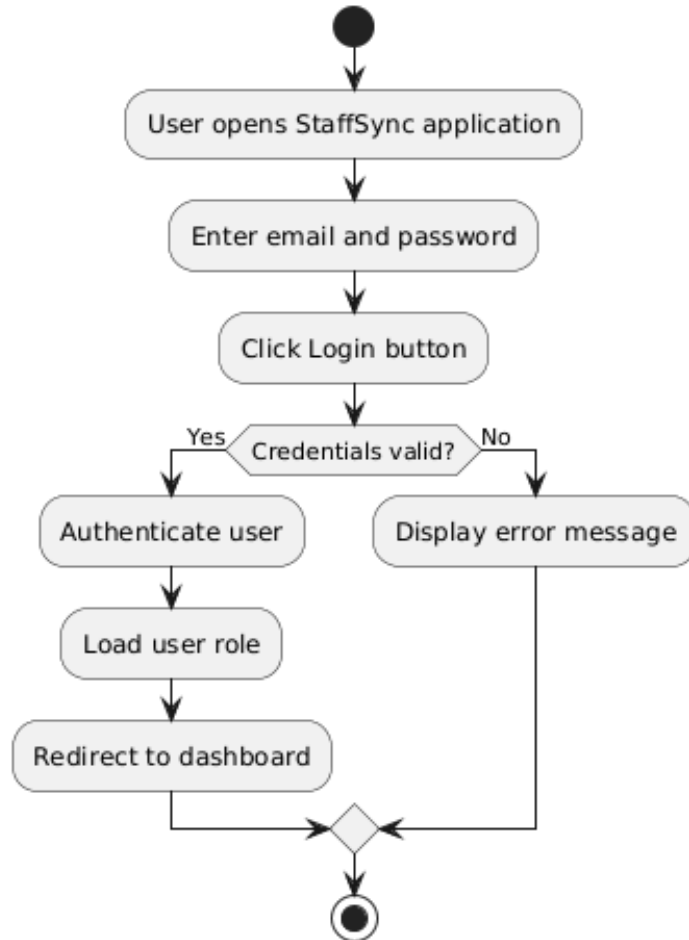


Figure 21 Activity Diagram – Login

Activity Diagram – Logout

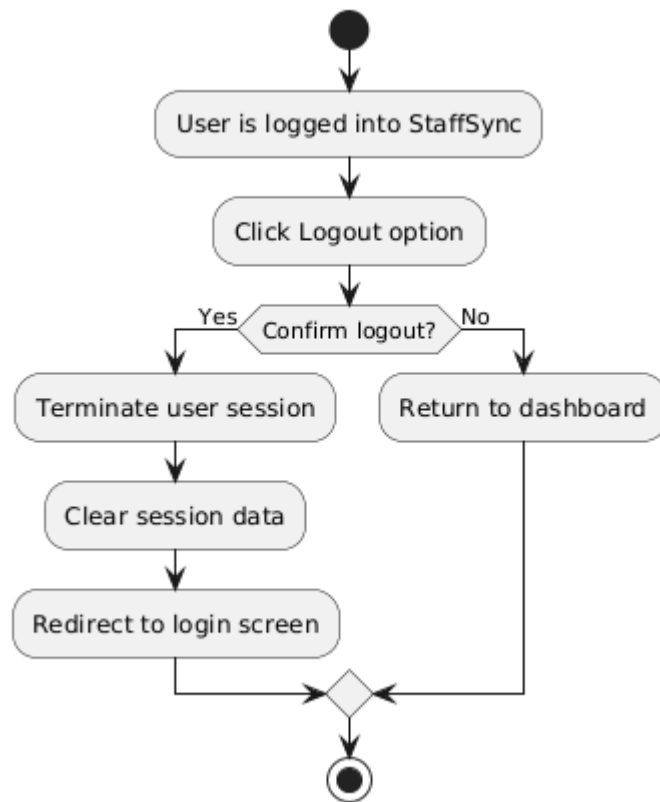


Figure 22 Activity Diagram – Logout

Activity Diagram – Manage Staff

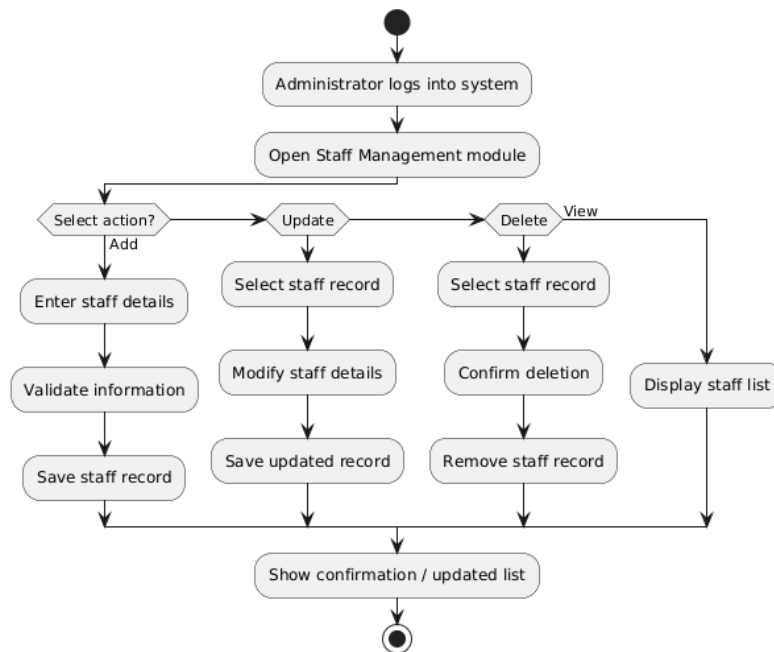


Figure 23 Activity Diagram – Manage Staff

Activity Diagram – Mark Attendance

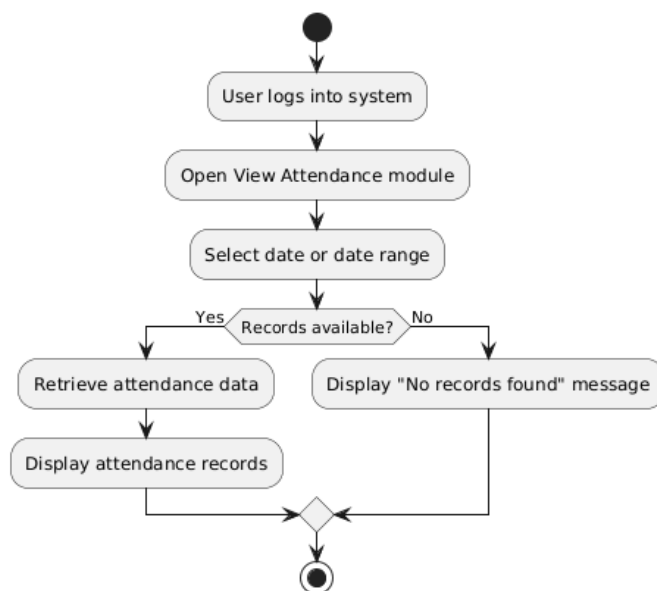


Figure 24 Activity Diagram – Mark Attendance

Activity Diagram – View Attendance

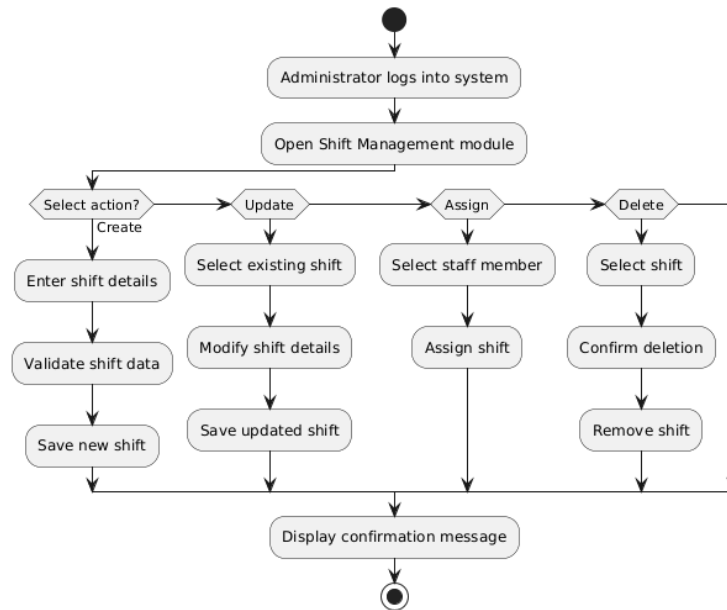


Figure 25 Activity Diagram – View Attendance

Activity Diagram – Manage Shifts

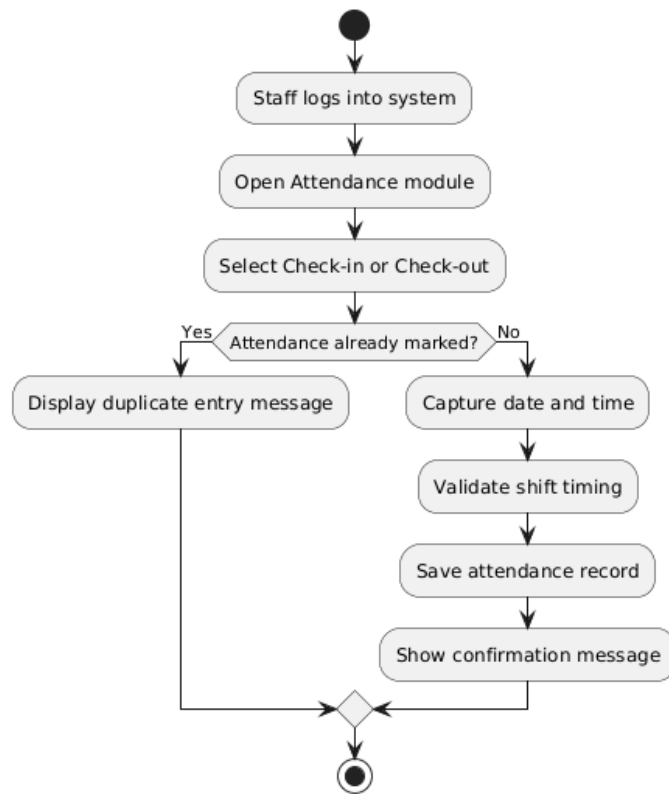


Figure 26 Activity Diagram – Manage Shifts

Activity Diagram – Apply for Leave

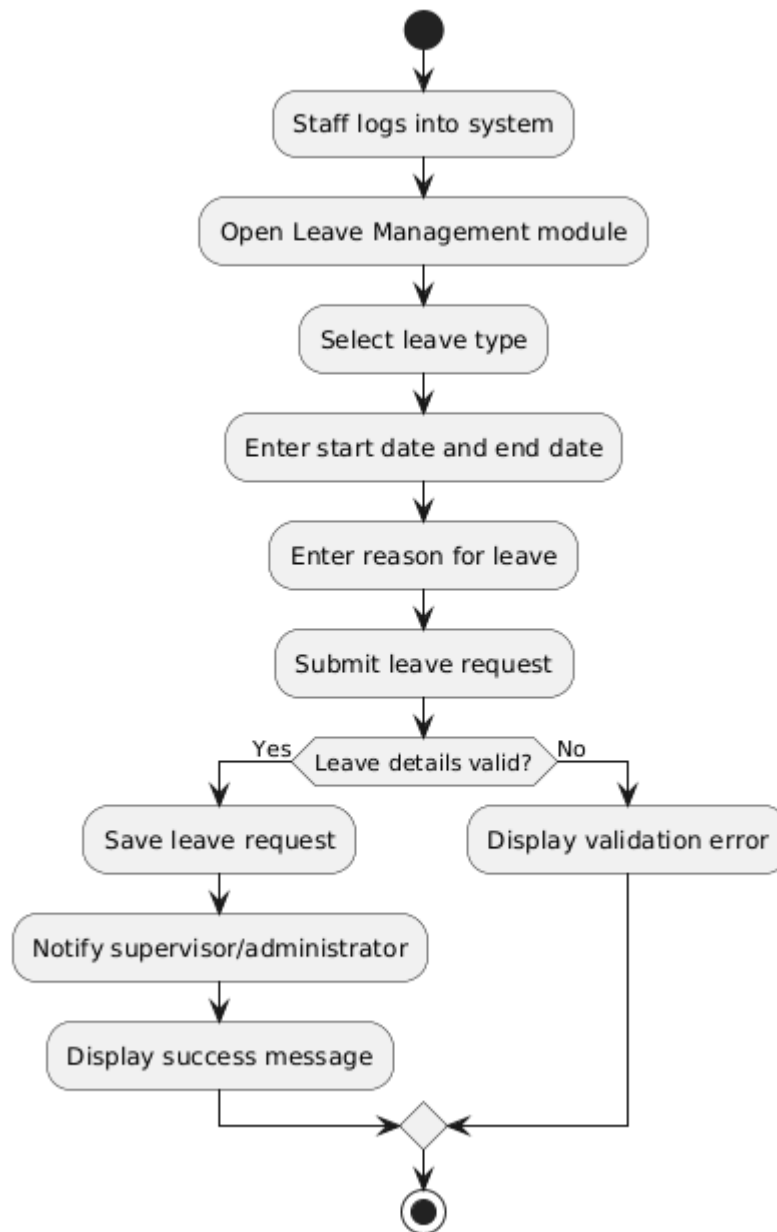


Figure 27 Activity Diagram – Apply for Leave

Activity Diagram 8: Approve / Reject Leave

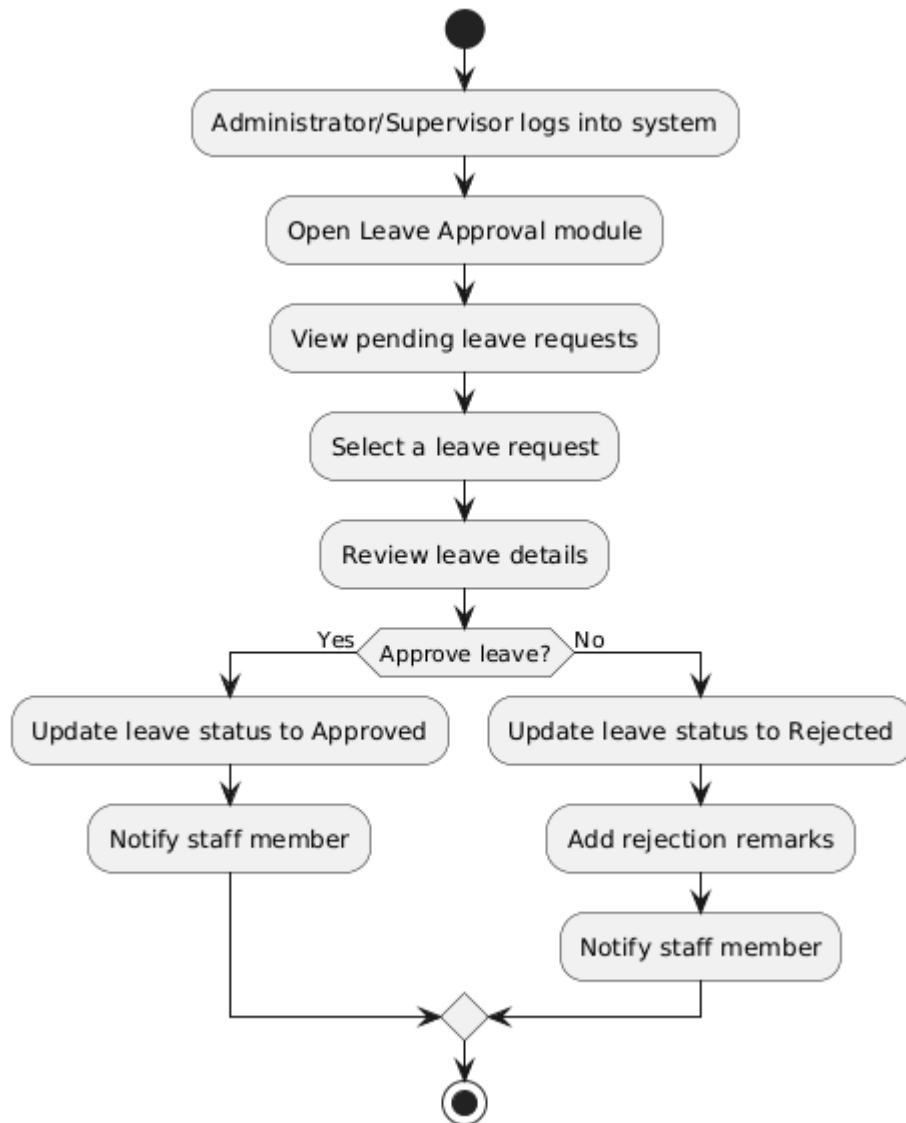


Figure 28 Activity Diagram 8: Approve / Reject Leave

Activity Diagram – Assign Task

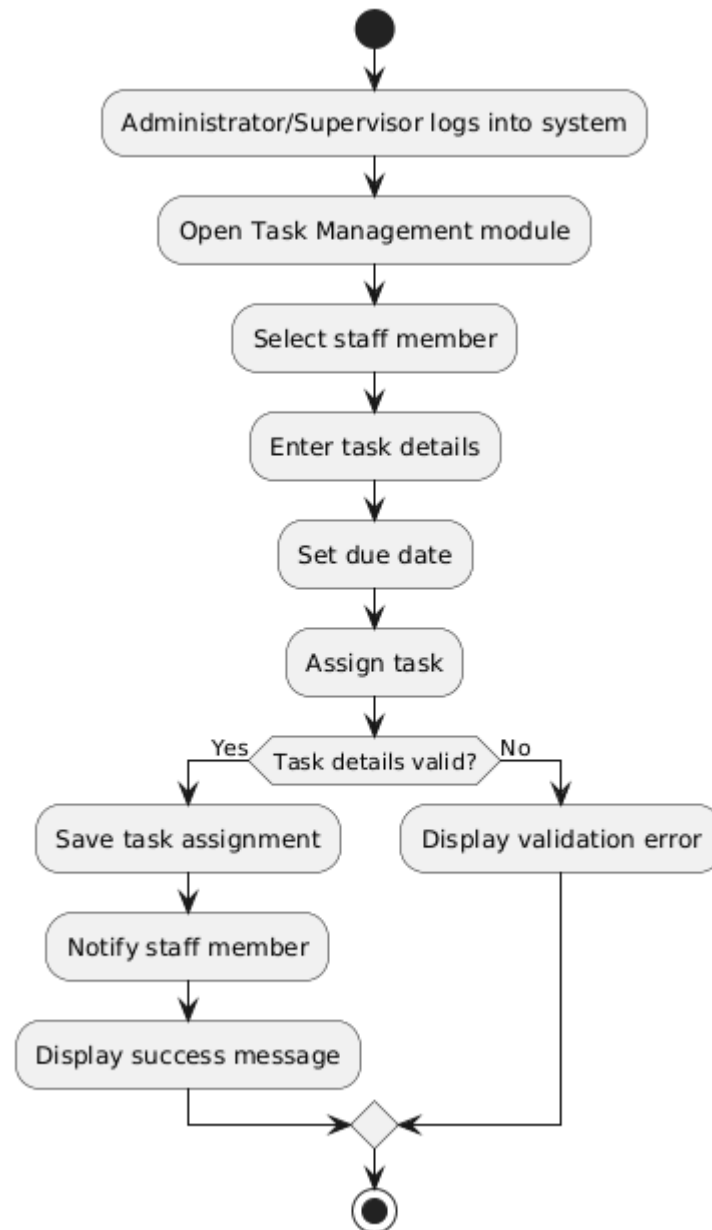


Figure 29 Activity Diagram – Assign Task

Activity Diagram – Verify Task

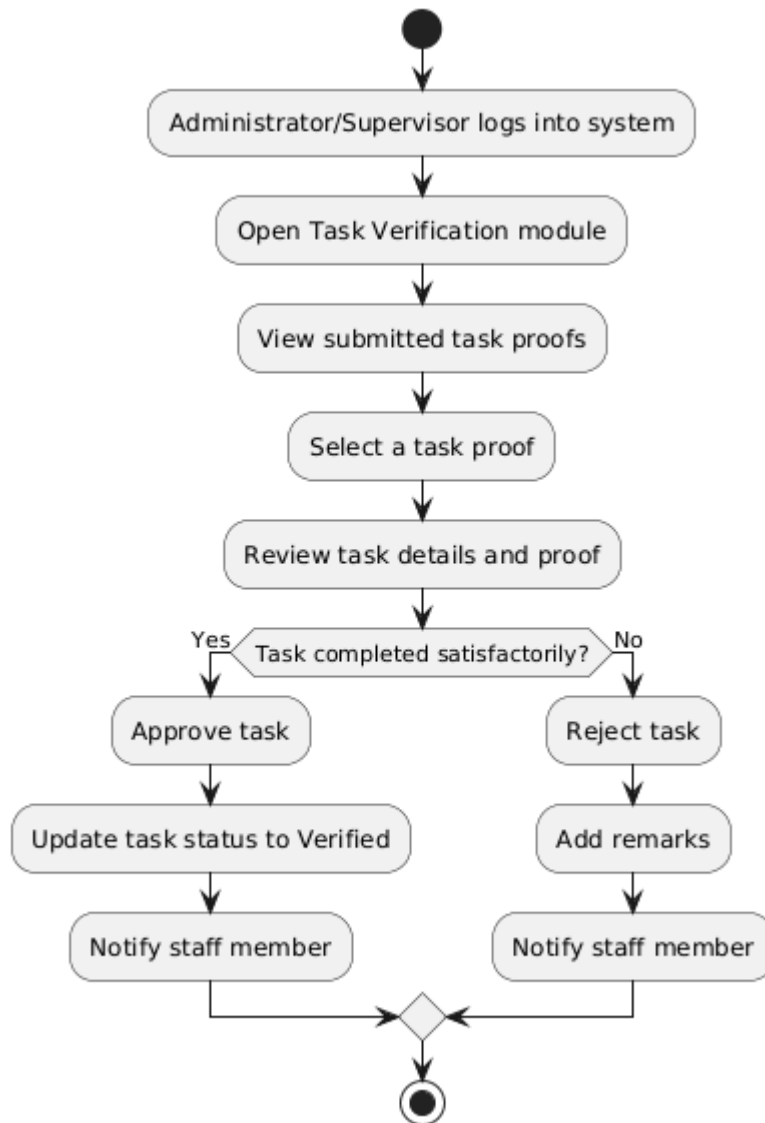


Figure 30 Activity Diagram – Verify Task

Activity Diagram – Manage Payroll

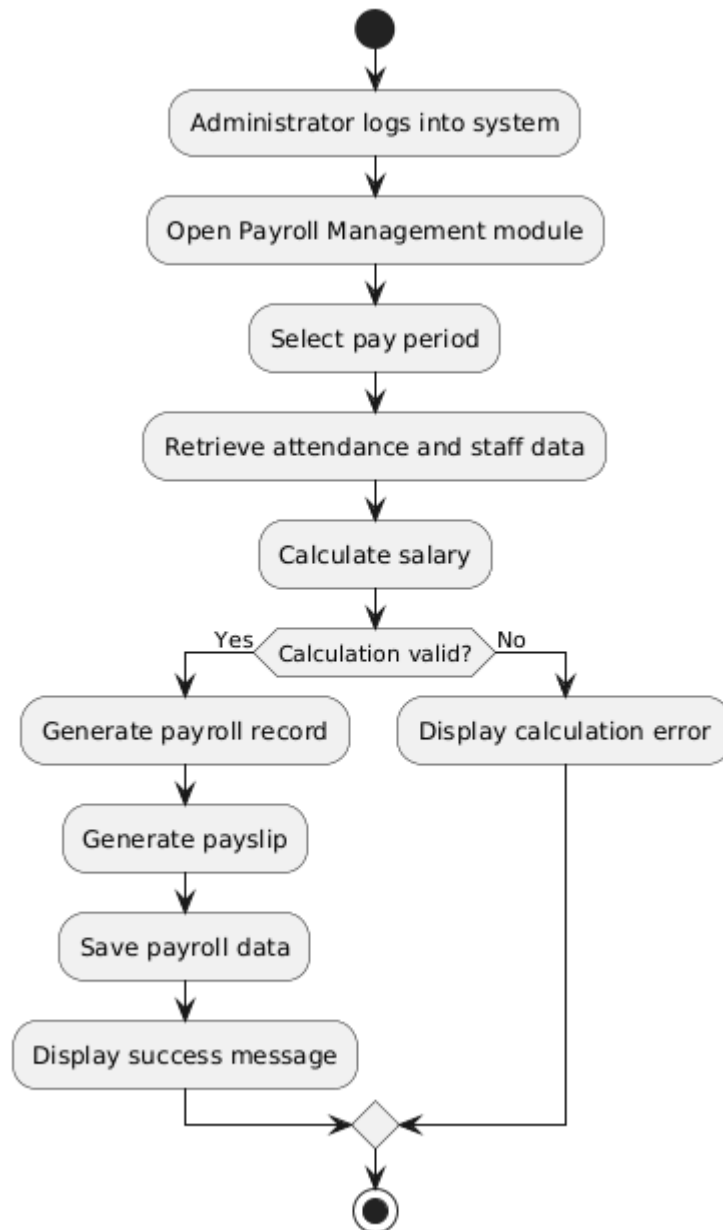


Figure 31 Activity Diagram – Manage Payroll

Activity Diagram – Generate Reports

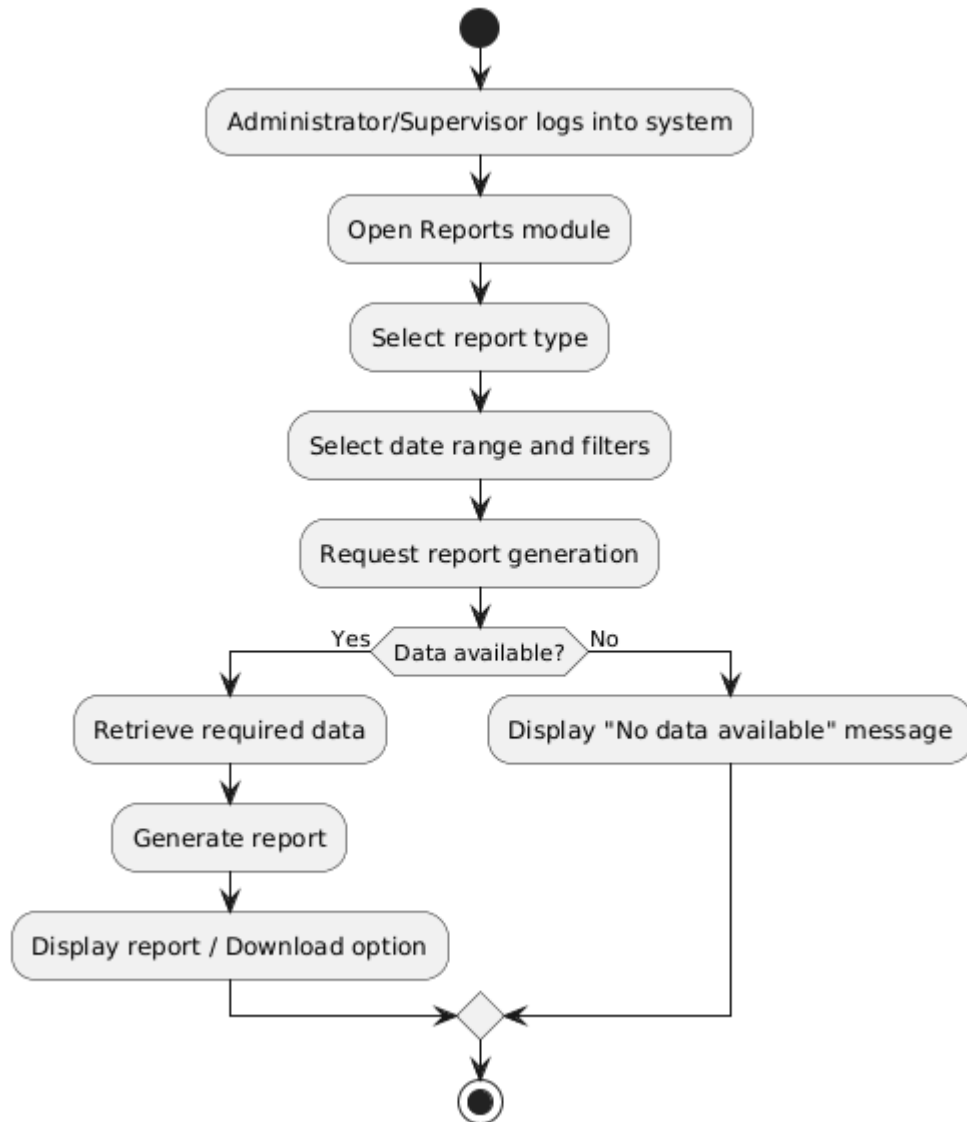


Figure 32 Activity Diagram – Generate Reports

Activity Diagram – Send Notifications

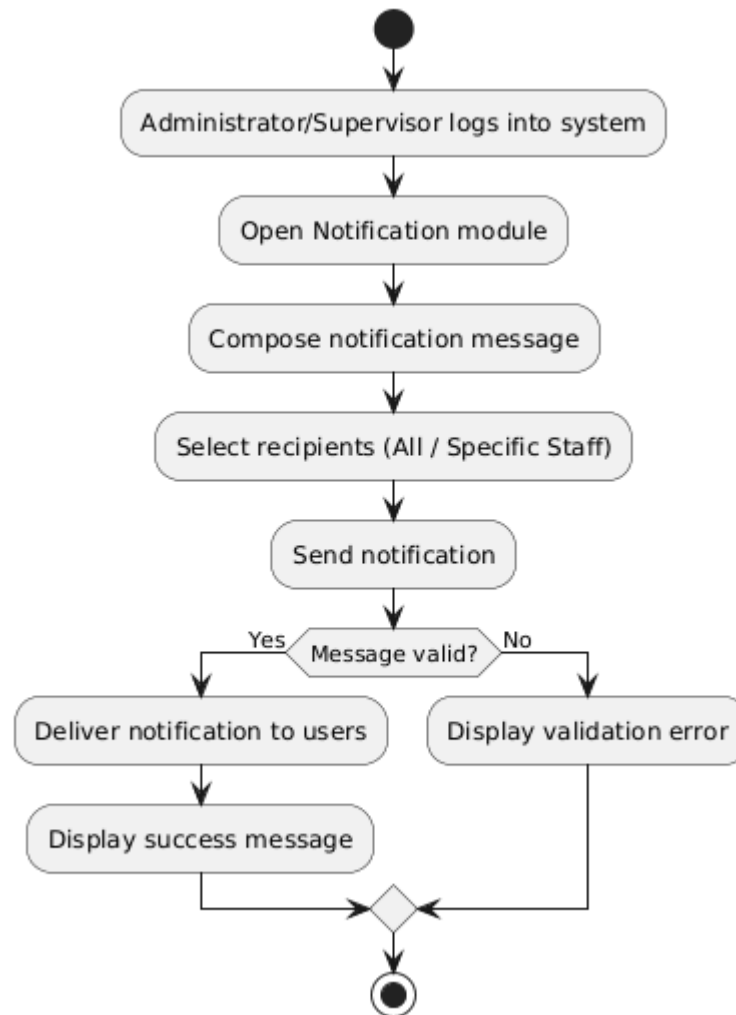


Figure 33 Activity Diagram – Send Notifications

Activity Diagram 14: View Dashboard

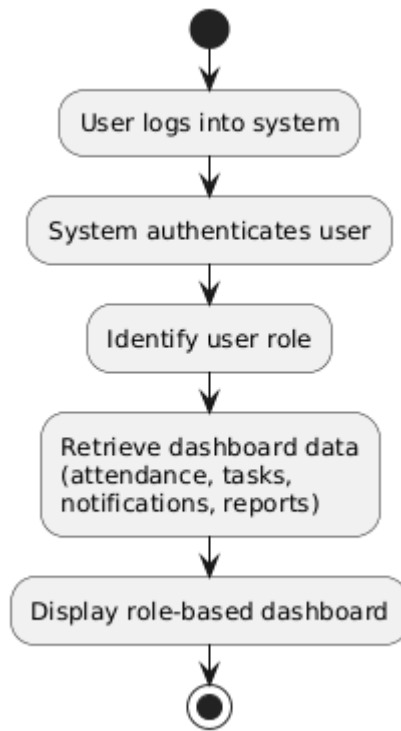


Figure 34 Activity Diagram 14: View Dashboard

4.6 Sequence Diagram

Sequence diagrams illustrate the interaction between system components in a time-ordered sequence. They show how users, system modules, and the database communicate to complete a specific operation.

In StaffSync, sequence diagrams are used to model critical system interactions such as authentication, attendance marking, and leave approval. These diagrams help in understanding system behavior and message flow during runtime.

Sequence Diagram – Login

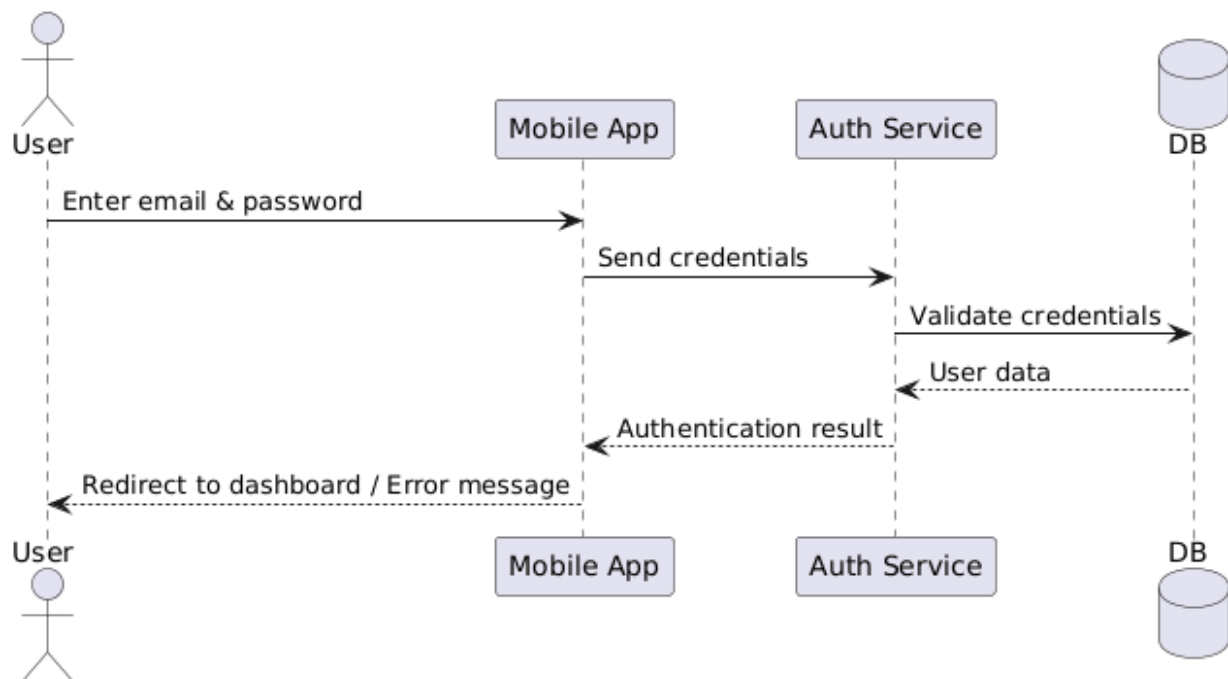


Figure 35 Sequence Diagram for Login Process

Sequence Diagram – Logout

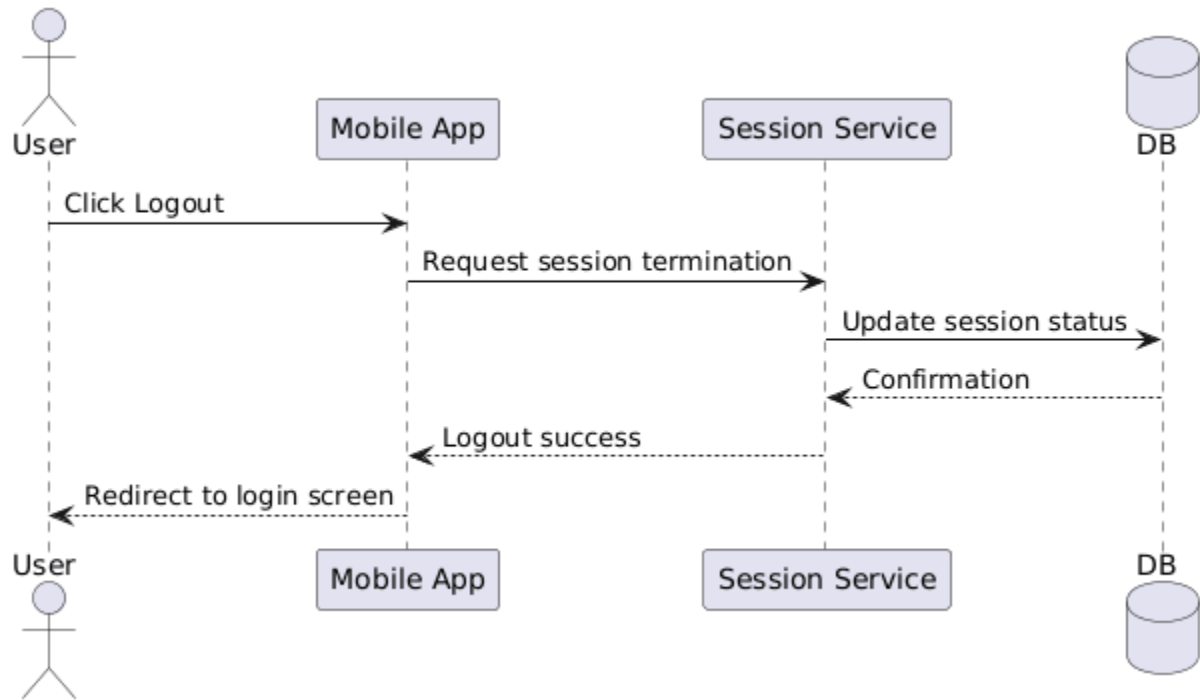


Figure 36 Sequence Diagram for Logout Process

Sequence Diagram – Manage Staff

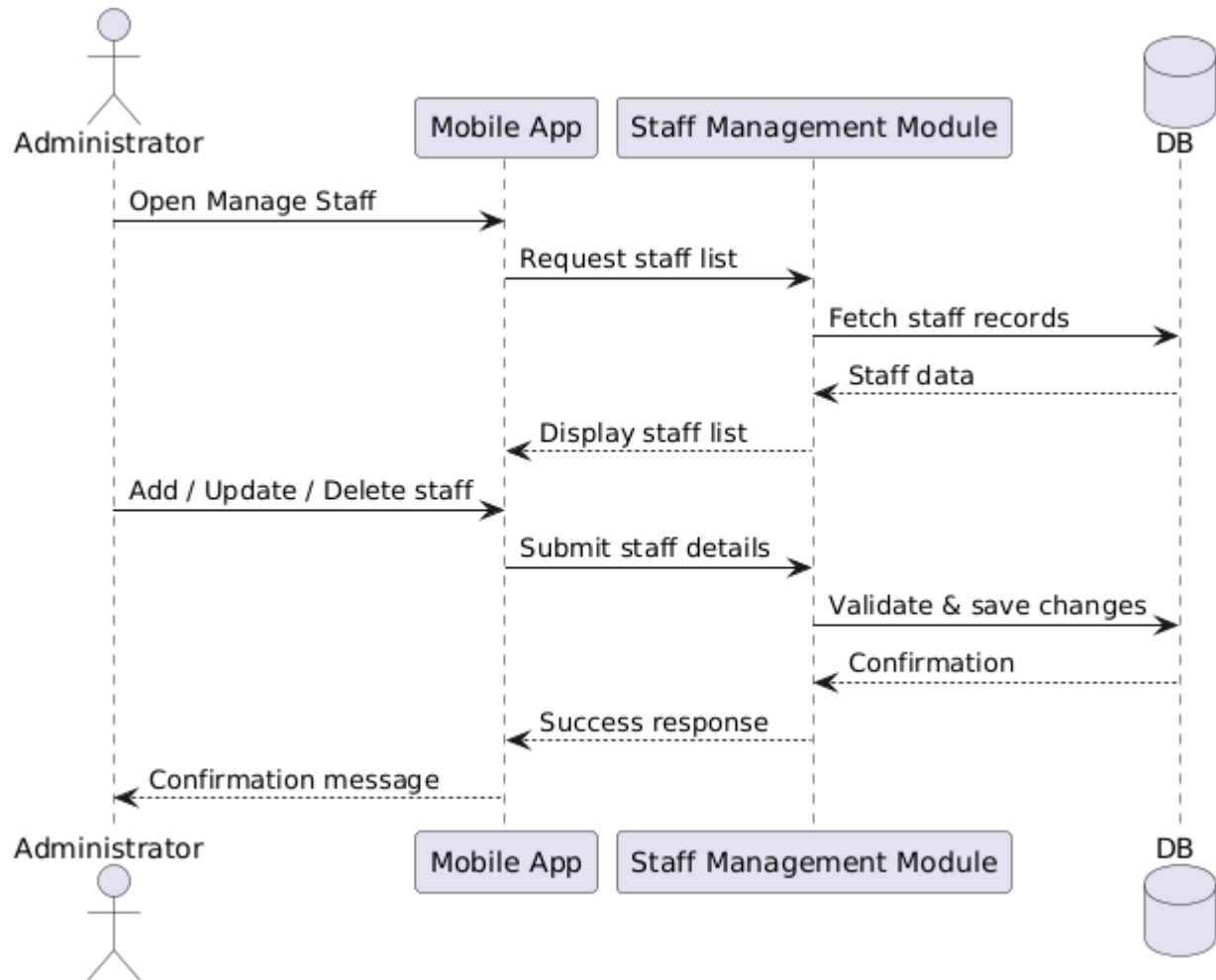


Figure 37 Sequence Diagram for Manage Staff

Sequence Diagram – Mark Attendance

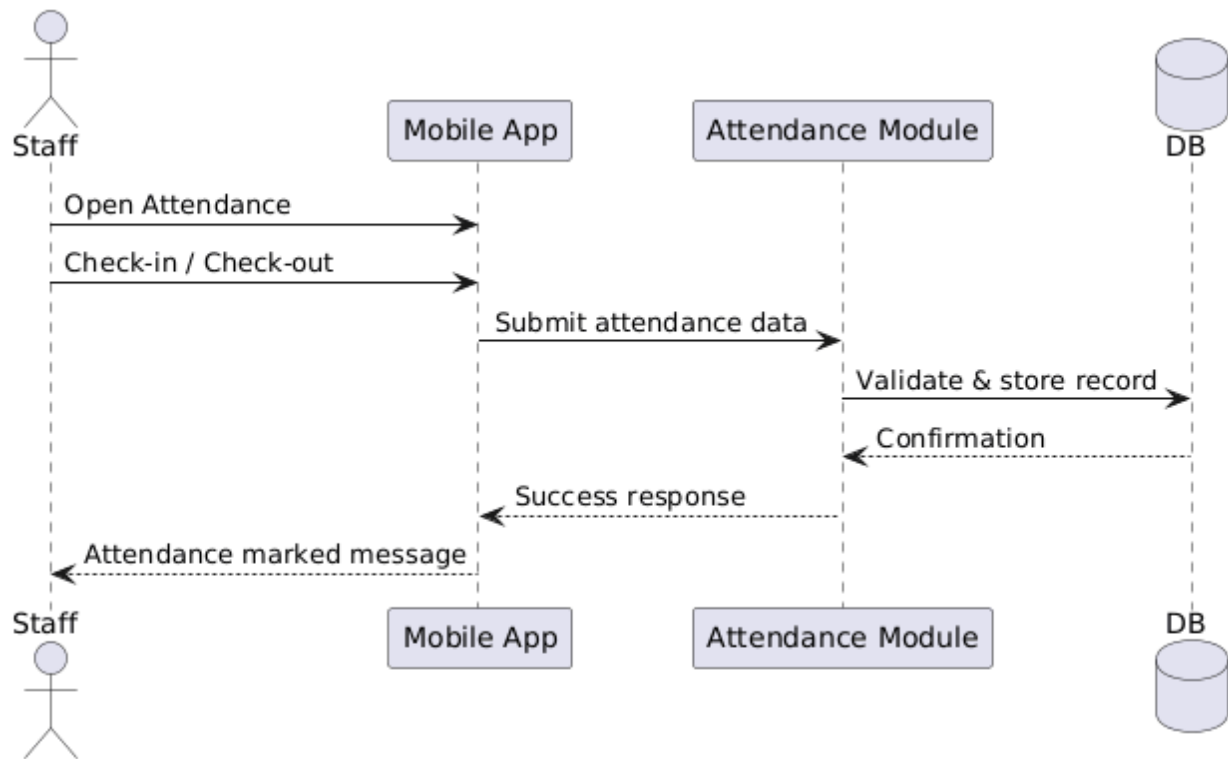


Figure 38 Sequence Diagram for Mark Attendance

Sequence Diagram – View Attendance

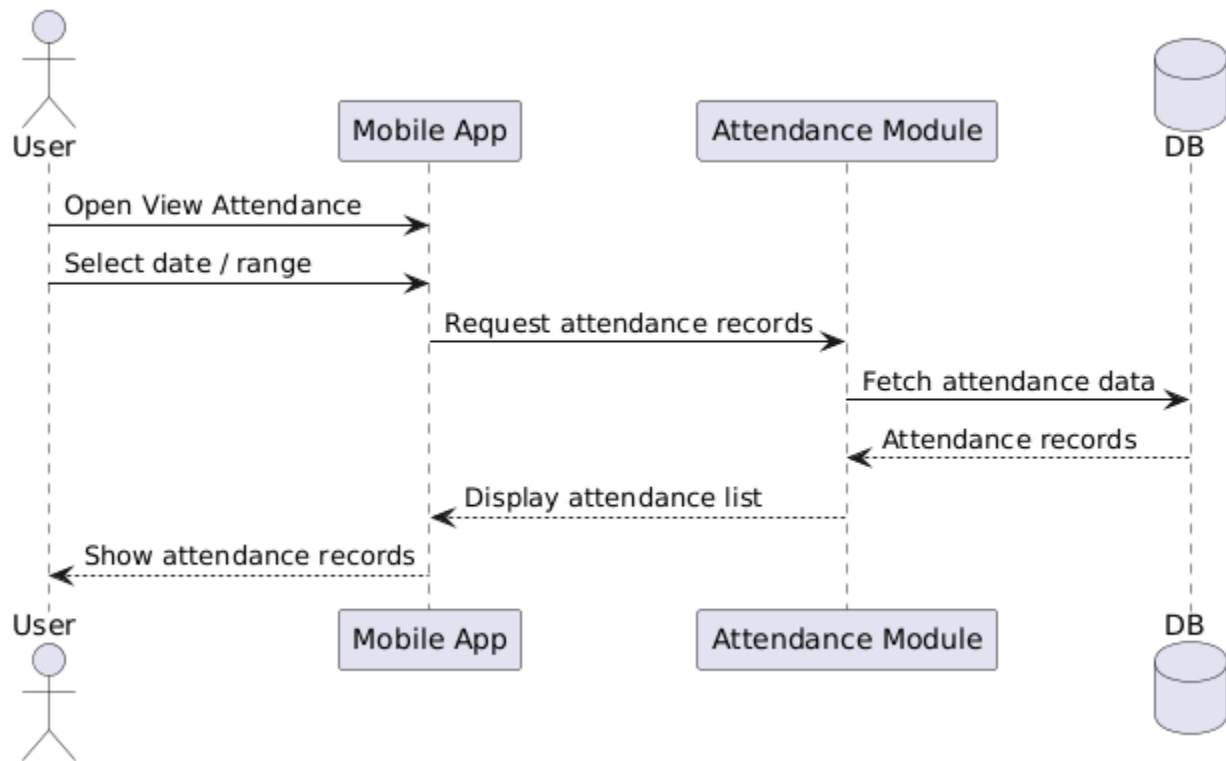


Figure 39 Sequence Diagram for View Attendance

Sequence Diagram – Manage Shifts

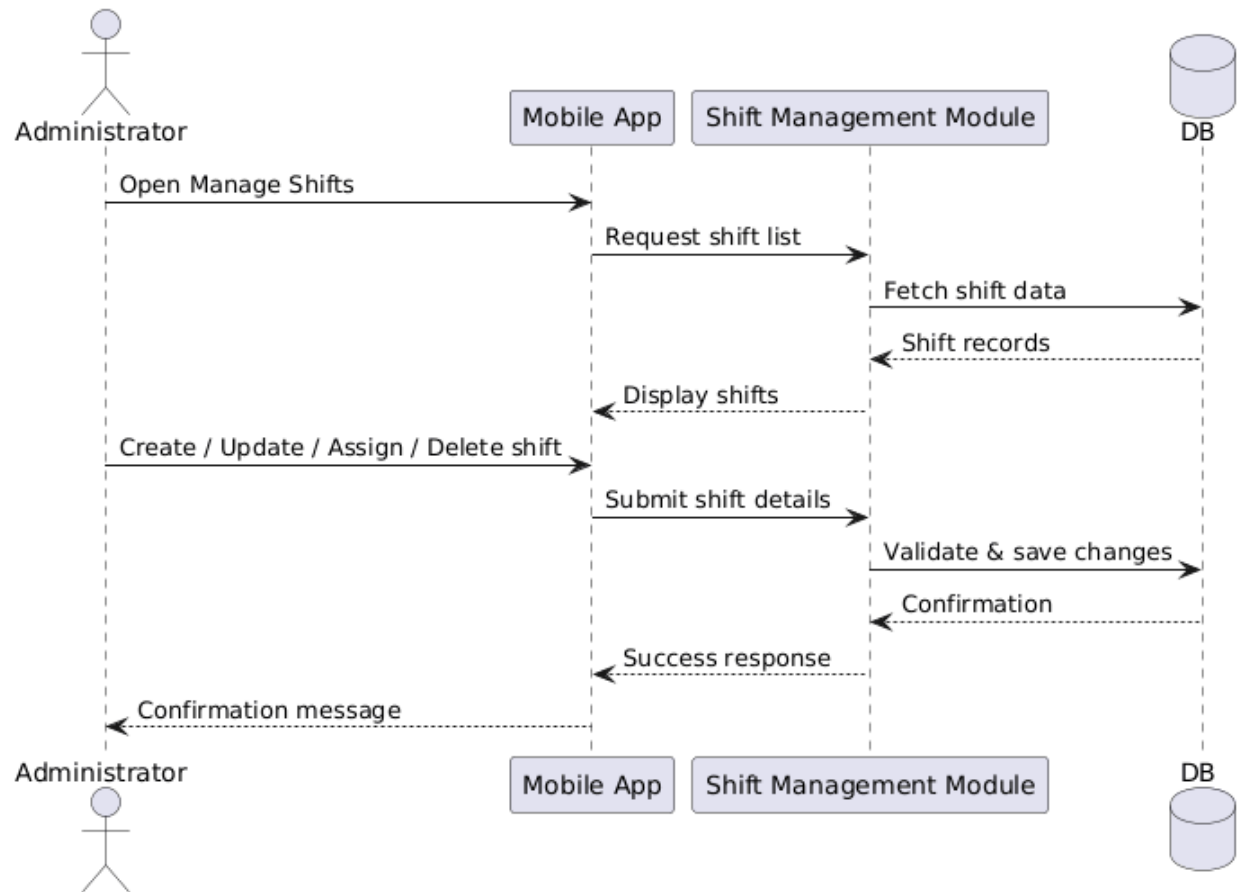


Figure 40 Sequence Diagram for Manage Shifts

Sequence Diagram – Apply for Leave

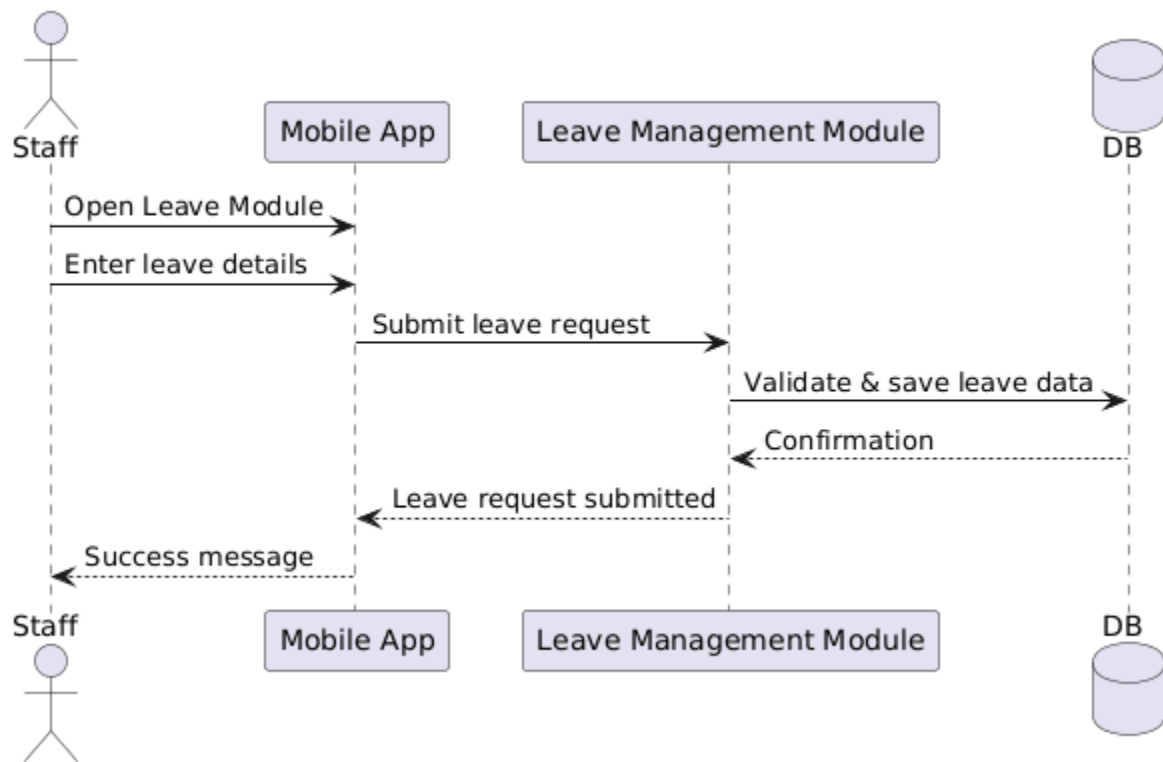


Figure 41 Sequence Diagram for Apply For Leave

Sequence Diagram – Approve / Reject Leave

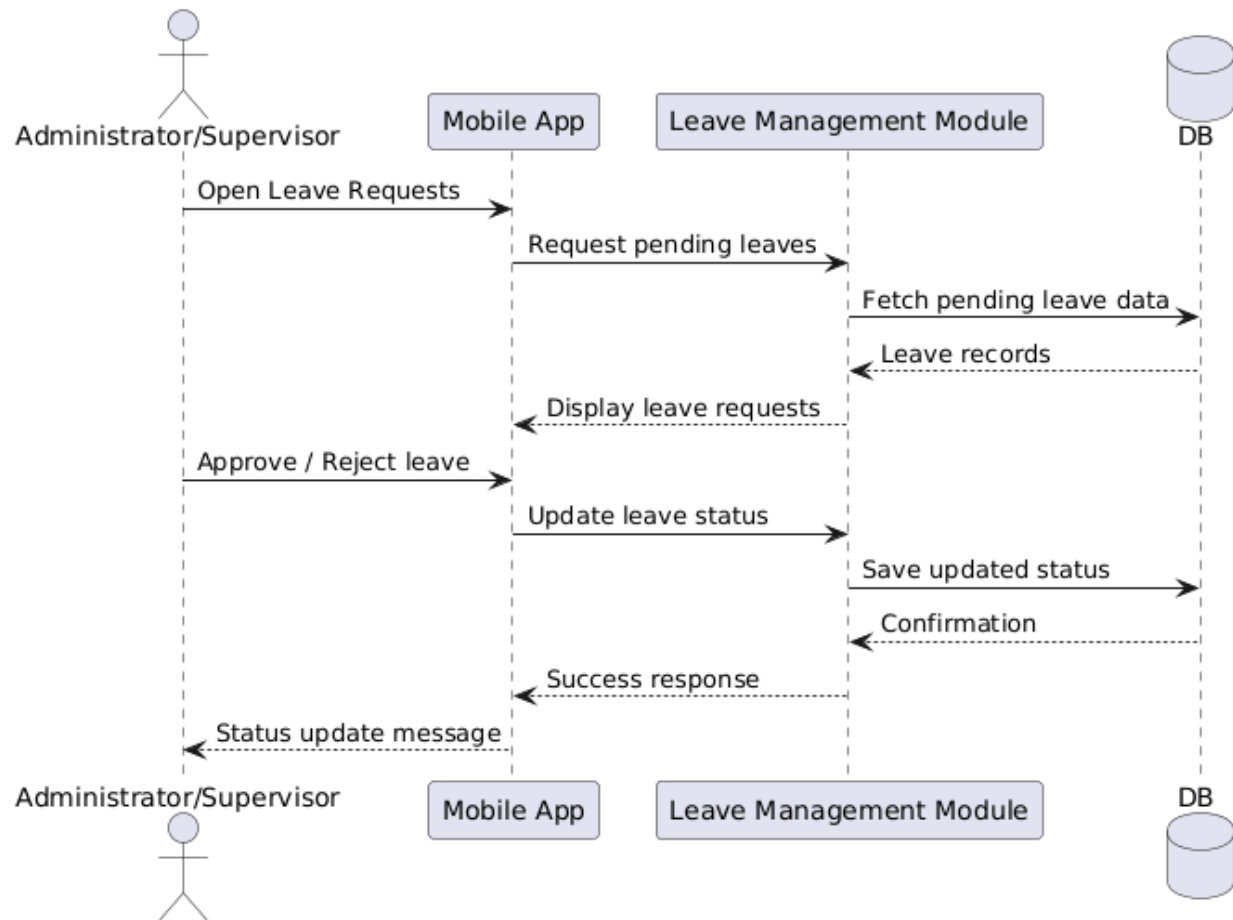


Figure 42 Sequence Diagram for Approve / Reject Leave

Sequence Diagram – Assign Task

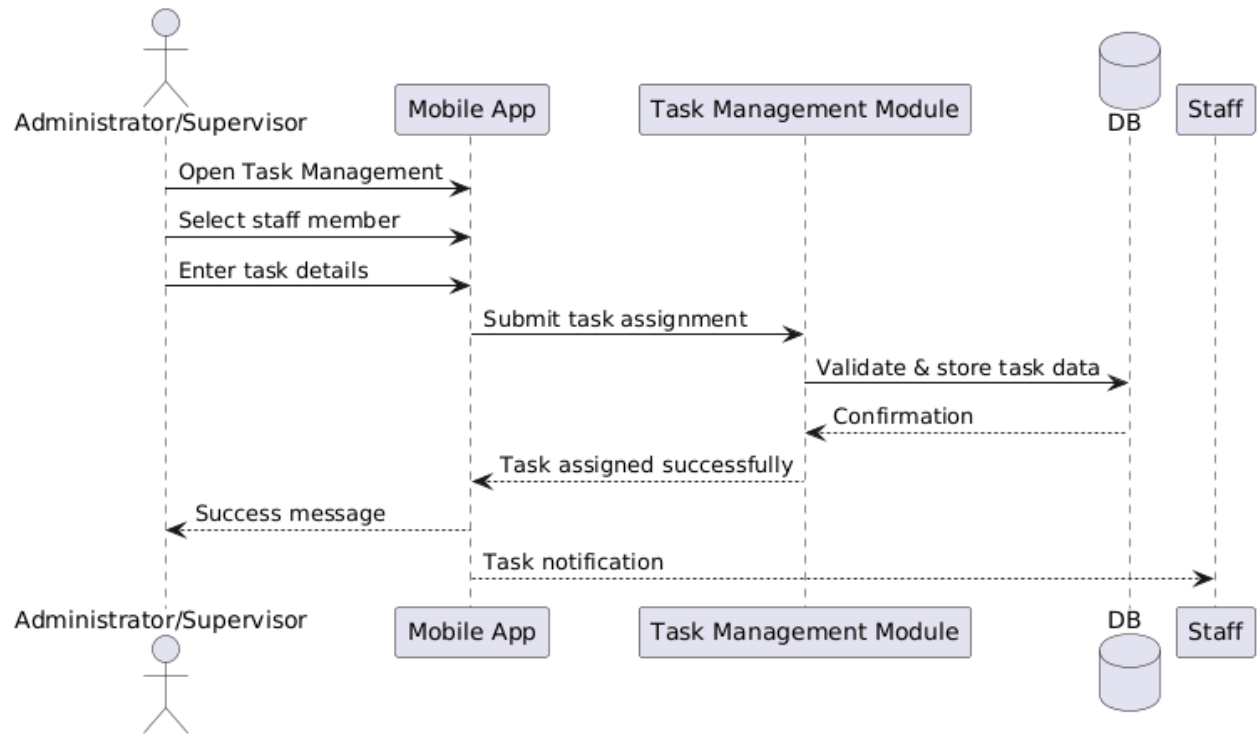


Figure 43 Sequence Diagram for Assign Task

Sequence Diagram – Verify Task

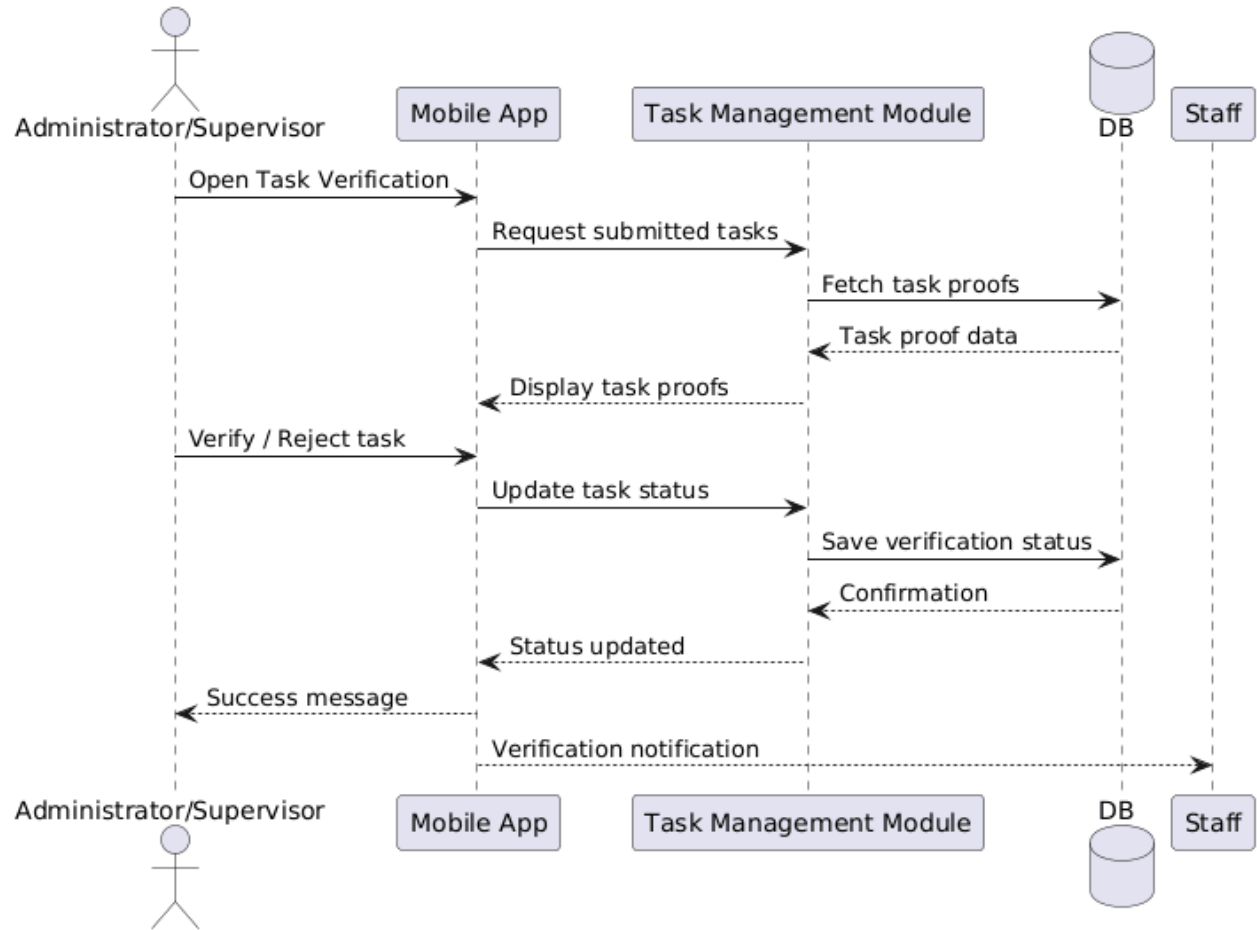


Figure 44 Sequence Diagram for Verify Task

Sequence Diagram – Manage Payroll

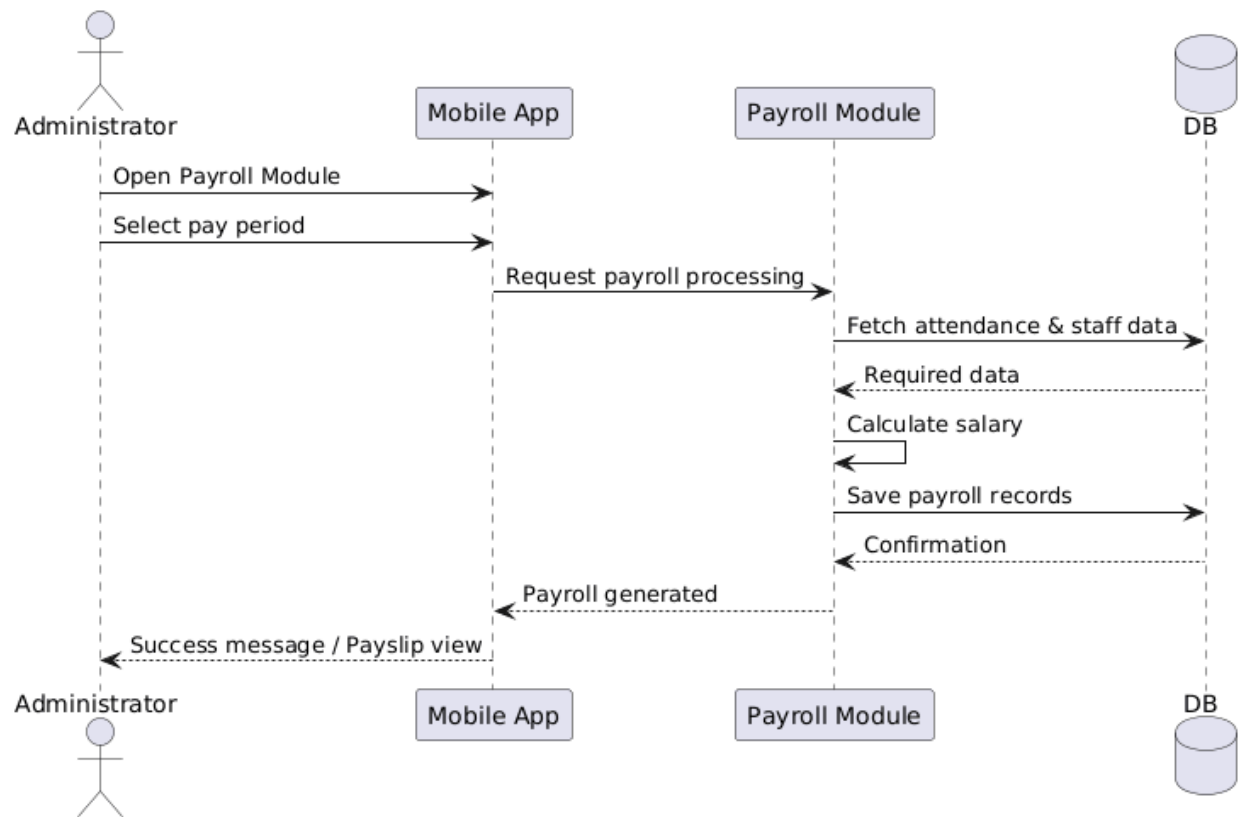


Figure 45 Sequence Diagram for Manage Payrolls

Sequence Diagram – Generate Reports

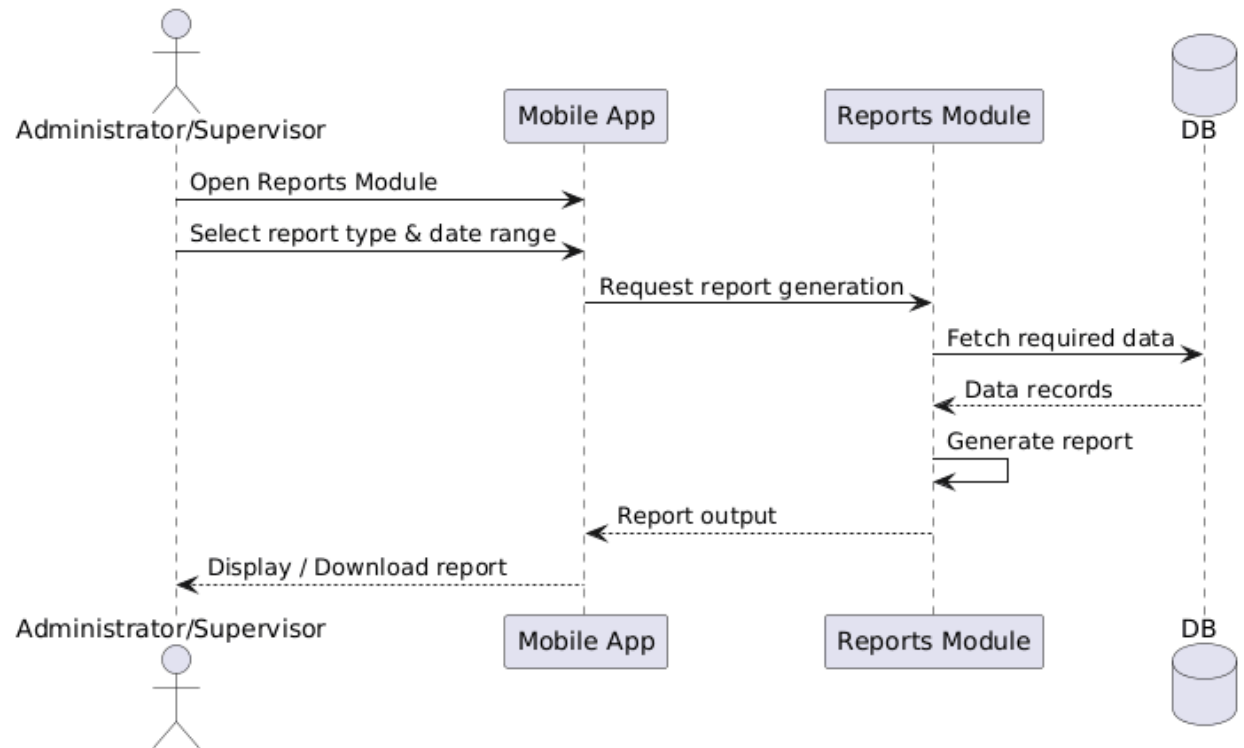


Figure 46 Sequence Diagram for Generate Reports

Sequence Diagram – Send Notifications

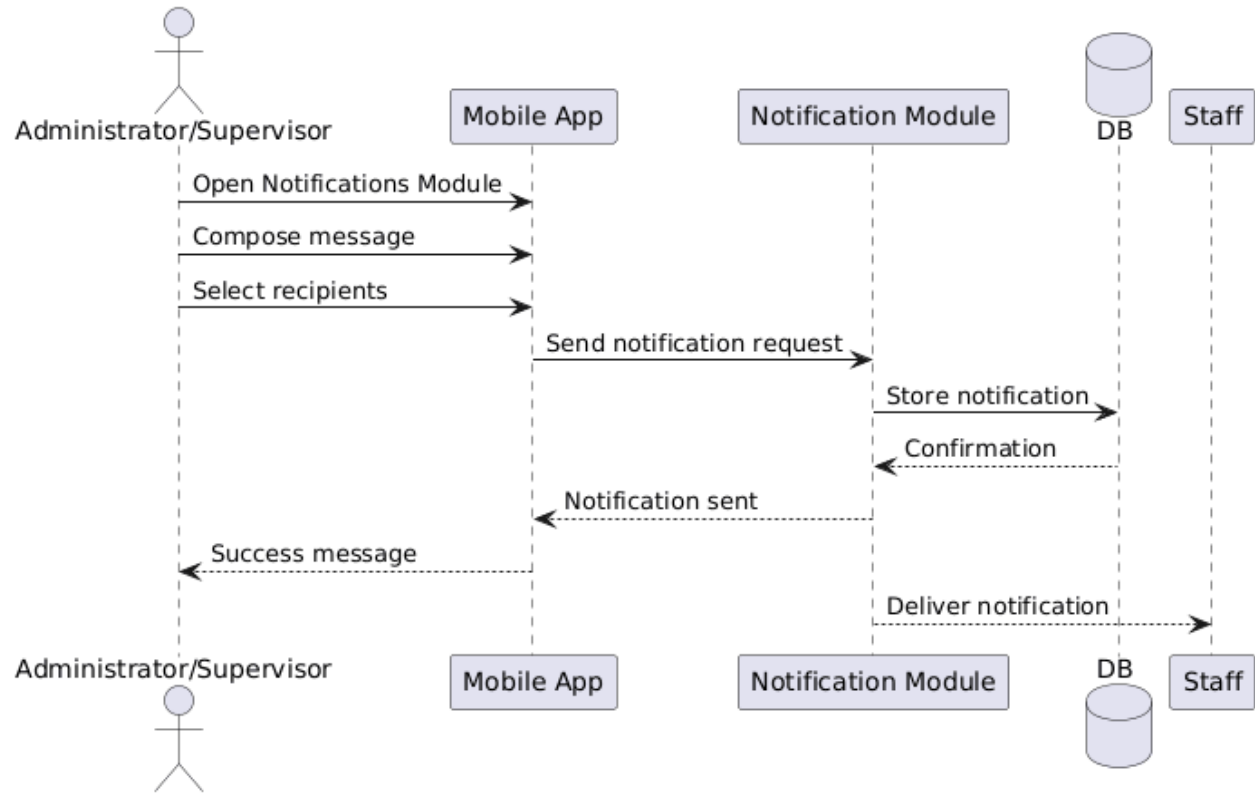


Figure 47 Sequence Diagram for Send Notifications

Sequence Diagram – View Dashboard

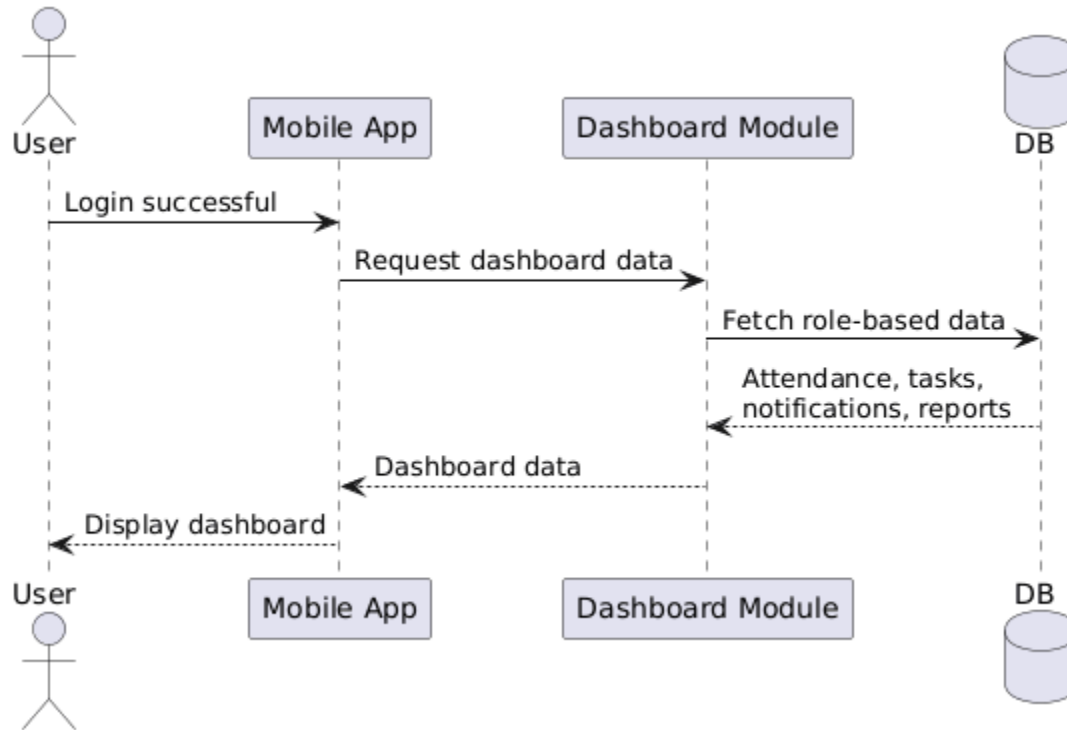


Figure 48 Sequence Diagram for View Dashboard

4.7 Collaboration Diagram

A collaboration diagram illustrates how objects interact with each other to perform a specific functionality. It focuses on the structural organization of objects and the messages exchanged between them rather than the time sequence.

In the StaffSync system, collaboration diagrams are used to show how users, system modules, and the database collaborate to complete system operations. This diagram complements sequence diagrams by emphasizing object relationships and communication links.

Collaboration Diagram – Login Process

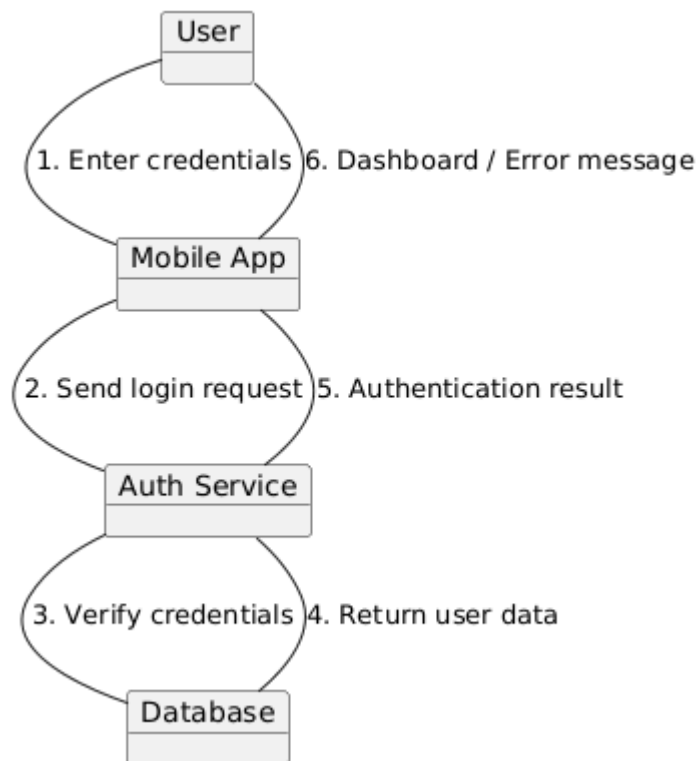


Figure 49 Collaboration Diagram For Login Process

Collaboration Diagram – Logout

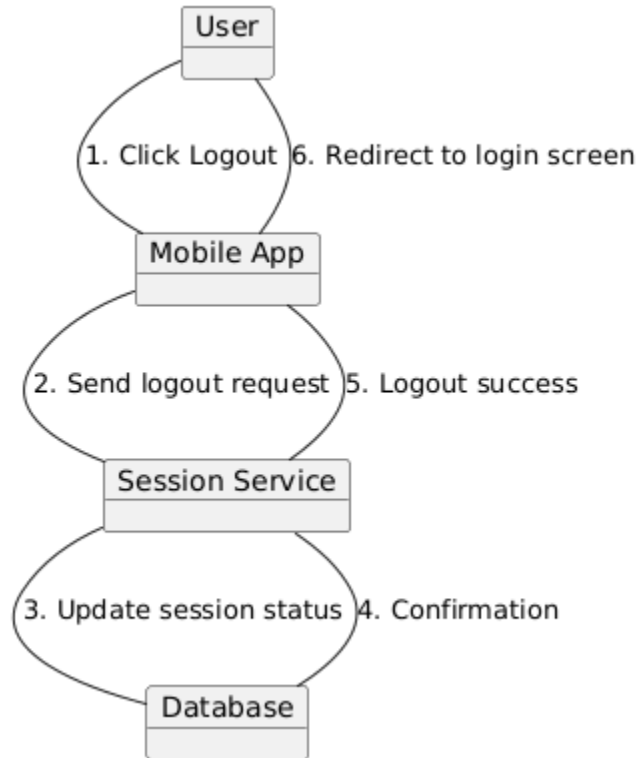


Figure 50 Collaboration Diagram For Logout Process

Collaboration Diagram – Manage Staff

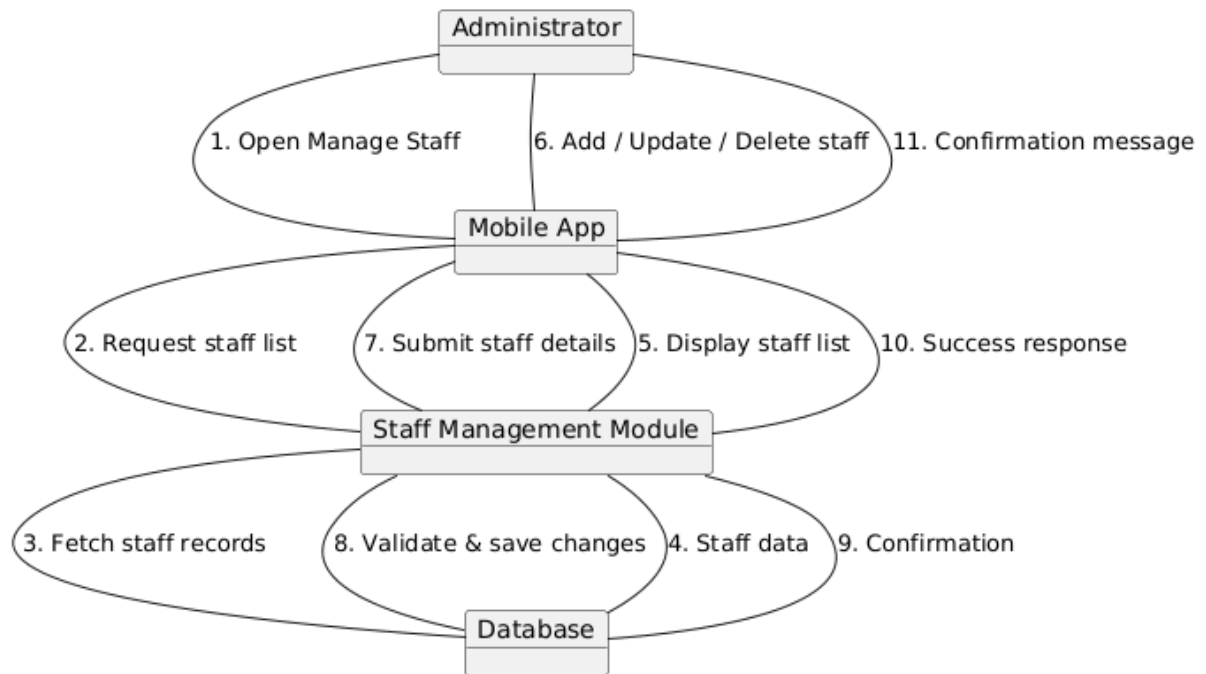


Figure 51 Collaboration Diagram For Manage staff

Collaboration Diagram – Mark Attendance

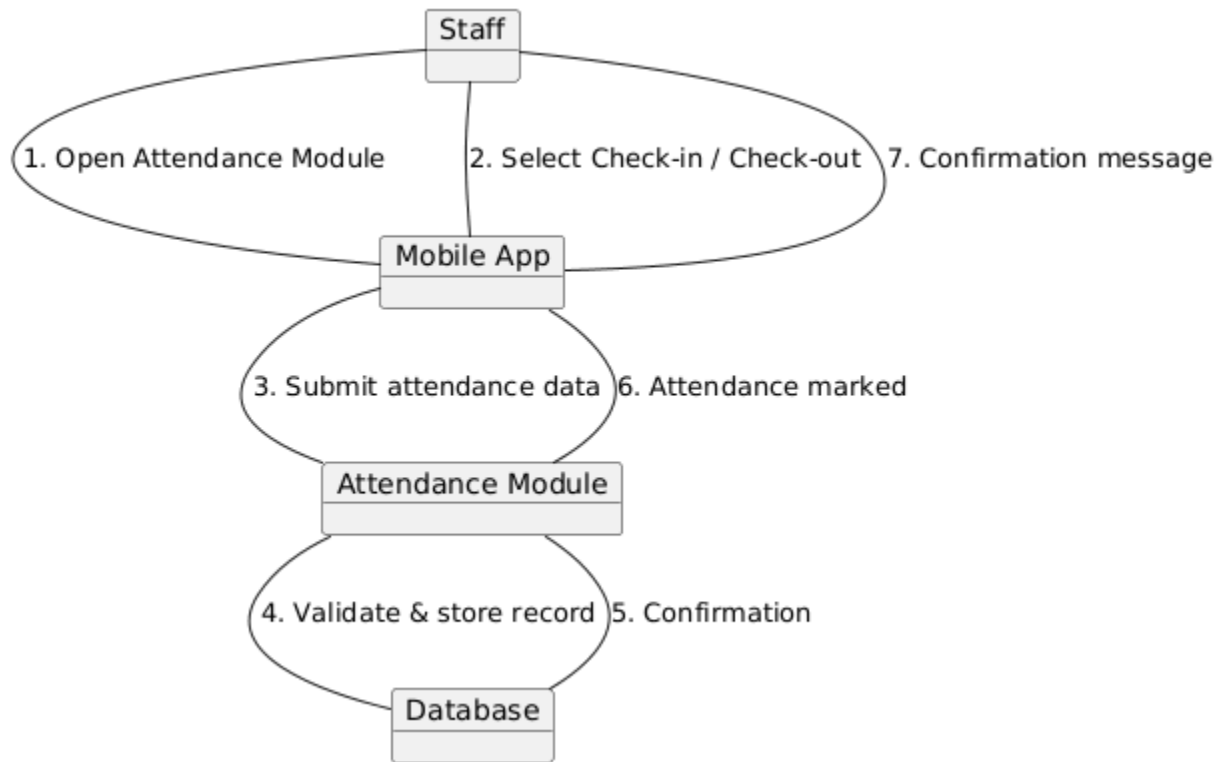


Figure 52 Collaboration Diagram For Mark Attendance

Collaboration Diagram – View Attendance

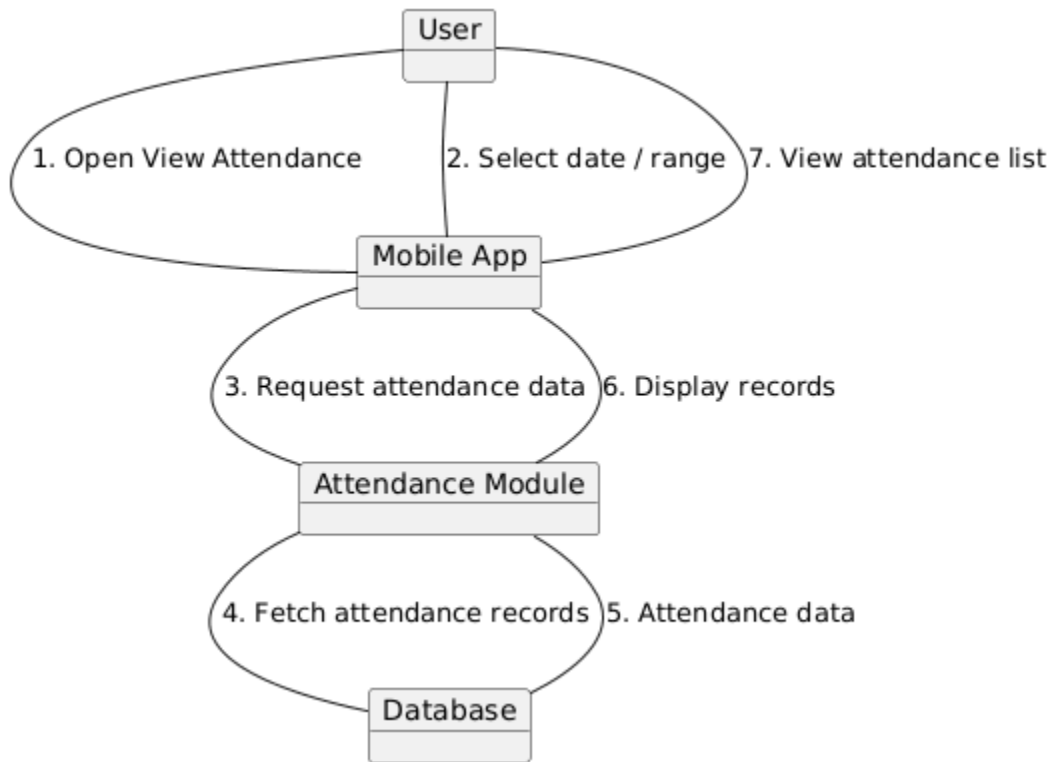


Figure 53 Collaboration Diagram for View Attendance

Collaboration Diagram – Manage Shifts

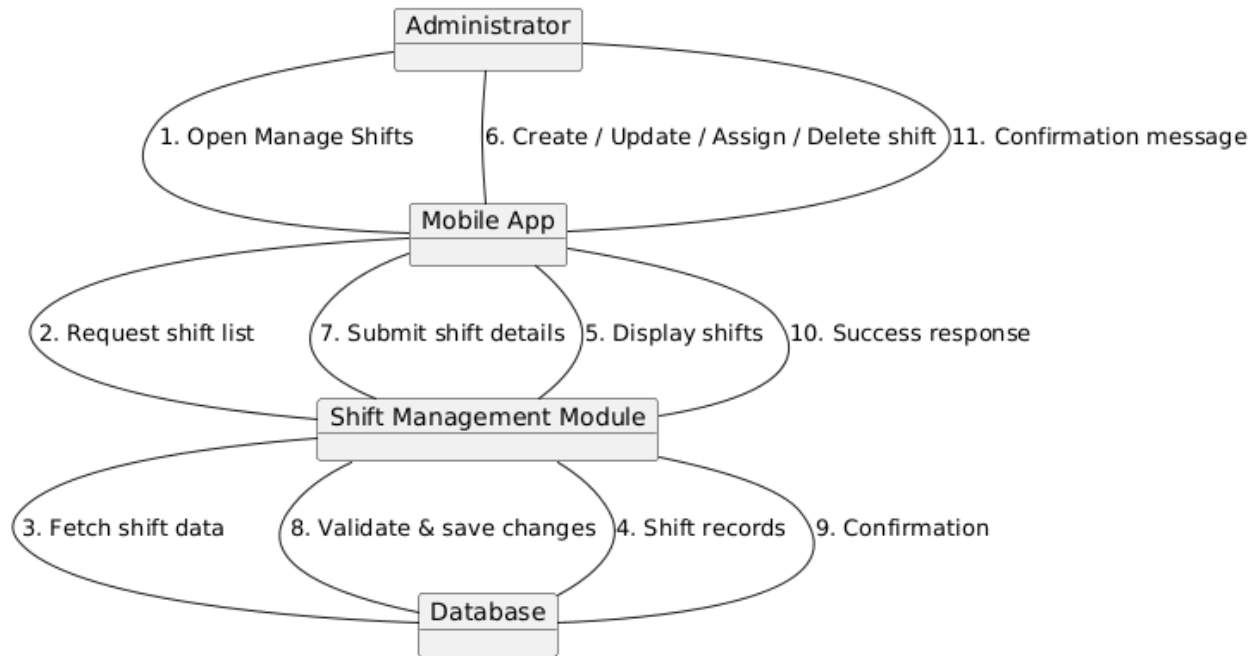


Figure 54 Collaboration Diagram for Manage Shift

Collaboration Diagram – Apply for Leave

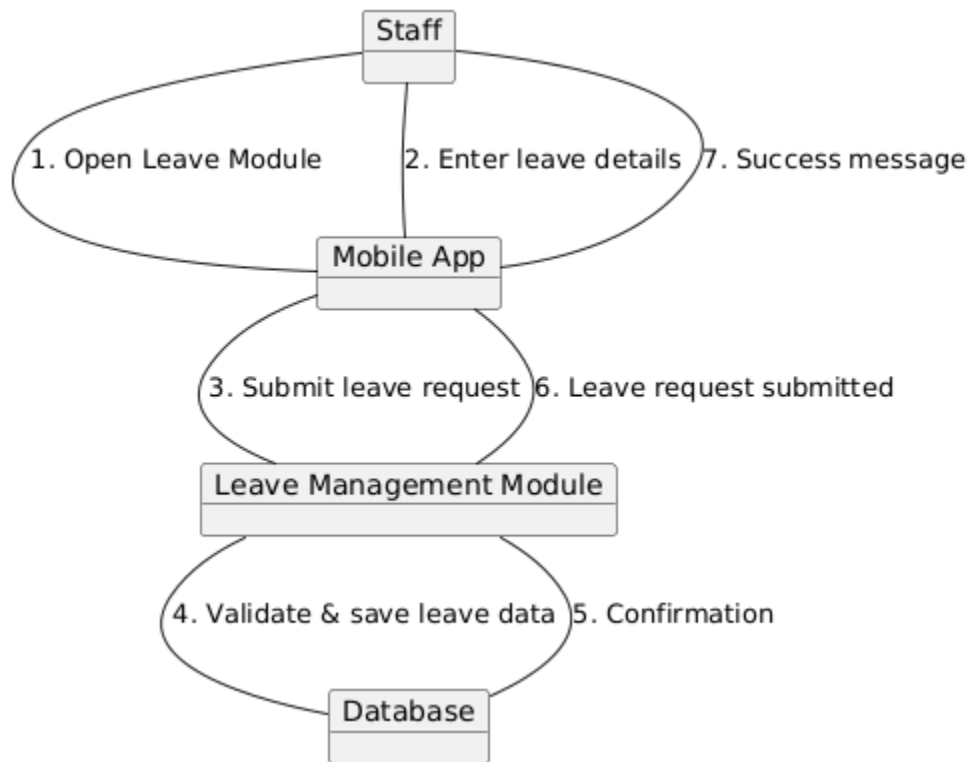


Figure 55 Collaboration Diagram for Apply Leave

Collaboration Diagram – Approve / Reject Leave

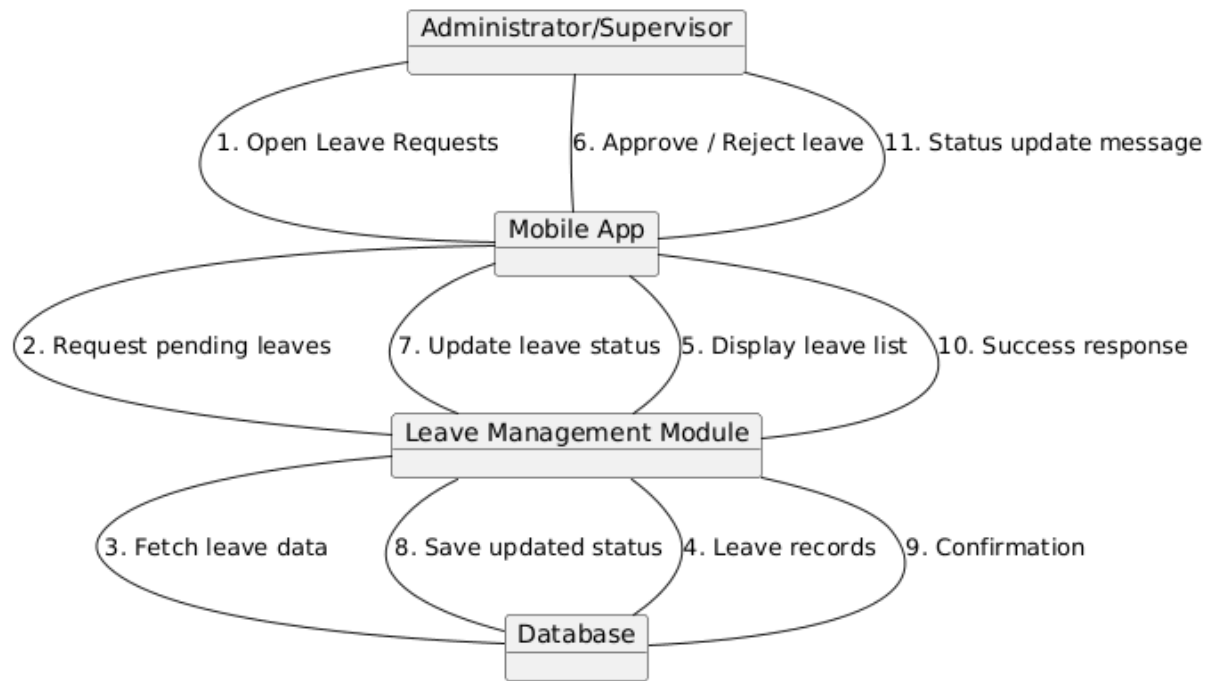


Figure 56 Collaboration Diagram for Approve / Leave Reject

Collaboration Diagram – Assign Task

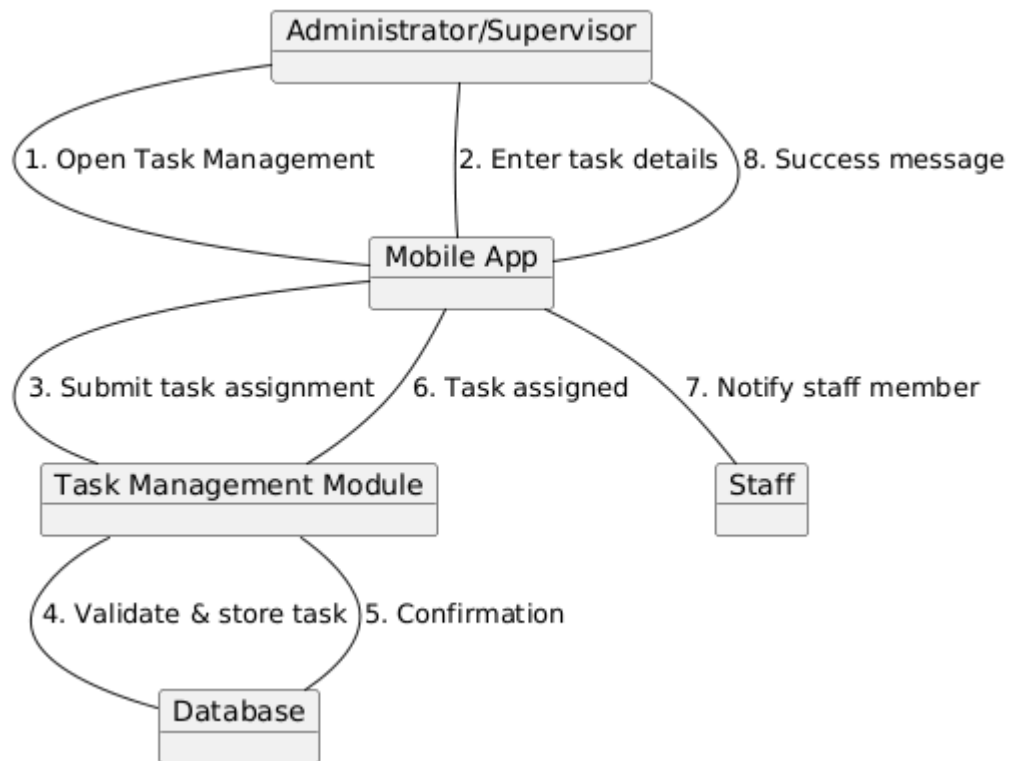


Figure 57 Collaboration Diagram for Assign Task

Collaboration Diagram – Verify Task

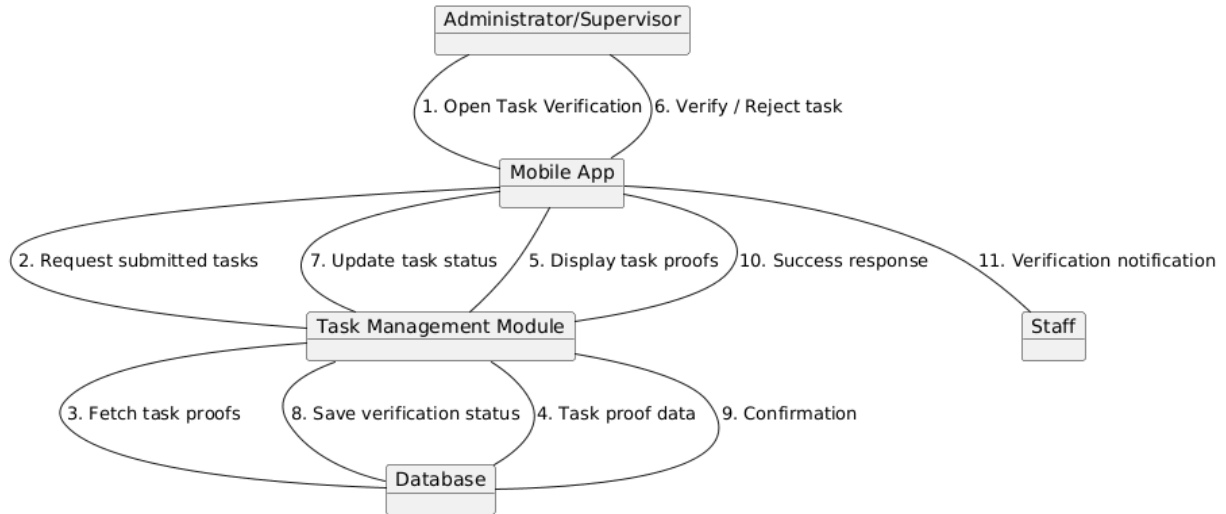


Figure 58 Collaboration Diagram for Verify Task

Collaboration Diagram – Manage Payroll

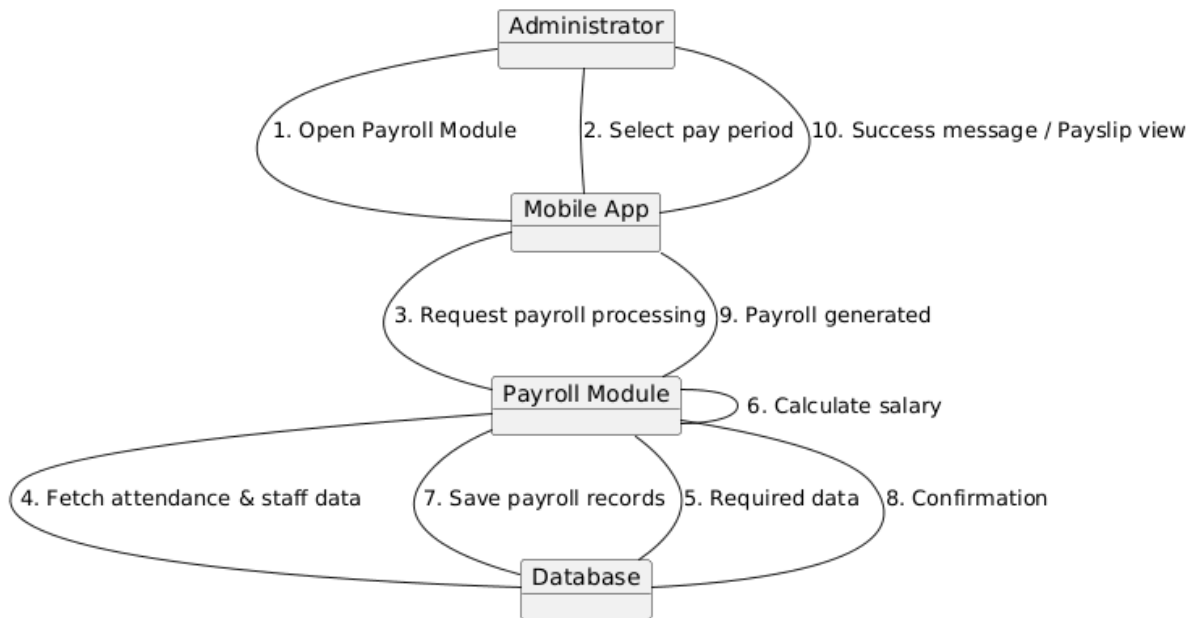


Figure 59 Collaboration Diagram for Manage Payroll

Collaboration Diagram – Generate Reports

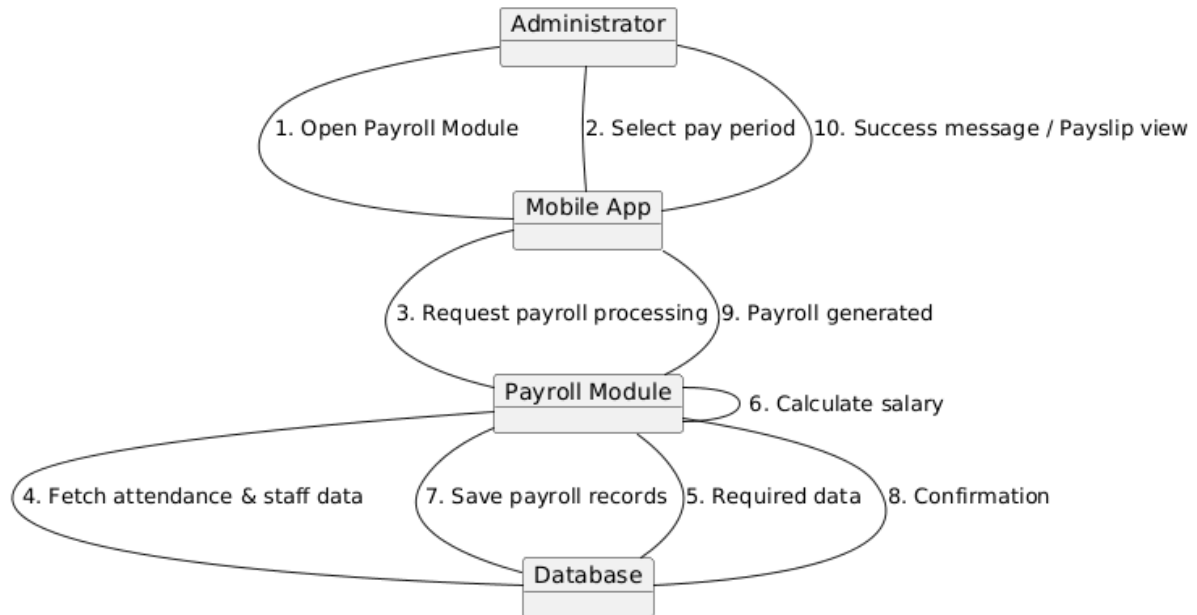


Figure 60 Collaboration Diagram for Generating Report

Collaboration Diagram – Send Notifications

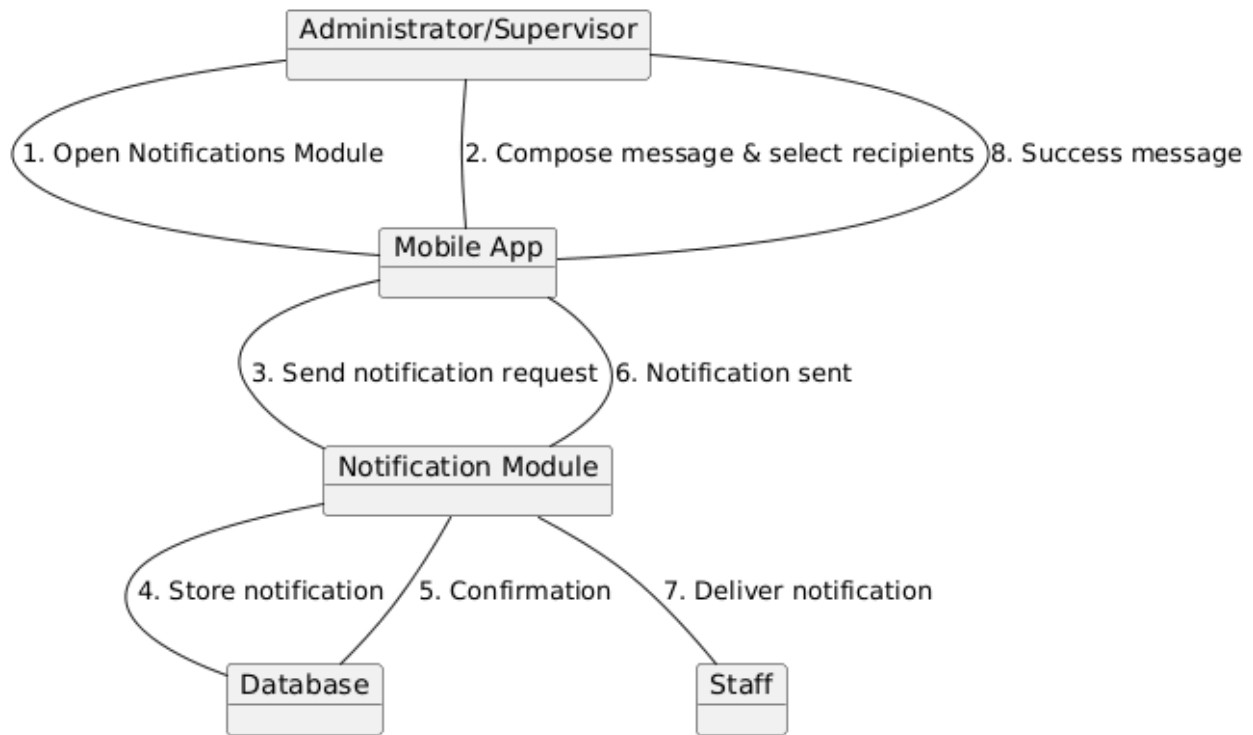


Figure 61 Collaboration Diagram for Sending Notifications

Collaboration Diagram – View Dashboard

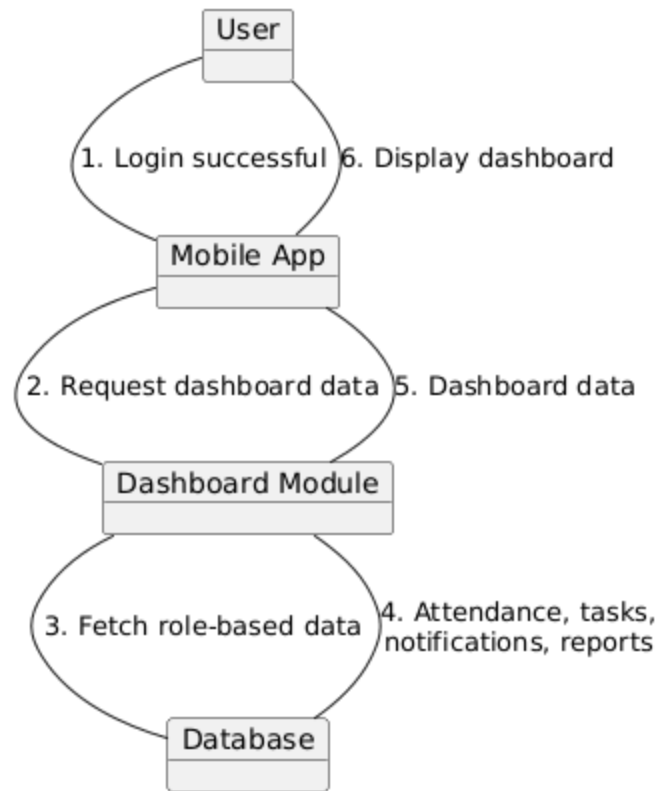


Figure 62 Collaboration Diagram for View Dashboard

Abbreviated Collaboration Diagram

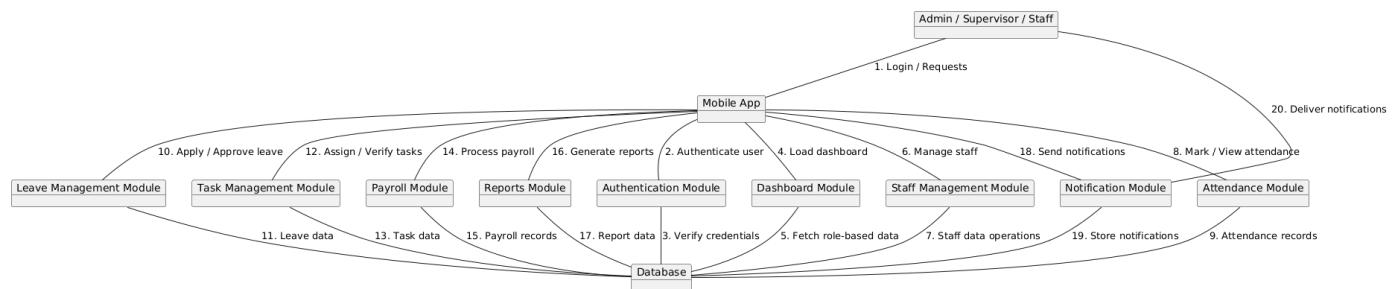


Figure 63 Abbreviated Collaboration Diagram

4.8 State Transition Diagram

A state transition diagram represents the dynamic behavior of a system by illustrating the different states an entity can be in and the transitions between those states based on events or actions. It helps in understanding how the system responds to internal and external inputs over time.

In the StaffSync system, state transition diagrams are used to model the lifecycle of key system operations such as authentication, attendance processing, leave management, task handling, payroll processing, and notification delivery. The diagram provides a high-level overview of how the system moves from one state to another during normal operation.

This section presents an **overall state transition diagram** that summarizes the behavior of major system components instead of showing individual state diagrams for each module, ensuring clarity and avoiding redundancy.

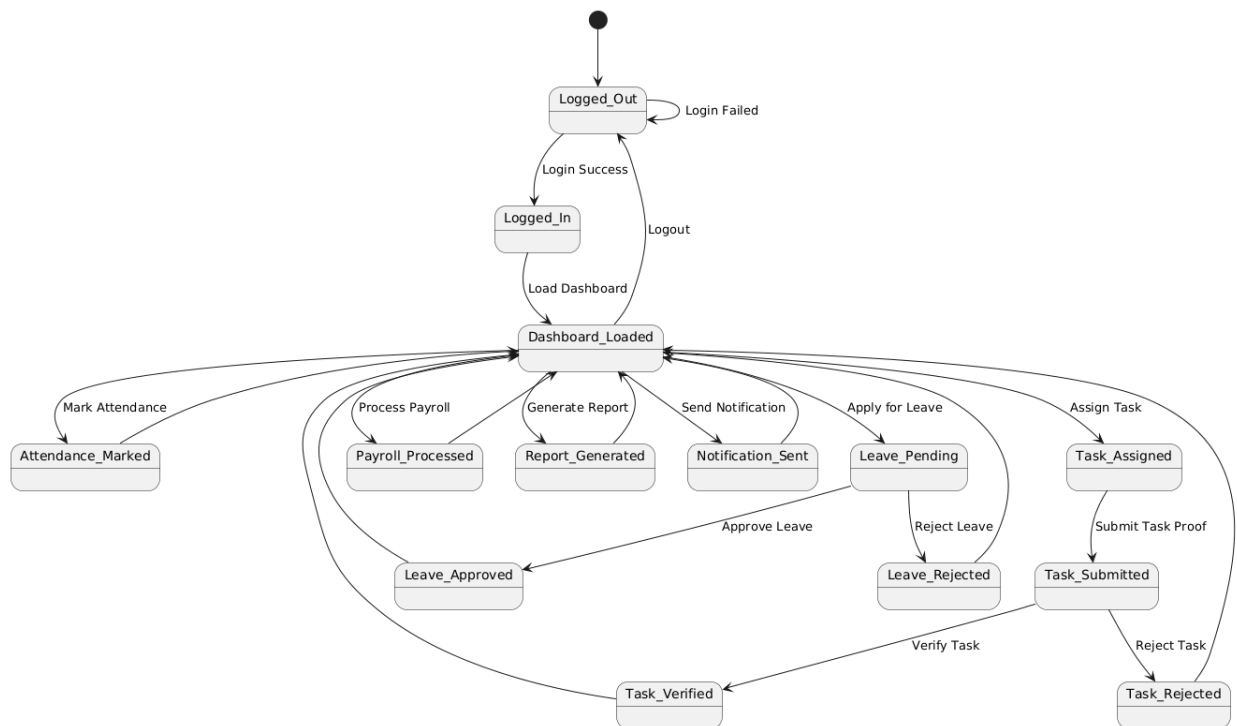


Figure 64 State Transition Diagram

4.9 Component Diagram

A component diagram represents the high-level organization of a system by showing how software components are structured and how they interact with each other through interfaces. It focuses on the modular decomposition of the system rather than internal implementation details.

In the StaffSync system, the component diagram is used to illustrate the overall system architecture by grouping related functionalities into reusable components such as authentication, attendance management, leave management, task handling, payroll processing, reporting, and notifications. This diagram helps in understanding how different system modules collaborate to deliver complete system functionality.

This section presents an overall component diagram that provides a consolidated view of all major system components and their interactions with the database, ensuring a clear understanding of the system's modular design.

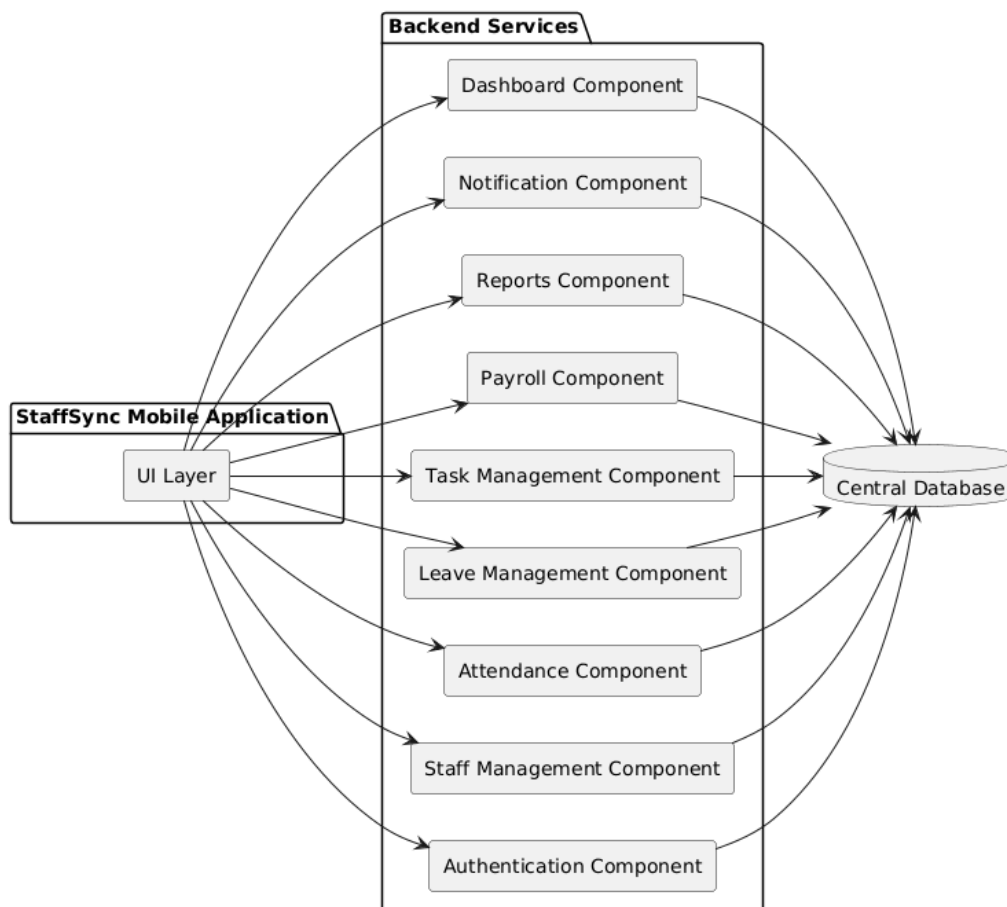


Figure 65 Component Diagram

4.10 Deployment Diagram

A deployment diagram represents the physical architecture of a system by showing how software components are deployed on hardware nodes and how these nodes communicate with each other during runtime. It helps in understanding the system's execution environment and infrastructure setup.

In the StaffSync system, the deployment diagram illustrates how the mobile application is deployed on user devices, how backend services are hosted on an application server, and how data is stored on a centralized database server. This diagram provides an overall view of the system's client-server architecture and deployment configuration.

This section presents an overall deployment diagram that summarizes the physical distribution of system components and their communication pathways

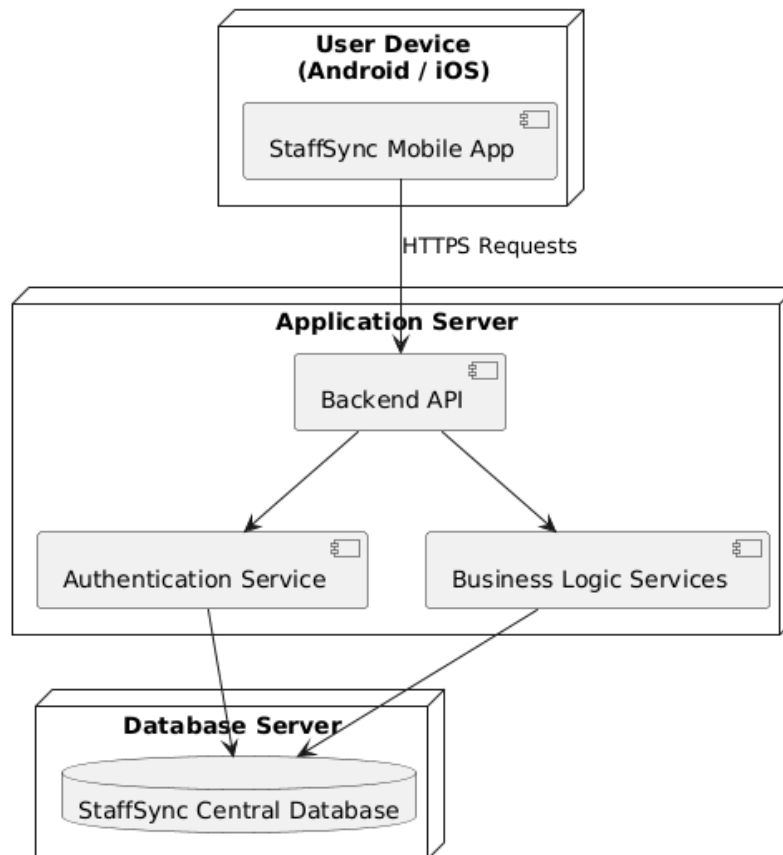


Figure 66 Deployment Diagram

Chapter 5: User Manual

5.1 Introduction

This chapter provides a comprehensive user manual for the StaffSync system. The purpose of this chapter is to guide users on how to effectively use the system's features and functionalities. The user manual explains system access, navigation, and operational steps for different user roles including administrator, supervisor, and staff members.

This chapter is designed to help users understand how to perform day-to-day tasks such as logging into the system, managing staff, marking attendance, applying for leave, handling tasks, generating reports, and logging out securely

5.2 Intended Audience

The StaffSync system is designed for the following types of users:

- **Administrator:** Responsible for managing staff records, shifts, payroll, reports, and system configurations.
- **Supervisor:** Responsible for approving leaves, assigning and verifying tasks, and monitoring staff performance.
- **Staff Member:** Responsible for marking attendance, applying for leave, viewing assigned tasks, and receiving notifications.

5.3 System Requirements

5.3.1 Hardware Requirements

- Smartphone (Android / iOS)
- Minimum 2 GB RAM
- Internet connectivity

5.3.2 Software Requirements

- Android OS / iOS
- StaffSync Mobile Application

5.4 User Roles and Access Levels

The StaffSync system implements role-based access control to ensure secure and authorized usage of system functionalities. Each user role has specific permissions that restrict access to system modules according to responsibilities

Role	Access Level
Administrator	Full system access
Supervisor	Limited administrative access
Staff	Basic operational access

Table 59 User Roles

5.5 System Login

5.5.1 Purpose

The login functionality allows registered users to securely access the StaffSync system according to their assigned roles. Only authenticated users can use system features such as attendance, leave management, task handling, and reporting.

5.5.2 Pre-requisites

Before logging into the system, ensure that:

- The user is registered in the system.
- A stable internet connection is available.
- The StaffSync mobile application is installed on the device.

5.5.3 Steps to Login

Follow the steps below to log into the StaffSync system:

1. Open the **StaffSync Mobile Application**.
2. The **Login Screen** will appear.
3. Enter your **registered email address** in the Email field.
4. Enter your **password** in the Password field.
5. Click on the **Login** button.
6. If the credentials are valid, the system will authenticate the user.
7. The user will be redirected to the **role-based dashboard**.

5.5.4 Input Fields Description

Field Name Description

Email Registered email address of the user

Field Name Description

Password Secure password associated with the account

5.5.5 System Response

- **Successful Login:**
The system grants access and redirects the user to the dashboard.
- **Failed Login:**
If incorrect credentials are entered, an error message is displayed indicating invalid email or password.

5.5.6 Error Handling

The system handles login errors gracefully by:

- Preventing access with invalid credentials.
- Displaying appropriate error messages.
- Ensuring account security.

5.5.7 Screenshot Placeholder

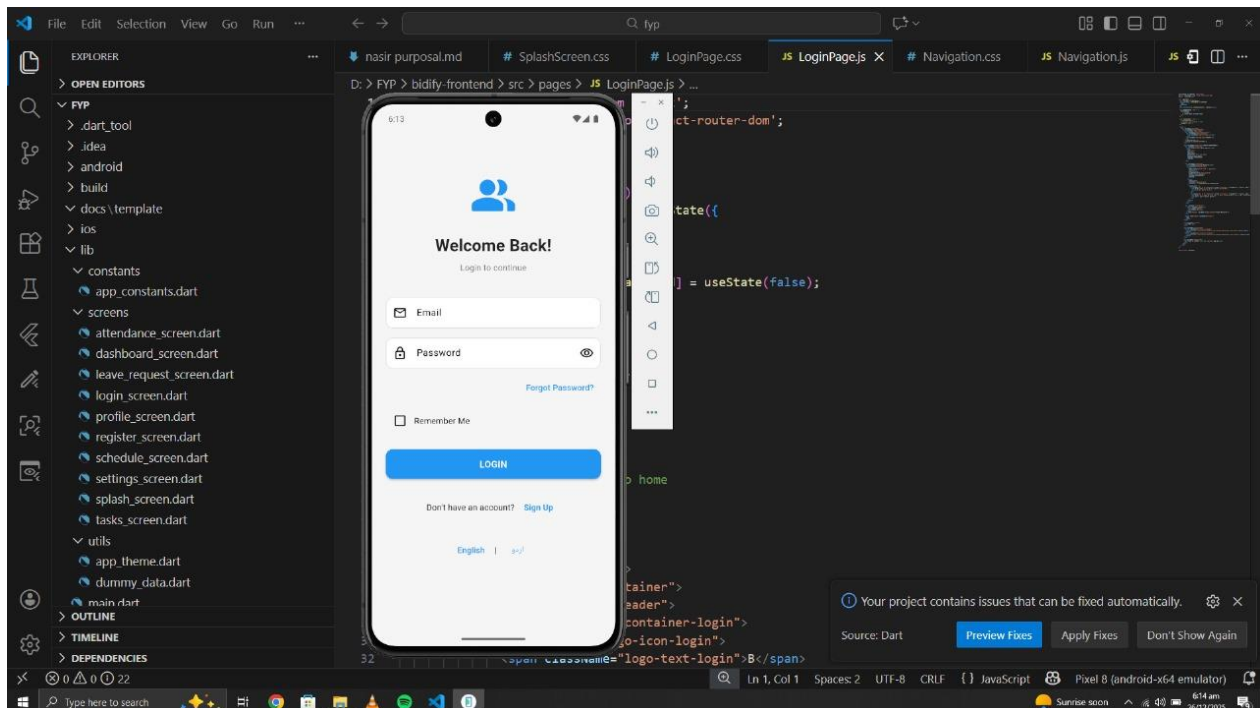


Figure 67 StaffSync Login Screen

5.6 Logout

5.6.1 Purpose

The logout functionality allows users to securely exit the StaffSync system. Logging out ensures that the user session is terminated properly and prevents unauthorized access to the system.

5.6.2 Pre-requisites

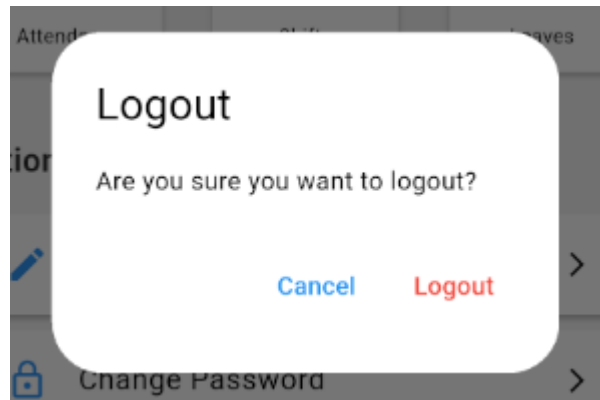


Figure 68 Logout

Before logging out, ensure that:

- The user is currently logged into the system.
- The system is running normally with an active internet connection.

5.6.3 Steps to Logout

Follow the steps below to log out of the StaffSync system:

1. From the **Dashboard**, locate the **Logout** option.
2. Click on the **Logout** button.
3. A confirmation prompt may appear.
4. Confirm the logout action.
5. The system will terminate the active session.
6. The user will be redirected to the **Login Screen**.

5.6.4 System Response

- **Successful Logout:**
The system ends the session and redirects the user to the login page.
- **Cancelled Logout:**
If the user cancels the logout action, the system returns to the dashboard.

5.6.5 Security Considerations

- Logout clears all session data.
- It is recommended to log out after completing system tasks, especially on shared devices.

5.6.6 Screenshot Placeholder

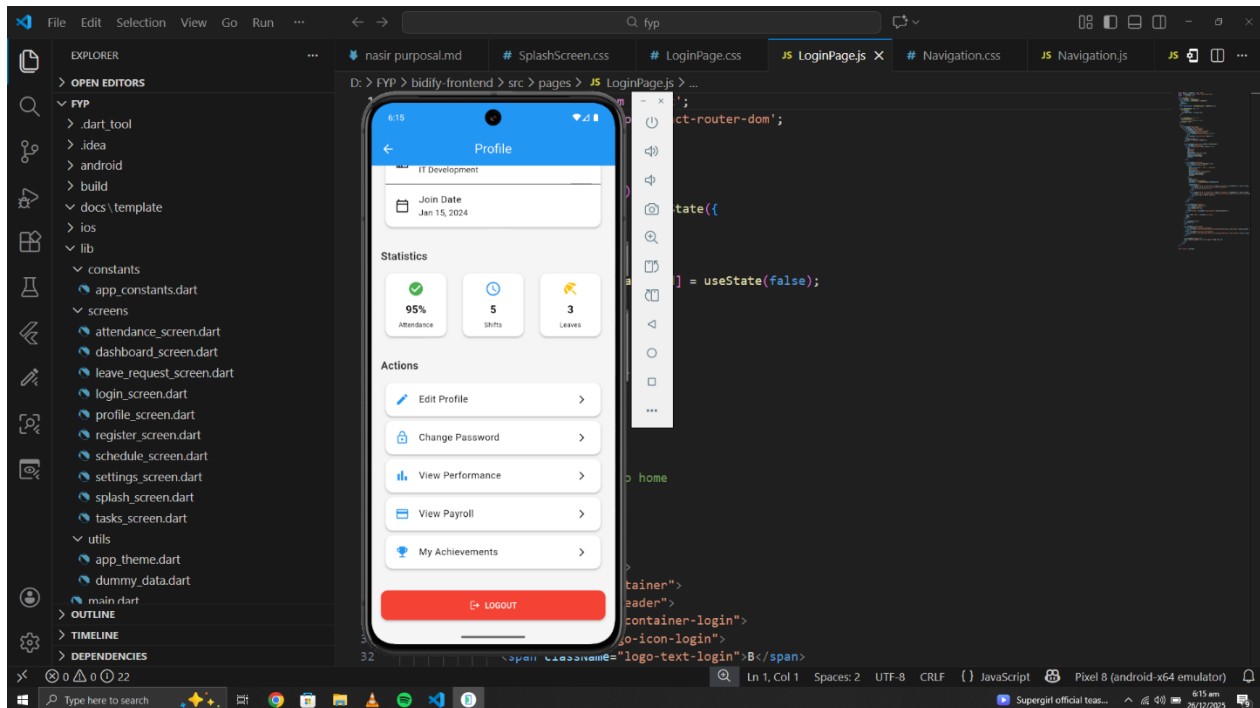


Figure 69 StaffSync Logout Screen

5.7 Dashboard Overview

5.7.1 Purpose

The dashboard serves as the main control panel of the StaffSync system. It provides users with a summarized view of important information such as attendance status, assigned tasks, leave updates, notifications, and quick access to system modules based on user roles.

5.7.2 Dashboard Access

- The dashboard is displayed immediately after a successful login.
- The content of the dashboard varies depending on the user role (Administrator, Supervisor, or Staff).

5.7.3 Dashboard Components

The StaffSync dashboard consists of the following components:

- **Attendance Summary:** Displays daily check-in and check-out status.
- **Task Overview:** Shows assigned or pending tasks.
- **Leave Status:** Displays leave requests and approval status.
- **Notifications Panel:** Shows system alerts and messages.
- **Quick Navigation Menu:** Provides access to system modules.

5.7.4 Role-Based Dashboard View

Administrator Dashboard

- Staff management shortcuts
- Attendance and payroll overview
- Reports and analytics access

Supervisor Dashboard

- Task assignment and verification
- Leave approval notifications

- Team performance summary

Staff Dashboard

- Attendance marking option
- Task list and submission status
- Leave application and status tracking

5.7.5 System Response

- Dashboard loads data dynamically from the system.
- Real-time updates are displayed when data changes.

5.7.6 Screenshot Placeholder

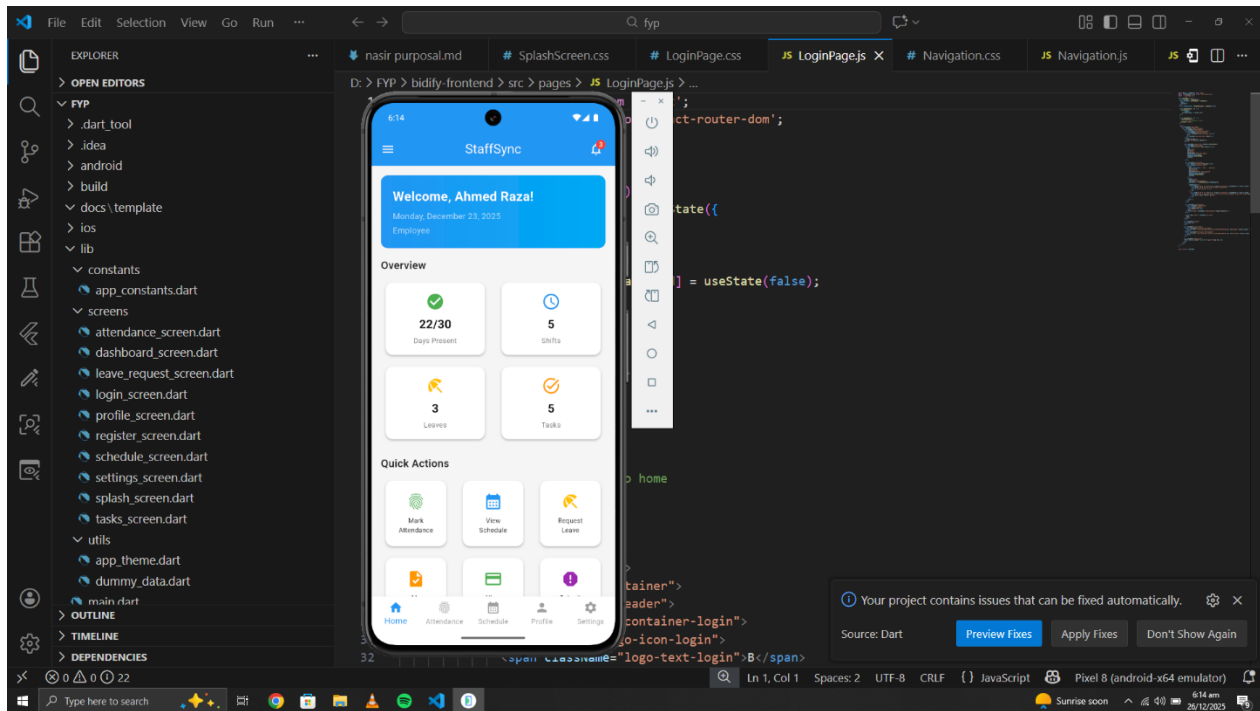


Figure 70 StaffSync Dashboard Screen

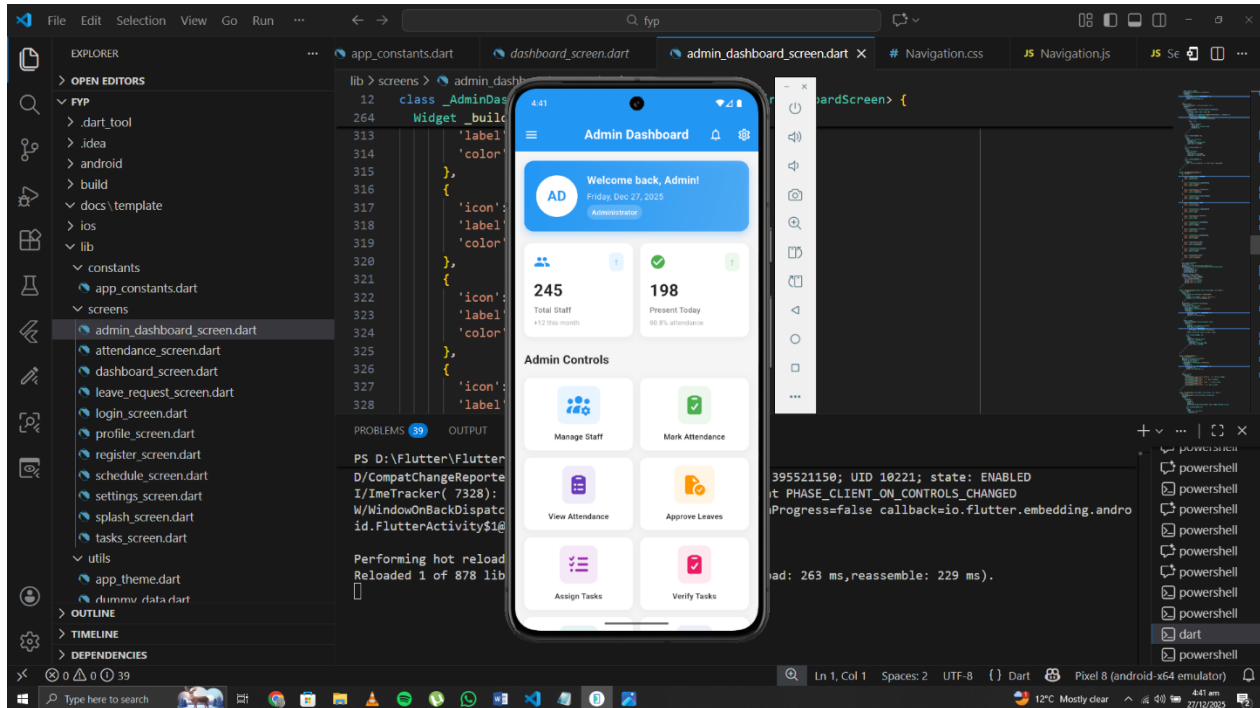


Figure 71 Admin StaffSync Dashboard Screen

5.8 Manage Staff

5.8.1 Purpose

The Manage Staff module allows the administrator to add, view, update, and delete staff records within the StaffSync system. This module helps in maintaining accurate staff information and ensuring proper system access control.

5.8.2 Access Rights

- **Accessible by:** Administrator only
- Other user roles do not have permission to access this module.

5.8.3 Steps to Add New Staff

1. Log in as an **Administrator**.
2. Navigate to the **Manage Staff** module from the dashboard.
3. Click on the **Add Staff** button.

4. Enter staff details such as name, email, role, and contact information.
5. Click the **Save** button.
6. The system validates the information and creates a new staff record.

5.8.4 Steps to Update Staff Information

1. Open the **Manage Staff** module.
2. Select a staff member from the list.
3. Click on the **Edit** option.
4. Update the required information.
5. Click **Update** to save changes.

5.8.5 Steps to Delete Staff

1. Select the staff member to be removed.
2. Click on the **Delete** button.
3. Confirm the deletion action.
4. The system removes the staff record permanently.

5.8.6 System Response

- Displays confirmation messages after successful operations.
- Shows error messages in case of invalid data or system issues.

5.8.7 Screenshot Placeholder

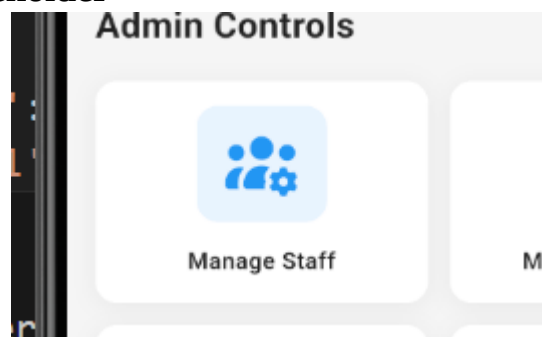


Figure 72 Manage Staff

5.9 Mark Attendance

5.9.1 Purpose

The Mark Attendance module allows staff members to record their daily attendance by checking in and checking out through the StaffSync system. This feature ensures accurate attendance tracking and supports payroll and reporting processes.

5.9.2 Access Rights

- **Accessible by:** Staff members
- Administrators and supervisors can view attendance records but cannot mark attendance on behalf of staff.

5.9.3 Steps to Mark Attendance

1. Log in to the StaffSync system as a **Staff Member**.
2. From the **Dashboard**, open the **Attendance** module.
3. Click on the **Check-In** button at the start of the workday.
4. The system records the current date and time.
5. At the end of the workday, click on the **Check-Out** button.
6. The system saves the attendance record successfully.

5.9.4 Attendance Validation

- The system prevents duplicate check-in or check-out entries.
- Attendance is validated against assigned shift timings, if applicable.

5.9.5 System Response

- **Successful Entry:**
A confirmation message is displayed indicating that attendance has been marked.
- **Invalid Entry:**
An error message is shown if attendance is already marked or if any issue occurs.

5.9.8 Screenshot Placeholder

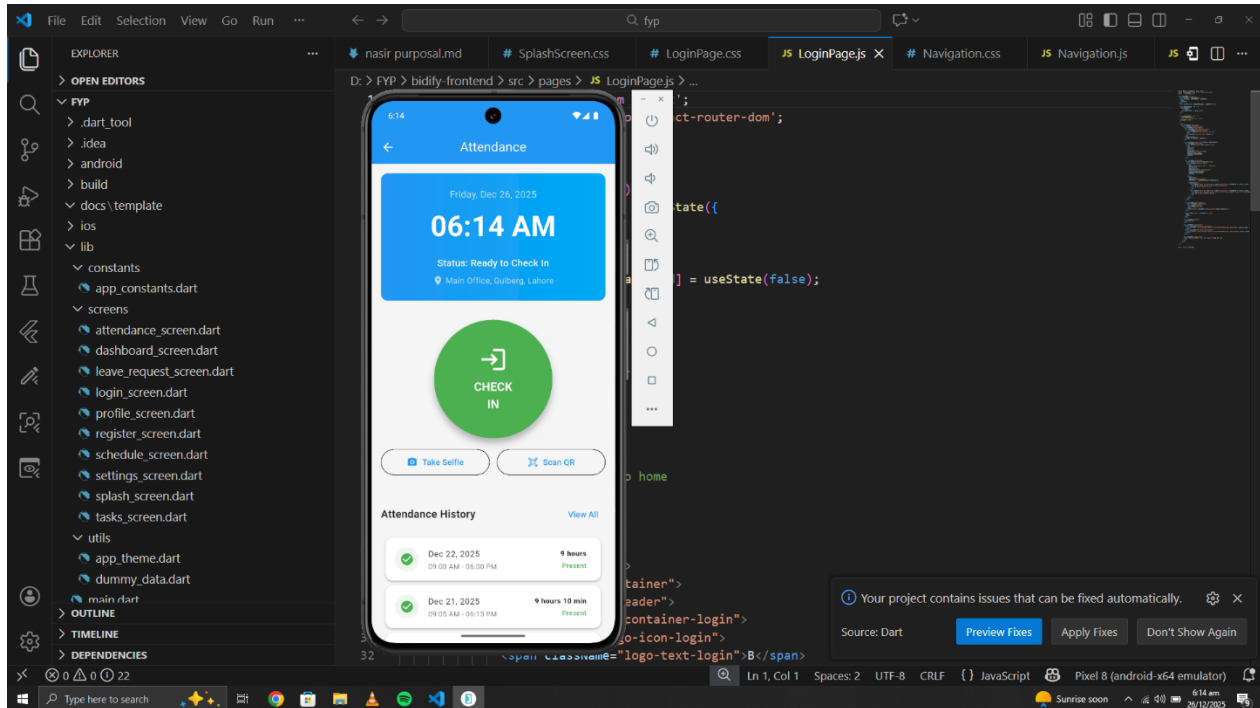


Figure 73 Attendance

5.10 View Attendance

5.10.1 Purpose

The View Attendance module allows users to review attendance records for a specific date or date range. This feature helps staff members track their attendance history, while supervisors and administrators can monitor attendance for management and reporting purposes.

5.10.2 Access Rights

- **Accessible by:**
 - Staff
 - Supervisor
 - Administrator
- **Staff:** Can view their own attendance records

- **Supervisor / Administrator:** Can view attendance records of all staff members

5.10.3 Steps to View Attendance

1. Log in to the StaffSync system.
2. From the **Dashboard**, navigate to the **View Attendance** module.
3. Select the required **date** or **date range**.
4. Click on the **View** or **Search** button.
5. The system retrieves and displays the attendance records.

5.10.4 Attendance Details Displayed

The following information is shown in the attendance list:

- Staff Name
- Date
- Check-in Time
- Check-out Time
- Attendance Status

5.10.5 System Response

- **Records Available:**
Attendance data is displayed in a structured format.
- **No Records Found:**
The system displays a message indicating no attendance data is available for the selected period.

5.10.6 Screenshot Placeholder

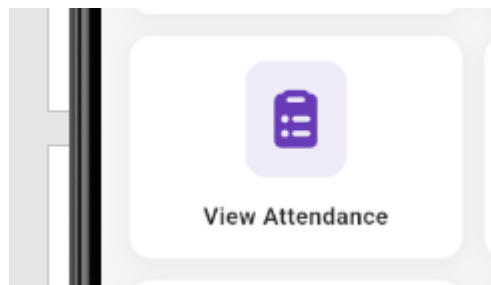


Figure 74 View Attendance

5.12 Apply for Leave

5.12.1 Purpose

The Apply for Leave module allows staff members to submit leave requests through the StaffSync system. This feature helps in managing leave records digitally and ensures a smooth approval process by supervisors or administrators.

5.12.2 Access Rights

- **Accessible by:** Staff members
- Supervisors and administrators can review and take action on submitted leave requests.

5.12.3 Steps to Apply for Leave

1. Log in to the StaffSync system as a **Staff Member**.
2. From the **Dashboard**, navigate to the **Leave Management** module.
3. Click on the **Apply for Leave** option.
4. Select the **leave type** (e.g., casual, sick, annual).
5. Choose the **start date** and **end date** of leave.
6. Enter the **reason for leave**.
7. Click on the **Submit** button.
8. The system submits the leave request for approval.

5.12.4 Leave Request Validation

- The system checks for overlapping leave dates.
- Ensures mandatory fields are filled before submission.
- Prevents submission of invalid leave periods.

5.12.5 System Response

- **Successful Submission:**
A confirmation message is displayed, and the leave status is set to *Pending*.

- **Submission Error:**
An error message is shown if required information is missing or invalid.

5.12.6 Screenshot Placeholder

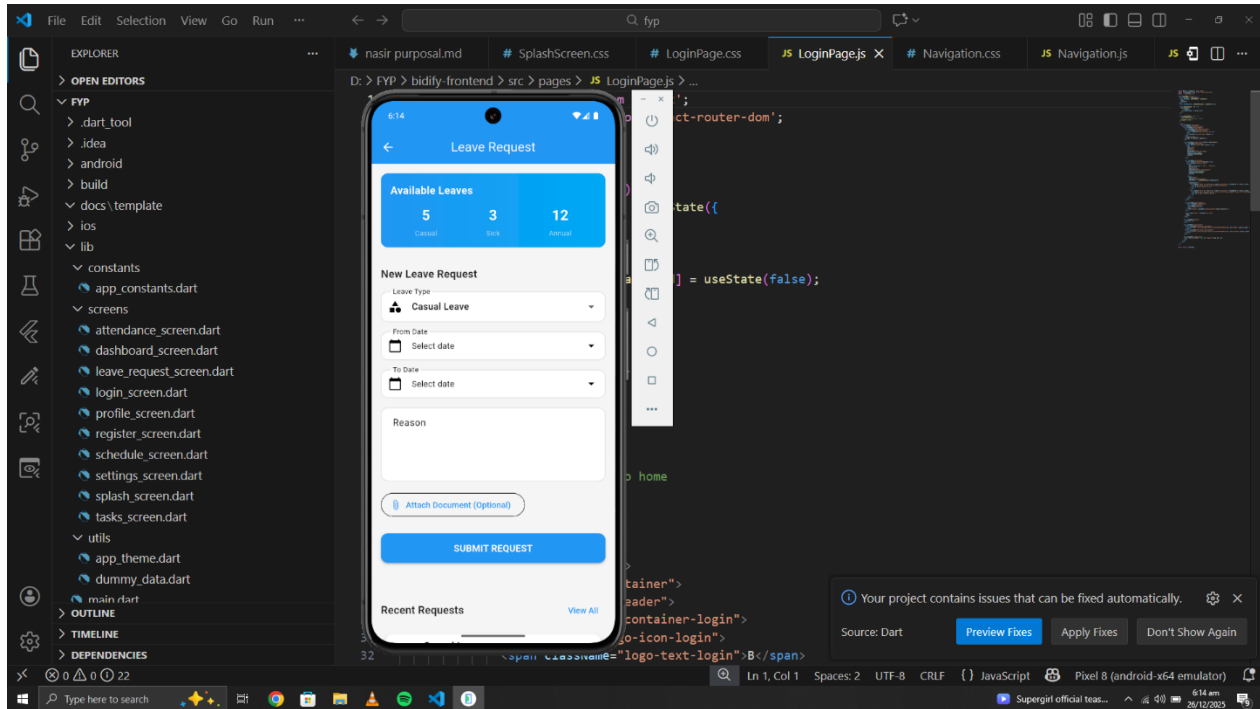


Figure 75 Apply for Leave Screen

5.13 Approve / Reject Leave

5.13.1 Purpose

The Approve / Reject Leave module allows supervisors or administrators to review leave requests submitted by staff members and take appropriate action. This ensures proper leave management and maintains workforce availability.

5.13.2 Access Rights

- **Accessible by:**
 - Administrator
 - Supervisor
- Staff members do not have permission to approve or reject leave requests.

5.13.3 Steps to Approve or Reject Leave

1. Log in to the StaffSync system as an **Administrator** or **Supervisor**.
2. From the **Dashboard**, navigate to the **Leave Management** or **Leave Requests** module.
3. View the list of **pending leave requests**.
4. Select a leave request to review details such as:
 - Staff name
 - Leave type
 - Leave duration
 - Reason for leave
5. Choose one of the following actions:
 - Click **Approve** to approve the leave request.
 - Click **Reject** to reject the leave request.
6. (Optional) Enter remarks while rejecting a request.
7. Submit the decision.

5.13.4 System Response

- **Leave Approved:**
The leave status is updated to *Approved*, and the staff member is notified.
- **Leave Rejected:**
The leave status is updated to *Rejected*, and the staff member receives a notification with remarks if provided.

5.13.5 Notifications

- The system automatically sends notifications to staff members regarding the approval or rejection of their leave requests.
- Leave status updates are reflected on the staff dashboard.

5.13.6 Screenshot Placeholder

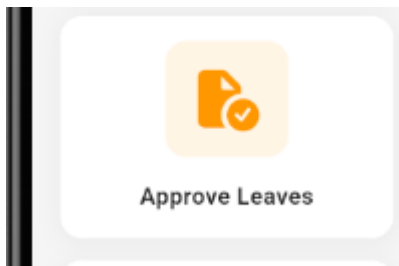


Figure 76 Approve / Reject Leave Screen

5.14 Assign Task

5.14.1 Purpose

The Assign Task module allows administrators or supervisors to assign tasks to staff members. This feature helps in managing daily activities, tracking work progress, and ensuring accountability within the organization.

5.14.2 Access Rights

- **Accessible by:**
 - Administrator
 - Supervisor
- Staff members can only view and complete assigned tasks.

5.14.3 Steps to Assign a Task

1. Log in to the StaffSync system as an **Administrator** or **Supervisor**.
2. From the **Dashboard**, navigate to the **Task Management** module.
3. Click on the **Assign Task** option.
4. Select the **staff member** from the list.
5. Enter the **task title** and **task description**.
6. Set the **due date** for task completion.
7. Click on the **Assign** button.
8. The system assigns the task to the selected staff member.

5.14.4 System Response

- **Successful Assignment:**
A confirmation message is displayed, and the task is added to the staff member's task list.
- **Assignment Error:**
An error message is shown if required task details are missing.

5.14.5 Notifications

- The system sends a notification to the assigned staff member.
- The task appears on the staff dashboard under assigned tasks.

5.14.6 Screenshot Placeholder

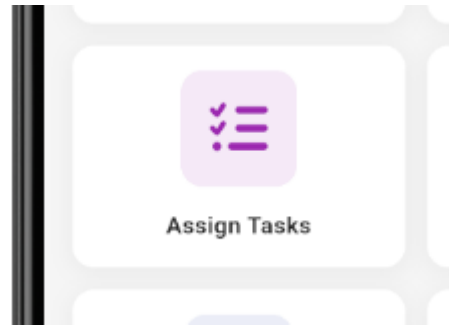


Figure 77 Assign Task Screen

5.15 Verify Task

5.15.1 Purpose

The Verify Task module allows administrators or supervisors to review tasks completed by staff members and verify whether the assigned work has been completed satisfactorily. This feature ensures task quality control and proper task closure.

5.15.2 Access Rights

- **Accessible by:**
 - Administrator
 - Supervisor
- Staff members do not have permission to verify tasks.

5.15.3 Steps to Verify a Task

1. Log in to the StaffSync system as an **Administrator** or **Supervisor**.
2. From the **Dashboard**, navigate to the **Task Verification** module.
3. View the list of **submitted tasks**.
4. Select a task to review its details and submitted proof (if any).
5. Choose one of the following actions:
 - Click **Verify** to approve the task.
 - Click **Reject** to reject the task.
6. (Optional) Enter remarks if the task is rejected.
7. Submit the decision.

5.15.4 System Response

- **Task Verified:**
The task status is updated to *Verified*, and the staff member is notified.
- **Task Rejected:**
The task status is updated to *Rejected*, and the staff member receives feedback for correction.

5.15.5 Notifications

- The system sends a notification to the staff member regarding task verification status.
- Task status updates are reflected on the staff dashboard.

5.15.6 Screenshot Placeholder

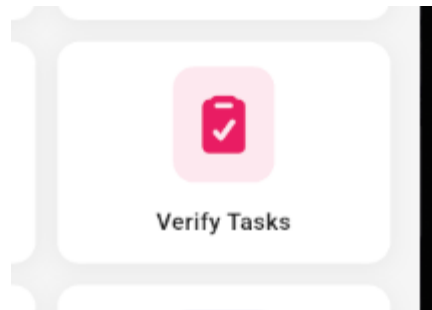


Figure 78 Verify Task Screen

5.16 Manage Payroll

5.16.1 Purpose

The Manage Payroll module allows the administrator to calculate salaries, generate payroll records, and manage staff payments based on attendance, shifts, and predefined salary rules. This module ensures accurate and timely payroll processing.

5.16.2 Access Rights

- **Accessible by:** Administrator only
- Supervisors and staff members do not have permission to manage payroll.

5.16.3 Steps to Process Payroll

1. Log in to the StaffSync system as an **Administrator**.
2. From the **Dashboard**, navigate to the **Payroll Management** module.
3. Select the **pay period** (monthly or weekly).
4. The system retrieves attendance and shift data automatically.
5. Review calculated salaries for each staff member.
6. Click on the **Generate Payroll** button.
7. The system generates payroll records and payslips.

5.16.4 Payroll Details Included

The payroll record may include:

- Staff name and designation
- Total working days
- Attendance summary
- Salary amount
- Deductions (if any)

5.16.5 System Response

- **Successful Processing:**
Payroll records are generated and saved successfully.
- **Processing Error:**
An error message is displayed if required data is missing or incorrect.

5.16.6 Screenshot Placeholder

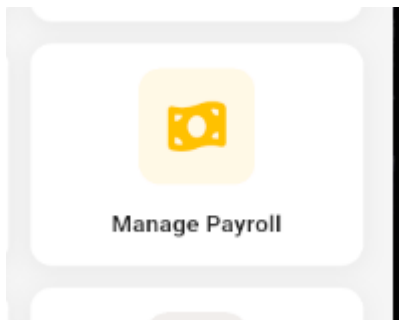


Figure 79 Manage Payroll Screen

5.17 Generate Reports

5.17.1 Purpose

The Generate Reports module allows administrators and supervisors to create system reports based on attendance, leave records, tasks, and payroll data. These reports help management in decision-making, performance evaluation, and record keeping.

5.17.2 Access Rights

- **Accessible by:**
 - Administrator
 - Supervisor
- Staff members do not have permission to generate system reports.

5.17.3 Steps to Generate Reports

1. Log in to the StaffSync system as an **Administrator** or **Supervisor**.
2. From the **Dashboard**, navigate to the **Reports** module.
3. Select the **report type** (Attendance, Leave, Tasks, Payroll, etc.).
4. Choose the required **date range** and filters.
5. Click on the **Generate Report** button.
6. The system processes the request and displays the report.

5.17.4 Report Output Options

- Reports can be **viewed on screen**.
- Reports may be **downloaded** in supported formats (e.g., PDF or Excel).
- Printed copies can be generated if required.

5.17.5 System Response

- **Report Generated Successfully:**
The requested report is displayed with accurate data.
- **No Data Available:**
The system displays a message indicating that no records were found for the selected criteria.

5.17.6 Screenshot Placeholder

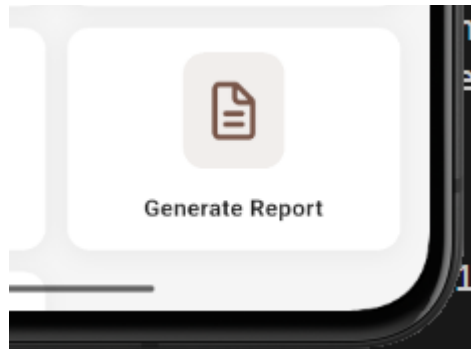


Figure 80Generate Reports Screen

5.18 Send Notifications

5.18.1 Purpose

The Send Notifications module allows administrators and supervisors to send important announcements, alerts, or updates to staff members through the StaffSync system. This feature ensures effective communication within the organization.

5.18.2 Access Rights

- **Accessible by:**
 - Administrator
 - Supervisor
- Staff members can **receive and view** notifications but cannot send them.

5.18.3 Steps to Send Notifications

1. Log in to the StaffSync system as an **Administrator** or **Supervisor**.
2. From the **Dashboard**, navigate to the **Notifications** module.
3. Click on the **Create Notification** option.
4. Enter the **notification title** and **message content**.
5. Select the **recipients** (all staff or specific staff members).
6. Click on the **Send** button.
7. The system delivers the notification to selected users.

5.18.4 Notification Delivery

- Notifications are displayed on the **staff dashboard**.
- Users may also receive alerts depending on system configuration.
- All notifications are stored for future reference.

5.18.5 System Response

- **Notification Sent Successfully:**
A confirmation message is displayed after successful delivery.
- **Delivery Error:**
An error message is shown if the notification could not be sent.

5.18.6 Screenshot Placeholder

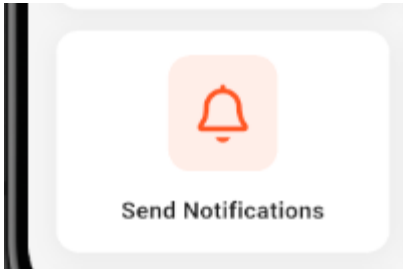


Figure 81 Send Notifications Screen

References

- **Flutter Team.** *Flutter Documentation.*
Available at: <https://docs.flutter.dev>
(Official framework documentation for mobile app development.)
- **Google.** *Firebase Documentation.*
Available at: <https://firebase.google.com/docs>
(Used for authentication, database, and backend services.)
- **PlantUML.** *PlantUML Language Reference Guide.*
Available at: <https://plantuml.com>
(Commonly used for UML diagram generation.)
- **IEEE Computer Society.** (2014). *IEEE Standard for Software Design Descriptions.*
(Used as a general guideline for system design documentation.)

Appendix

Appendix A: Tools and Technologies Used

The following tools and technologies were used during the development of the StaffSync system:

- Flutter – Mobile application development
- Firebase – Authentication and database services
- PlantUML – UML diagram generation
- Android Studio / VS Code – Development environment

Appendix B: User Roles

The StaffSync system supports the following user roles:

- **Administrator** – Manages staff, shifts, payroll, and reports
- **Supervisor** – Approves leaves, assigns and verifies tasks
- **Staff** – Marks attendance, applies for leave, completes tasks

Appendix C: System Screenshots

This appendix includes screenshots of key system screens for reference.

- Login Screen
- Dashboard
- Attendance Module
- Leave Management
- Task Management